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ATTACHMENT MECHANISMS FOR MODULAR BUILD PLATFORMS

Abstract

Systems, methods, and devices for additive manufacturing are provided. In some embodiments, an assembly for supporting 3D objects during an additive manufacturing process is provided. The assembly can include a plurality of build platforms, each build platform configured to support one or more 3D objects during the additive manufacturing process. The assembly can also include a carrier configured to support the plurality of build platforms. The assembly can further include an attachment mechanism configured to releasably couple the plurality of build platforms to the carrier during the additive manufacturing process such that the plurality of build platforms collectively form a build plane having a vertical deviation no greater than 500 μm.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] The present application claims the benefit of priority to U.S. Provisional Application No. 63/559,529, filed Feb. 29, 2024, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD
[0002] The present technology generally relates to manufacturing, and in particular, to attachment mechanisms for modular build platforms.
BACKGROUND
[0003] Additive manufacturing encompasses a variety of technologies that involve building up three-dimensional (3D) objects from multiple layers of material. Lithography-based additive manufacturing techniques generally involve curing a photoreactive resin by selectively exposing the resin to electromagnetic radiation, thereby forming a solid layer of cured material. This process can be repeated to build up a 3D object in a layer-by-layer manner. 3D objects that have been printed using lithography-based additive manufacturing techniques are typically subjected to finishing and post-processing steps. Usually, the printed objects are removed from the 3D printer together with the build platform onto which the objects are adhered, and the build platform serves as an aid for supporting and manipulating the objects during the finishing and post-processing steps. Although increasing the size of the build platform can increase the throughput of the additive manufacturing process by allowing more objects to be printed simultaneously, it may be challenging to use larger build platforms to support the objects during finishing and post-processing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale. Instead, emphasis is placed on illustrating clearly the principles of the present disclosure.
[0005] FIG. **1** is a flow diagram providing a general overview of a method for fabricating and post-processing an additively manufactured object, in accordance with embodiments of the present technology.
[0006] FIG. **2** is a partially schematic diagram providing a general overview of a lithography-based additive manufacturing process, in accordance with embodiments of the present technology.
[0007] FIG. **3** is a partially schematic diagram of a system for lithography-based additive manufacturing configured in accordance with embodiments of the present technology.
[0008] FIG. **4** is a side view of a modular build substrate for additive manufacturing, in accordance with embodiments of the present technology.
[0009] FIG. **5** is a partially schematic diagram of an additive manufacturing system including a modular build substrate, in accordance with embodiments of the present technology.
[0010] FIG. **6** is a partially schematic diagram of another additive manufacturing system including a modular build substrate, in accordance with embodiments of the present technology.

[0011] FIG. **7**A is a side view of a modular build substrate configured in accordance with embodiments of the present technology.
[0012] FIG. **7**B is a top view of the modular build substrate of FIG. **7**A.
[0013] FIG. **8**A is a side view of another modular build substrate configured in accordance with embodiments of the present technology.
[0014] FIG. **8**B is a top view of the modular build substrate of FIG. **8**A.
[0015] FIG. **9**A is a side view of yet another modular build substrate configured in accordance with embodiments of the present technology.
[0016] FIG. **9**B is a top view of the modular build substrate of FIG. **9**A.
[0017] FIG. **10**A is a side view of another modular build substrate configured in accordance with embodiments of the present technology.
[0018] FIG. **10**B is a top view of the modular build substrate of FIG. **10**A.
[0019] FIGS. **11**A and **11**B are side and top views, respectively, of a build platform including empty recesses in accordance with embodiments of the present technology.
[0020] FIGS. **11**C and **11**D are side and top views, respectively, showing placement of a prefabricated element into the recess of the build platform, in accordance with embodiments of the present technology.
[0021] FIGS. **11**E and **11**F are side and top views, respectively, showing a 3D object printed onto the prefabricated element in the build platform, in accordance with embodiments of the present technology.
[0022] FIGS. **11**G and **11**H are side and top views, respectively, showing removal of the final structure from the build platform, in accordance with embodiments of the present technology.
[0023] FIG. **12** is a side view of a build platform including an interface layer, in accordance with embodiments of the present technology.
[0024] FIG. **13**A is a side cross-sectional view of a portion of a modular build substrate including a carrier and a build platform, in accordance with embodiments of the present technology.
[0025] FIG. **13**B is a bottom view of a build platform including a plurality of pins, in accordance with embodiments of the present technology.
[0026] FIG. **13**C is a bottom view of a build platform including a plurality of ribs, in accordance with embodiments of the present technology.
[0027] FIG. **14** is a perspective view of a modular build substrate including a plurality of blocks, in accordance with embodiments of the present technology.
[0028] FIG. **15** is a side cross-sectional view of a modular build substrate including an external spring clip, in accordance with embodiments of the present technology.
[0029] FIG. **16**A is a side cross-sectional view of a modular build substrate including an internal spring clip, in accordance with embodiments of the present technology.
[0030] FIG. **16**B is a bottom view of a build platform of the modular build substrate of FIG. **16**A, in accordance with embodiments of the present technology.
[0031] FIG. **16**C is a top view of a carrier of the modular build substrate of FIG. **16**A, in accordance with embodiments of the present technology.
[0032] FIG. **17** is a side cross-sectional view of a modular build substrate including a plurality of spring clips, in accordance with embodiments of the present technology.
[0033] FIG. **18** is a side cross-sectional view of a modular build substrate including a plurality of spring clips, in accordance with embodiments of the present technology.
[0034] FIGS. **19**A and **19**B illustrate a process for coupling a build platform to a carrier using a placement tool, in accordance with embodiments of the present technology.
[0035] FIGS. **20**A and **20**B illustrate a process for decoupling a build platform from a carrier using a removal tool, in accordance with embodiments of the present technology.
[0036] FIG. **21**A is a perspective view of a modular build substrate including spring clips and rotatable clips, in accordance with embodiments of the present technology.

[0037] FIG. **21**B is a side view of a portion of the modular build substrate of FIG. **21**A, in accordance with embodiments of the present technology.
[0038] FIG. **21**C is a perspective view of a portion of the modular build substrate of FIG. **21**A, in accordance with embodiments of the present technology.
[0039] FIG. **22**A is a perspective view of a modular build substrate including a plurality of laterally rotatable clips, in accordance with embodiments of the present technology.
[0040] FIG. **22**B is a close-up perspective view of the modular build substrate of FIG. **22**A with the laterally rotatable clips in an open configuration, in accordance with embodiments of the present technology.
[0041] FIG. **22**C is a close-up perspective view of the modular build substrate of FIG. **22**A with the laterally rotatable clips in a closed configuration, in accordance with embodiments of the present technology.
[0042] FIG. **23** is a side cross-sectional view of a modular build substrate including a fixed clip and a rotatable clip, in accordance with embodiments of the present technology.
[0043] FIG. **24**A is a top view of a modular build substrate including a plurality of fixed clips and a plurality of rotatable clips, in accordance with embodiments of the present technology.
[0044] FIG. **24**B is a perspective view of a fixed clip of the modular build substrate of FIG. **24**A, in accordance with embodiments of the present technology.
[0045] FIG. **24**C is a perspective view of a portion of a build platform of the modular build substrate of FIG. **24**A, in accordance with embodiments of the present technology.
[0046] FIG. **24**D is a top view of a rotatable clip and a build platform of the modular build substrate of FIG. **24**A, in accordance with embodiments of the present technology.
[0047] FIG. **24**E is a perspective view of a rotatable clip of the modular build substrate of FIG. **24**A, in accordance with embodiments of the present technology.
[0048] FIG. **24**F is a perspective view of a rotatable clip of the modular build substrate of FIG. **24**A in an open configuration, in accordance with embodiments of the present technology.
[0049] FIG. **24**G is a perspective view of a rotatable clip of the modular build substrate of FIG. **24**A in a closed configuration, in accordance with embodiments of the present technology.
[0050] FIG. **25** is a side cross-sectional view of a modular build substrate including a plurality of magnets, in accordance with embodiments of the present technology.
[0051] FIG. **26** is a side cross-sectional view of a modular build substrate including a spring clip and a plurality of magnets, in accordance with embodiments of the present technology.
[0052] FIG. **27** is a side cross-sectional view of a portion of a modular build substrate including a carrier, a build platform, and an attachment mechanism including a bolt, in accordance with embodiments of the present technology.
[0053] FIG. **28** is a side cross-sectional view of a portion of a modular build substrate including a carrier, a build platform, and an attachment mechanism including a bolt, in accordance with embodiments of the present technology.
[0054] FIG. **29** is a side cross-sectional view of a modular build substrate including a carrier, a build platform, and registration features, in accordance with embodiments of the present technology.
[0055] FIG. **30** is a flow diagram illustrating a method for fabricating additively manufactured objects, in accordance with embodiments of the present technology.
[0056] FIG. **31** is a flow diagram illustrating a method for fabricating additively manufactured objects, in accordance with embodiments of the present technology.
[0057] FIG. **32**A illustrates a representative example of a tooth repositioning appliance configured in accordance with embodiments of the present technology.
[0058] FIG. **32**B illustrates a tooth repositioning system including a plurality of appliances, in accordance with embodiments of the present technology.
[0059] FIG. **32**C illustrates a method of orthodontic treatment using a plurality of appliances, in

accordance with embodiments of the present technology.
[0060] FIG. **33** illustrates a method for designing an orthodontic appliance, in accordance with embodiments of the present technology.
[0061] FIG. **34** illustrates a method for digitally planning an orthodontic treatment and/or design or fabrication of an appliance, in accordance with embodiments of the present technology.

DETAILED DESCRIPTION

I. Overview of Technology

[0062] The present technology relates to systems, devices, and methods for additive manufacturing of 3D objects, such as dental appliances. In some embodiments, for example, an assembly for supporting 3D objects during an additive manufacturing process is provided. The assembly (which may also be referred to herein as a “build substrate”) can include a plurality of build platforms, each build platform configured to support one or more 3D objects during the additive manufacturing process. The assembly can also include a carrier configured to support the plurality of build platforms, and an attachment mechanism configured to releasably couple the plurality of build platforms to the carrier during the additive manufacturing process. In some embodiments, the build platforms are coupled to the carrier such that the plurality of build platforms collectively form a flat build plane, e.g., a build plane having a vertical deviation no greater than 500 μm, or a vertical deviation within a range from 0 μm to 1 mm, 500 μm to 1 mm, etc.
[0063] The present technology can provide various advantages compared to conventional additive manufacturing techniques. For instance, the use of modular build platforms that can be individually coupled and decoupled from a carrier allows multiple build platforms to be assembled to form a single build substrate with a larger build plane for printing, which may be advantageous for producing a larger number of objects in a single printing operation and/or may be needed to accommodate certain types of additive manufacturing systems and processes. The build platforms can be attached to the carrier in a manner that ensures that the resulting build plane is sufficiently flat to avoid vertical deviations that could detrimentally affect adhesion of the printed objects to the build platforms, cause the printer to become misaligned with the build platforms, and/or reduce the accuracy of the printed objects. Moreover, the build platforms can be designed to accommodate excess material (e.g., resin) that may accumulate on the build platform and/or carrier during printing without disrupting the flat build plane.
[0064] After the additive manufacturing process, the build platforms can subsequently be removed from the carrier along with their respective objects, and may serve as a support for handling the objects during post-processing operations. The individual build platforms can be sufficiently small to be placed within post-processing devices (e.g., centrifuges, solvent baths, post-curing ovens) that are not capable of accommodating larger build platforms. The modular nature of the build platforms also allows for replacement of individual build platforms if a particular build platform becomes damaged or fouled, and also allows the type, geometry, and/or arrangement of the build platforms to be customized to accommodate different sizes, shapes, and/or arrangements of objects to be printed.
[0065] In some embodiments, 3D objects that have been printed via a lithography-based additive manufacturing method may be subjected to finishing and/or post-processing steps, such as removing uncured resin adhering to the object, removing support structures from the 3D printed objects, removing the 3D printed objects from the build platform (e.g., via a blade or other removal mechanism), and/or post-curing 3D objects via energy (e.g., UV light or heat-induced curing), in order to reach the objects' final specifications. Usually, the printed objects are removed from the additive manufacturing system together with the build platform onto which the objects are adhered. The build platform can be used as a carrier for supporting, handling, and/or manipulating the objects during the finishing and/or post-processing steps. However, it may be challenging to use larger build platforms as a carrier for the 3D printed objects during finishing and post-processing steps.

[0066] Therefore, it is an object of the present technology to improve lithography-based additive manufacturing methods so as to facilitate finishing and postprocessing steps in cases where a relatively large build platform is used. In some embodiments, the present technology provides a method of producing a plurality of 3D objects by lithography-based additive manufacturing. Further, the present technology can provide a 3D printer for carrying out this method.
[0067] The present technology in a first aspect thereof provides a method of producing a plurality of 3D objects by lithography-based additive manufacturing, comprising: [0068] providing a 3D printer comprising a carrier and a plurality of build platforms releasably fixed on the carrier, each build platform defining a build plane for building at least one 3D object thereon, the 3D printer further comprising a light engine for selectively curing layers of a light-polymerizable resin on the build platforms; [0069] building a plurality of 3D objects with the 3D printer, wherein at least one of said plurality of 3D objects is built on each build platform; [0070] removing the build platforms with said at least one 3D object placed thereon from the 3D printer; and [0071] subjecting the 3D objects, while being arranged on their respective build platform, to at least one post-processing step after the build platforms have been separated from the carrier.
[0072] Thus, this present technology is based on the idea to divide a large build platform into a plurality of smaller build platforms that are releasably fixed on a carrier. Due to their smaller size, each build platform can be easier to handle, and the individual build platforms can be handled separately from each other for subjecting the objects thereon to post-processing steps. Further, using a plurality of smaller build platforms instead of a single platform allows an even larger building area than in conventional approaches. Using a plurality of smaller build platforms can also provide a modular system, in which the size, arrangement and/or number of build platforms may be adapted to the requirements of the specific print job.
[0073] The build platforms may be removed from the 3D printer together with the carrier or after having been separated from the carrier. If the build platforms are removed from the 3D printer while still fixed to the carrier, the build platforms can be separated from the carrier before subjecting the 3D objects to the at least one post-processing step. In any case, for carrying out the at least one post-processing step, the build platforms may have been removed and be separated from the carrier. Removing the build platforms from the carrier may be undertaken manually or automatically. In the automatic embodiment, the 3D printer may comprise releasing means for releasing a holding force, such as, e.g., a magnetic or mechanical holding force.
[0074] Each build platform from the plurality of build platforms carries at least one 3D object. In some embodiments, a plurality of 3D objects is built on each build platform, such as five or more, 10 or more, or 20 or more objects.
[0075] The carrier may hold at least two build platforms. In some embodiments, a larger number of build platforms is releasably fixed on the carrier, such as three, four, or more build platforms, 20 or more build platforms, or 50 or more build platforms.
[0076] In some embodiments, the plurality of build platforms are arranged on the carrier to cover an area of at least 2000 cm.sup.2, such as at least 2500 cm.sup.2. In some instances, a single build platform having a size of, e.g., 100 cm×30 cm, or 150 cm×50 cm, does not easily fit in, e.g., centrifuges, solvent cleaning devices, or UV or thermal furnaces. Therefore, the segmentation of a large build platform into smaller segments that are easier to handle can be a straightforward way to facilitate the subsequent post-processing steps after printing is finished.
[0077] The build platforms may all have the same size and be arranged in a regular grid. Alternatively, the build platforms may have different sizes and shapes. The build platforms may be made of any kind of material that has a sufficient stiffness so as to be inherently stable to support the 3D objects printed thereon and to be handled during the post-processing steps. Suitable materials include metal, ceramic, glass, wood, or paper. For example, a simple sheet metal plate may be used as a build platform. Suitable metals are, for example, aluminum or steel. Further, the build platforms may have different surface finishes or a rough pattern, which can promote the

adhesion of the 3D object onto the build platform.
[0078] In some embodiments, the build platforms may have recesses, such as holes, drills, hollows, or gaps, which can promote resin to flow through or in said recesses. In some cases, the resin flowing through said recesses can be collected below the build platforms to be reused in a later printing job.
[0079] In some embodiments, the build platforms may be cooled. In other embodiments, the build platforms may be heated. The build platforms may be heated to a surface temperature at the build plane of 20° C. to 200° C., such as 30° C. to 90° C.
[0080] The build platforms can be releasably fixable on the carrier so that, on the one hand, their position on the carrier can reliably be kept stable during the printing process, and that, on the other hand, they may easily be detached from the carrier upon completion of the printing process. The build platforms may be releasably fixed to the carrier by means of mechanical clamping, electromagnetic forces, magnetic forces, vacuum, and/or a form-fit engagement. By being releasably fixable to the carrier, the build platforms can constitute exchangeable parts that may be reused for a plurality of production cycles.
[0081] In some embodiments, the build platforms are arranged and fixable on the carrier so as to provide a build plane that is common to all build platforms.
[0082] Due to the modular system provided by the use of a plurality of exchangeable build platforms, the build platforms may advantageously be adapted to the footprint of the one or the plurality of 3D objects to be printed onto the respective build platform.
[0083] In some embodiments, prefabricated elements can be fixed on and/or attached to one or more build platforms. Said fixed or attached prefabricated elements can then be manipulated by printing, e.g., on their surfaces.
[0084] The method of the present technology may be carried out by stereolithography manufacturing principles, e.g., by using 3D printers comprising a light engine for selectively curing layers of a light-polymerizable resin on the plurality of build platforms.
[0085] As used herein, “light” may include any electromagnetic radiation that is able to induce polymerization of a light-polymerizable resin. The term “light” needs not be restricted to visible light, e.g., the portion of the spectrum that can be perceived by the human eye. The radiation may have a wavelength in the range of 10 nm to 10,000 nm, such as 100 nm to 500 nm.
[0086] The term “light-polymerizable resin” may refer to a material that conforms into a hardened polymeric material through a curing process. A light-polymerizable resin may include, but is not limited to, a mixture of monomers, oligomers, and photoinitiators. A light-polymerizable resin may also be referred to as an uncured photopolymer.
[0087] Typically, a light-polymerizable resin may include (e.g., consist of) optionally at least one (reactive) oligomer, optionally at least one (reactive) diluent, at least one photoinitiator, optionally additives, and/or optionally fillers. Reactive groups may be unsaturated chemical bonds or cyclic chemical structures. Examples of reactive groups include alkenes, alkynes, vinyl compounds, (meth)acrylates, acrylamides, allyl compounds, norbornene, vinyl ethers, vinyl esters, epoxides, oxetanes, maleimides, thiols, and so forth. Oligomers may be optionally reactive group-functionalized and may comprise all kinds of polymerisates, polycondensation and polyaddition products, e.g., epoxies, polyesters, polyurethanes, copolymers, homopolymers, polyamides, polycarbonates, polythioethers, polythioesters, silicones, and many more. Reactive diluents may be monofunctional or multifunctional low molecular weight reactive monomers that serve as a reactive solvent in order to adjust process viscosity of the photoreactive resins and mechanical properties of the final photopolymer. Additives may include defoamers, wetting agents, leveling agents, flame retardants, UV stabilizers, UV absorbers, IR absorbers, thermal stabilizers, and/or thermal initiators. Examples for fillers may include metals, metallic alloys, ceramics, glass, polymers, natural fabrics, salts, and many more. Photoinitiators may form reactive species when exposed to radiation of certain wavelength(s) that trigger off the polymerization. Typical reactive

species include radicals, cations, anions, or activated catalytic species. Photoinitiators can also act in combination with catalysts, coinitiators, and/or sensitizers.

[0088] "Curing" the light-polymerizable resin may be a process, wherein the light-polymerizable resin is polymerized or cross-linked as a result of being irradiated by light.

[0089] As used herein, a "light engine" may be a device that is able to generate dynamic light information according to a predetermined pattern. As an example, liquid crystal displays, digital light processing, other active mask projection systems, and/or laser-scanner based systems may be used to selectively project light information on the surface of the light-polymerizable resin.

[0090] According to some embodiments of the present technology, the post-processing step is selected from removing uncured resin (e.g., by centrifuging the 3D object), washing the 3D object with fluids, removing solvents from the 3D object, subjecting the 3D object to pressurized air, drying the 3D object, removing support structures from the 3D object, removing the 3D object from the build platform, post-curing the 3D object by means of UV light, and/or heat-curing and/or microwave-curing the 3D object. In some embodiments, dual cure systems utilize at least one subsequent activation step (e.g., heat or microwave) after 3D printing to trigger off a second reaction to reach the final desired material properties.

[0091] In order to facilitate the separation of the 3D object from the build platform after the post-processing step, an interface layer may each be arranged on the build platforms, on top of which the 3D objects are built, wherein the at least one post-processing step comprises destabilizing, eliminating, or removing the interface layer, thereby causing the 3D object to be detached from the build platform. Removing or destabilizing the interface layer may comprise subjecting the interface layer to a physical and/or chemical process that causes the interface layer to disintegrate or lose its stability, such as by means of dissolving, etching, melting, or other chemical or physical means.

[0092] As to the nature of the interface layer, any material may be used that differs from the cured light-polymerizable resin in at least one physical and/or chemical property, such as the melting point, the solubility, the boiling point, etc. In some embodiments, the interface layer is made from a material that is polymerizable and, in its polymerized and pre-polymerized state, can be dissolved or swollen in a solvent. Such a material may therefore include (e.g., consist of) at least one polymerizable group, such as acrylates, methacrylates, acrylamides, vinyl ethers, vinyl esters, maleimides, cyclic ethers, isocyanates, amines, or other polymerizable unsaturated or saturated groups, and may optionally further comprise at least one hydrophilic or oleophilic group. The polymerized material may be dissolvable or swellable in a solvent, such as water, alcohol, oil, or other organic solvents. Such materials, for example, may comprise hydroxyl, carbonyl, and/or carboxyl groups, and/or derivatives with other electronegative hetero atoms, amines, ionic liquids and salts, for example, hydroxyethylacrylate (HEA), hydroxyethylmethacrylate (HEMA), 2-(2-ethoxyethoxy)ethyl acrylate (EOEOEA), acryloyl morpholine (ACMO), polyethylene glycol derivates, polyethers, hydroxy ethylene, or lauryl acrylates. The interface layer may be applied by a spray coating.

[0093] Alternatively, a film may each be arranged on the build platforms and the 3D objects are built on top of the film, and wherein the at least one post-processing step comprises peeling the film off from the build platform, thereby detaching the 3D object from the build platform.

[0094] In order to integrate prefabricated elements into the 3D printed object, some embodiments provide that, before printing the 3D object, a prefabricated element is placed and/or mounted onto the build platform or into a recess of the build platform and at least one of the layers of light-polymerizable resin is bonded to the prefabricated element during the printing of the 3D object. The prefabricated element can be made of a material that is different from the light-polymerizable resin. In this way multi-material combinations are possible. Further, a combination of parts may be realized that cannot be printed in the same process.

[0095] In accordance with lithography-based additive manufacturing methods, the 3D object can be built on the building platform layer-by-layer to obtain a stack of structured layers, wherein each structured layer is obtained by the steps of: [0096] providing an unstructured layer of light-polymerizable resin; and [0097] selectively projecting light onto the unstructured layer according to a desired pattern, thereby curing the light-polymerizable resin to obtain the structured layer that is structured according to the pattern.

[0098] The use of a plurality of smaller build platforms instead of a single, larger platform is particularly useful in a 3D printing method disclosed in International Publication Nos. WO 2021/130654, WO 2021/130657, and WO 2021/130661, which may be characterized by a significantly larger build area than in conventional embodiments. Therefore, some embodiments of the present technology provide that at least one of the light engine and the carrier is driven for relative movement to one another while selectively curing a layer of the light-polymerizable resin, so that an exposure field of the light engine sweeps across said plurality of build platforms.

[0099] In some embodiments, the light engine is configured for the dynamic patterning of light in the exposure field of said light engine, wherein pattern data is fed to the light engine so that a light pattern is scrolled in the exposure field at a rate that corresponds to the relative movement speed of the light engine and the carrier.

[0100] According to a second aspect, the present technology provides a 3D printer for carrying out a method according to the first aspect of the present technology, comprising a carrier and a plurality of build platforms releasably fixed on the carrier, each build platform defining a build plane for building at least one 3D object thereon, the 3D printer further comprising a light engine for selectively curing layers of a light-polymerizable resin on the build platforms.

[0101] In some embodiments, fixing means are provided for fixing the position and/or orientation of the build platforms on the carrier, the fixing means being selected from mechanical clamping means, electromagnetic holding means, magnetic holding means, vacuum means, and form-fit engagement means.

[0102] In some embodiments, the carrier comprises means for collecting excess resin that might drop through the gaps between neighboring build platforms. The excess resin may be collected in a suitable vessel, wherein the collected material may be recycled for being reused in another printing process.

[0103] In some embodiments, the build platforms and/or the carrier may be equipped with machine-readable identification means, such as RFID chips or engraved patterns readable by a camera system, in order that the individual build platforms can be automatically identified.

[0104] In some embodiments, at least one of the light engine and the carrier is driven for relative movement to one another while selectively curing a layer of the light-polymerizable resin, for an exposure field of the light engine to sweep across said plurality of build platforms.

[0105] According to some embodiments, the printer may comprise means for applying an unstructured layer of light-polymerizable resin onto the build platform or on the partially built object, wherein the light engine is designed for the patterning of light onto the unstructured layer of light-polymerizable resin, the light engine being adapted to cure the light-polymerizable resin to obtain the structured layer that is structured according to the pattern.

[0106] In some embodiments, the light engine is designed for the dynamic patterning of light in the exposure field of said light engine, wherein pattern data is fed to the light engine so that a light pattern is scrolled in the exposure field at a rate that corresponds to the relative movement speed of the light engine and the carrier.

[0107] The present technology may be used for manufacturing various types of 3D objects. Examples for applications of 3D printed objects include electronics, electric and electro-mechanic components, connectors, housings, automobile and aerospace sectors, electric mobility, communication technology, computer technology, military technology, medical devices, medical technology, consumer goods, sports industry, energy industry, printed electronics, and dental and orthodontic applications. Orthodontic applications comprise, without limitations, aligners, retainers, brackets and wires, whitening trays, mouth trays and guards, aligners for drug delivery, aligners for the detection of substances, night guards, anti-bruxing or anti-grinding devices, tongue thrust devices, palatal expanders, oral appliance therapy for treatment of malocclusion, sleep apnea, anti-snoring devices, attachment templates, mandibular advancement devices, prefabricated attachment templates, etc., each of which includes the methods and processes for these purposes.

[0108] Embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings in which like numerals represent like elements throughout the several figures, and in which example embodiments are shown. Embodiments of the claims may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. The examples set forth herein are non-limiting examples and are merely examples among other possible examples.

[0109] As used herein, the terms “vertical,” “lateral,” “upper,” “lower,” “left,” “right,” etc., can refer to relative directions or positions of features of the embodiments disclosed herein in view of the orientation shown in the Figures. For example, “upper” or “uppermost” can refer to a feature positioned closer to the top of a page than another feature. These terms, however, should be construed broadly to include embodiments having other orientations, such as inverted or inclined orientations where top/bottom, over/under, above/below, up/down, and left/right can be interchanged depending on the orientation.

[0110] The headings provided herein are for convenience only and do not interpret the scope or meaning of the claimed present technology. Embodiments under any one heading may be used in conjunction with embodiments under any other heading.

II. Modular Build Platforms for Additive Manufacturing

A. Additive Manufacturing Technology

[0111] FIG. **1** is a flow diagram providing a general overview of a method **100** for fabricating and post-processing an additively manufactured object, in accordance with embodiments of the present technology. The method **100** can be used to produce many different types of additively manufactured objects, such as orthodontic appliances (e.g., aligners, palatal expanders, retainers, attachment placement devices, attachments), restorative objects (e.g., crowns, veneers, implants), and/or other dental appliances and devices (e.g., oral sleep apnea appliances, mouth guards). Additional examples of dental appliances and associated methods that are applicable to the present technology are described in Section III below.

[0112] The method **100** begins at block **102** with fabricating an object on a build platform using an additive manufacturing process. Additive manufacturing (also referred to herein as “3D printing”) includes a variety of technologies which fabricate 3D objects directly from digital models through an additive process. In some embodiments, additive manufacturing includes depositing a precursor material onto a build platform. The build platform can be one of a plurality of modular build platforms that are releasably coupled to a carrier, as described in Section II.B below. The precursor material can be cured, polymerized, melted, sintered, fused, and/or otherwise solidified to form a portion of the object and/or to combine the portion with previously formed portions of the object. In some embodiments, the additive manufacturing techniques provided herein build up the object geometry in a layer-by-layer fashion, with successive layers being formed in discrete build steps. Alternatively or in combination, the additive manufacturing techniques described herein can allow for continuous build-up of an object geometry.

[0113] The additive manufacturing process can implement any suitable technique known to those of skill in the art. Examples of additive manufacturing techniques include, but are not limited to, the following: (1) vat photopolymerization, in which an object is constructed from a vat or other bulk source of liquid photopolymer resin, including techniques such as stereolithography (SLA), digital light processing (DLP), continuous liquid interface production (CLIP), two-photon induced photopolymerization (TPIP), and volumetric additive manufacturing; (2) material jetting, in which material is jetted onto a build platform using either a continuous or drop on demand (DOD) approach; (3) binder jetting, in which alternating layers of a build material (e.g., a powder-based material) and a binding material (e.g., a liquid binder) are deposited by a print head; (4) material extrusion, in which material is drawn though a nozzle, heated, and deposited layer-by-layer, such as fused deposition modeling (FDM) and direct ink writing (DIW); (5) powder bed fusion, including techniques such as direct metal laser sintering (DMLS), electron beam melting (EBM), selective heat sintering (SHS), selective laser melting (SLM), and selective laser sintering (SLS); (6) sheet lamination, including techniques such as laminated object manufacturing (LOM) and ultrasonic additive manufacturing (UAM); and (7) directed energy deposition, including techniques such as laser engineering net shaping, directed light fabrication, direct metal deposition, and 3D laser cladding. Optionally, an additive manufacturing process can use a combination of two or more additive manufacturing techniques.

[0114] For example, the additively manufactured object can be fabricated using a vat photopolymerization process in which light is used to selectively cure a vat or other bulk source of a curable material (e.g., a polymeric resin). Each layer of curable material can be selectively exposed to light in a single exposure (e.g., DLP) or by scanning a beam of light across the layer (e.g., SLA). Vat polymerization can be performed in a "top-down" or "bottom-up" approach, depending on the relative locations of the material source, light source, and build platform.

[0115] As another example, the additively manufactured object can be fabricated using high temperature lithography (also known as "hot lithography"). High temperature lithography can include any photopolymerization process that involves heating a photopolymerizable material (e.g., a polymeric resin). For example, high temperature lithography can involve heating the material to a temperature of at least 30° C., 40° C., 50° C., 60° C., 70° C., 80° C., 90° C., 100° C., 110° C., or 120° C. In some embodiments, the material is heated to a temperature within a range from 50° C. to 120° C., from 90° C. to 120° C., from 100° C. to 120° C., from 105° C. to 115° C., or from 105° C. to 110° C. The heating can lower the viscosity of the photopolymerizable material before and/or during curing, and/or increase reactivity of the photopolymerizable material. Accordingly, high temperature lithography can be used to fabricate objects from highly viscous and/or poorly flowable materials, which, when cured, may exhibit improved mechanical properties (e.g., stiffness, strength, stability) compared to other types of materials. For example, high temperature lithography can be used to fabricate objects from a material having a viscosity of at least 5 Pa-s, 10 Pa-s, 15 Pa-s, 20 Pa-s, 30 Pa-s, 40 Pa-s, or 50 Pa-s at 20° C. Representative examples of high-temperature lithography processes that may be incorporated in the methods herein are described in International Publication Nos. WO 2015/075094, WO 2016/078838, WO 2018/032022, WO 2020/070639, WO 2021/130657, and WO 2021/130661, the disclosures of each of which are incorporated herein by reference in their entirety.

[0116] In some embodiments, the additively manufactured object is fabricated using continuous liquid interphase production (also known as "continuous liquid interphase printing") in which the object is continuously built up from a reservoir of photopolymerizable resin by forming a gradient of partially cured resin between the building surface of the object and a polymerization-inhibited "dead zone." In some embodiments, a semi-permeable membrane is used to control transport of a photopolymerization inhibitor (e.g., oxygen) into the dead zone in order to form the polymerization gradient. Representative examples of continuous liquid interphase production processes that may be incorporated in the methods herein are described in U.S. Patent Publication Nos. 2015/0097315, 2015/0097316, and 2015/0102532, the disclosures of each of which are incorporated herein by reference in their entirety.

[0117] As another example, a continuous additive manufacturing method can achieve continuous build-up of an object geometry by continuous movement of the build platform (e.g., along the vertical or Z-direction) during the irradiation phase, such that the hardening depth of the irradiated photopolymer is controlled by the movement speed. Accordingly, continuous polymerization of material on the build surface can be achieved. Such methods are described in U.S. Pat. No. 7,892,474, the disclosure of which is incorporated herein by reference in its entirety. In another example, a continuous additive manufacturing method can involve extruding a composite material composed of a curable liquid material surrounding a solid strand. The composite material can be extruded along a continuous 3D path in order to form the object. Such methods are described in U.S. Pat. No. 10,162,624 and U.S. Patent Publication No. 2014/0061974, the disclosure of which is incorporated herein by reference in its entirety. In yet another example, a continuous additive manufacturing method can utilize a "heliolithography" approach in which the liquid photopolymer is cured with focused radiation while the build platform is continuously rotated and raised. Accordingly, the object geometry can be continuously built up along a spiral build path. Such methods are described in U.S. Pat. No. 10,162,264 and U.S. Patent Publication No. 2014/0265034, the disclosures of which are incorporated herein by reference in their entirety.

[0118] In a further example, the additively manufactured object can be fabricated using a volumetric additive manufacturing (VAM) process in which an entire object is produced from a 3D volume of resin in a single print step, without requiring layer-by-layer build up. During a VAM process, the entire build volume is irradiated with energy, but the projection patterns are configured such that only certain voxels will accumulate a sufficient energy dosage to be cured. Representative examples of VAM processes that may be incorporated into the present technology include tomographic volumetric printing, holographic volumetric printing, multiphoton volumetric printing, and xolography. For instance, a tomographic VAM process can be performed by projecting 2D optical patterns into a rotating volume of photosensitive material at perpendicular and/or angular incidences to produce a cured 3D structure. A holographic VAM process can be performed by projecting holographic light patterns into a stationary reservoir of photosensitive material. A xolography process can use photoswitchable photoinitiators to induce local polymerization inside a volume of photosensitive material upon linear excitation by intersecting light beams of different wavelengths. Additional details of VAM processes suitable for use with the present technology are described in U.S. Pat. No. 11,370,173, U.S. Patent Publication No. 2021/0146619, U.S. Patent Publication No. 2022/0227051, International Publication No. WO 2017/115076, International Publication No. WO 2020/245456, International Publication No. WO 2022/011456, and U.S. Provisional Patent Application No. 63/181,645, the disclosures of each of which are incorporated herein by reference in their entirety.

[0119] In yet another example, the additively manufactured object can be fabricated using a powder bed fusion process (e.g., selective laser sintering) involving using a laser beam to selectively fuse a layer of powdered material according to a desired cross-sectional shape in order to build up the object geometry. As another example, the additively manufactured object can be fabricated using a material extrusion process (e.g., fused deposition modeling) involving selectively depositing a thin filament of material (e.g., thermoplastic polymer) in a layer-by-layer manner in order to form an object. In yet another example, the additively manufactured object can be fabricated using a material jetting process involving jetting or extruding one or more materials onto a build surface in order to form successive layers of the object geometry.

[0120] The additively manufactured object can be made of any suitable material or combination of materials. As discussed above, in some embodiments, the additively manufactured object is made partially or entirely out of a polymeric material, such as a curable polymeric resin. The resin can be composed of one or more monomer components that are initially in a liquid state. The resin can be in the liquid state at room temperature (e.g., 20° C.) or at an elevated temperature (e.g., a temperature within a range from 50° C. to 120° C.). When exposed to energy (e.g., light), the monomer components can undergo a polymerization reaction such that the resin solidifies into the desired object geometry. Representative examples of curable polymeric resins and other materials suitable for use with the additive manufacturing techniques herein are described in International Publication Nos. WO 2019/006409, WO 2020/070639, and WO 2021/087061, the disclosures of each of which are incorporated herein by reference in their entirety.

[0121] Optionally, the additively manufactured object can be fabricated from a plurality of different materials (e.g., at least two, three, four, five, or more different materials). The materials can differ from each other with respect to composition, curing conditions (e.g., curing energy wavelength), material properties before curing (e.g., viscosity), material properties after curing (e.g., stiffness, strength, transparency), and so on. In some embodiments, the additively manufactured object is formed from multiple materials in a single manufacturing step. For instance, a multi-tip extrusion apparatus can be used to selectively dispense multiple types of materials from distinct material supply sources in order to fabricate an object from a plurality of different materials. Examples of such methods are described in U.S. Pat. Nos. 6,749,414 and 11,318,667, the disclosures of which are incorporated herein by reference in their entirety. Alternatively or in combination, the additively manufactured object can be formed from multiple materials in a plurality of sequential manufacturing steps. For instance, a first portion of the object can be formed from a first material in accordance with any of the fabrication methods herein, then a second portion of the object can be formed from a second material in accordance with any of the fabrication methods herein, and so on, until the entirety of the object has been formed.

[0122] After the additively manufactured object is fabricated, the object can undergo one or more additional process steps, also referred to herein as “post-processing.” As described in detail below with respect to blocks **104-108**, post-processing can include removing residual material from the object, curing the object, and/or separating the object from the build platform.

[0123] For example, at block **104**, the method **100** continues with removing residual material from the object. The residual material can include excess precursor material (e.g., uncured resin) and/or other unwanted material (e.g., debris) that remains on or within the object after the additive manufacturing process. The residual material can be removed in many different ways, such as by exposing the object to a solvent (e.g., via spraying, immersion), heating or cooling the object, applying a vacuum to the object, blowing a pressurized gas onto the object, applying mechanical forces to the object (e.g., vibration, agitation, centrifugation, tumbling, brushing), and/or other suitable techniques. Optionally, the residual material can be collected and/or processed for reuse.

[0124] At block **106**, the method **100** can optionally include curing the object. This additional curing step (also known as “post-curing”) can be used in situations where the object is still in a partially cured “green” state after fabrication. For example, the energy used to fabricate the object in block **102** may only partially polymerize the precursor material forming the object. Accordingly, the post-curing step may be needed to fully cure (e.g., fully polymerize) the object to its final, usable state. Post-curing can provide various benefits, such as improving the mechanical properties (e.g., stiffness, strength) and/or temperature stability of the object. Post-curing can be performed by heating the object, applying radiation (e.g., UV, visible, microwave) to the object, or suitable combinations thereof. In other embodiments, however, the post-curing process of block **106** is optional and can be omitted.

[0125] At block **108**, the method **100** can include separating the object from the build platform. The build platform can mechanically support the object during the additive manufacturing and/or the post-processing steps described herein. In some embodiments, the build platform is coupled to a carrier during additive manufacturing, and is removed from the carrier during post-processing. Additional details and examples of build platforms that may be used in the method **100** are described in Section II.B below.

[0126] The method **100** illustrated in FIG. **1** can be modified in many different ways. For example, although the above steps of the method **100** are described with respect to a single object, the method **100** can be used to sequentially or concurrently fabricate and post-process any suitable number of objects, such as tens, hundreds, or thousands of additively manufactured objects. As another example, the ordering of the processes shown in FIG. **1** can be varied (e.g., the process of block **108** can be performed before and/or concurrently with the processes of blocks **104** and/or **106**). Some of the processes of the method **100** can be omitted, such as the process of block **106**.

[0127] Additionally, the method **100** can include processes not shown in FIG. **1**, such as cleaning

the object (e.g., washing), annealing, trimming the object to remove structures that are not intended to be present in the final product (e.g., residual parts of the support structures), and/or packaging the object for shipment. Optionally, the method **100** can include modifying at least one surface of the object. The surface modifications can be applied to some or all of the surfaces of the object (e.g., the exterior and/or interior surfaces) to alter one or more surface characteristics, such as the surface finish (e.g., roughness, waviness, lay), porosity, visual appearance (e.g., gloss, transparency, visibility of print lines), hydrophobicity, and/or chemical reactivity. In some embodiments, the surface modifications include removing material from the object, e.g., by polishing, abrading, blasting, etc. Alternatively or in combination, the surface modifications can include applying an additional material to the object. For example, the additional material can be a coating, such as a polymeric coating. The coating can be applied to one or more surfaces of the object for various purposes, including, but not limited to: providing a smooth surface finish, which can be beneficial for aesthetics and/or to improve user comfort if the object is intended to be in contact with the user's body (e.g., an orthodontic appliance worn on the teeth); coloring and/or applying other aesthetic features to the object; improving scratch resistance and/or other mechanical properties; providing antimicrobial properties; and incorporating therapeutic agents into the object for controlled release.
[0128] In some embodiments, the present technology utilizes a lithography-based additive manufacturing process. Lithography-based additive manufacturing generally refers to methods in which a curable material (e.g., a photoreactive resin) is selectively exposed to energy (e.g., electromagnetic radiation) and cures upon exposure to the energy, thereby forming a solid layer of cured material. In some embodiments, the very first layer adheres to a build platform and shows sufficient bonding during the manufacturing process to support the rest of the object. Subsequent layers of cured material are repeatedly added upon the already cured layer, thus generating a 3D object. Examples of lithography-based additive manufacturing processes include SLA, DLP, liquid crystal display (LCD) printing, two-photon polymerization (2PP) (also known as two-photon induced polymerization (TPIP)), inkjet printing (e.g., MultiJet printing), volumetric 3D printing, and other suitable radiation-curable technologies, as well as their combinations and/or combinations of other manufacturing approaches. Compared to other additive manufacturing technologies, lithography-based additive manufacturing process can yield geometrically complex, highly resolved objects with exceptional surface finish.
[0129] Traditional lithography-based additive manufacturing processes use large photopolymer resin vats, in which a build platform and the layers of the object already printed on the build platform are submerged during the printing process. In these systems, new layers are added on top of each other at the surface of the liquid resin bath. Various light sources are typically used in order to induce photopolymerization of the liquid photopolymer resin layer. As an example, DLP, other active mask projection systems, and/or laser-scanner based systems may be used to selectively project light information on the surface of the photopolymer resin. These printing concepts advantageously allow use of large resin vats and often result in large building areas.
[0130] However, generating a thin layer of resin between a submerged structure and the free surface of the liquid resin bath may be limited in accuracy (e.g., regarding the liquid layer thickness) due to a variety of factors, including the viscosity and/or surface tension phenomena of the resin formulation used. Further, feature accuracy is typically limited when large building areas are used, even if laser-scanner based systems are used. Optical limitations of the scanner lens construction, timing limitations of the traditionally used pulse laser sources, and/or large deviation angles of the scanning field may result in accuracy limitations of the whole printing process, and/or accuracy shifts between the center and the edge of the building area. Another issue is the need for significant amounts of photopolymer material before a printing job can be started (e.g., vat filling procedure). As photopolymer resins can become chemically unstable, resin storage and degradation as well as cleaning a large resin vat can become an economical problem and limits the process

stability over time.

[0131] Some lithography-based approaches use vat-based concepts, where a liquid resin is filled into a transparent material vat. According to these approaches, a layer of the liquid resin is irradiated by selective light information from below, e.g., through the bottom of the material vat, so that the printed components are generated upside-down, sticking to a so-called build platform. These systems present some advantages, such as the possibility of mechanically adjusting the resin layer height by lowering the building platform into the resin vat. By doing so, layers of resin with desired thicknesses (e.g., thin layers of resin) and/or products with features of desired resolutions (e.g., products with high feature resolution have become possible.

[0132] Further, with certain process adaptions like heating or thin film coating (see, e.g., European Patent No. EP3284583 and European Patent No. EP3418033, the disclosures of which are incorporated by reference herein in their entirety), highly viscous materials (e.g., resins) can be printed, which may not be capable of processing with any other additive manufacturing technique. Highly viscous materials can lead to advanced mechanical properties of parts printed with stereolithography. Dynamic lithography-based manufacturing technologies have massively increased the available build area and consequently significantly increased the production rate of small, highly resolved objects. These novel technologies are based on a light engine that is driven to travel across a large build platform while depositing a layer of light-polymerizable resin by means of an endless carrier foil and curing said layer. Examples of such technologies are provided in International Publication Nos. WO 2021/130654, WO 2021/130657, and WO 2021/130661, the disclosures of which are incorporated by reference herein in their entirety.

[0133] FIG. **2** is a partially schematic diagram providing a general overview of a lithography-based additive manufacturing process, in accordance with embodiments of the present technology. In the illustrated embodiment, an object **202** is fabricated on a build platform **204** from a series of cured material layers, with each layer having a geometry corresponding to a respective cross-section of the object **202**. To fabricate an individual object layer, a layer of curable material **206** (e.g., polymerizable resin) is brought into contact with the build platform **204** (when fabricating the first layer of the object **202**) or with the previously formed portion of the object **202** on the build platform **204** (when fabricating subsequent layers of the object **202**). In some embodiments, the curable material **206** is formed on and supported by a substrate (not shown), such as a film. Energy **208** (e.g., light) from an energy source **210** (e.g., a laser, projector, or light engine) is then applied to the curable material **206** to form a cured material layer **212** on the build platform **204** or on the object **202**. The remaining curable material **206** can then be moved away from the build platform **204** (e.g., by lowering the build platform **204**, by moving the build platform **204** laterally, by raising the curable material **206**, and/or by moving the curable material **206** laterally), thus leaving the cured material layer **212** in place on the build platform **204** and/or object **202**. The fabrication process can then be repeated with a fresh layer of curable material **206** to build up the next layer of the object **202**.

[0134] The illustrated embodiment shows a “top down” configuration in which the energy source **210** is positioned above and directs the energy **208** down toward the build platform **204**, such that the object **202** is formed on the upper surface of the build platform **204**. Accordingly, the build platform **204** can be incrementally lowered relative to the energy source **210** as successive layers of the object **202** are formed. In other embodiments, however, the additive manufacturing process of FIG. **2** can be performed using a “bottom up” configuration in which the energy source **210** is positioned below and directs the energy **208** up toward the build platform **204**, such that the object **202** is formed on the lower surface of the build platform **204**. Accordingly, the build platform **204** can be incrementally raised relative to the energy source **210** as successive layers of the object **202** are formed.

[0135] FIG. **3** illustrates a representative example of a system **300** for lithography-based additive manufacturing configured in accordance with embodiments of the present technology. The system

300 can be used to fabricate any embodiment of the objects described herein. For example, the system **300** can be used to produce an object in accordance with block **102** of the method **100** of FIG. **1**.

[0136] The system **300** includes a printer assembly **302** configured to fabricate an additively manufactured object **304** ("object **304**") using any of the additive manufacturing processes described herein, such as a lithography-based additive manufacturing process. The printer assembly **302** is configured to deposit a curable material **306** (e.g., a polymerizable resin or other solidifiable precursor material) on a build platform **308** (e.g., a tray, plate, film, sheet, or other planar substrate) to form the object **304**. In the illustrated embodiment, the printer assembly **302** includes a carrier film **310** configured to deliver the curable material **306** to the build platform **308**. The carrier film **310** can be a flexible loop of material having an outer surface and an inner surface. The outer surface of the carrier film **310** can adhere to and carry a thin layer of the curable material **306**. The inner surface of the carrier film **310** can contact one or more rollers **312** that rotate to move the carrier film **310** in a continuous loop trajectory, e.g., along the direction indicated by arrows **314**.

[0137] The printer assembly **302** can also include a material source **316** (shown schematically) configured to apply the curable material **306** to the carrier film **310**. In the illustrated embodiment, the material source **316** is located at the upper portion of the printer assembly **302**. In other embodiments, however, the material source **316** can be at a different location in the printer assembly **302**. The material source **316** can include nozzles, ports, vats, reservoirs, etc., that deposit the curable material **306** onto the outer surface of the carrier film **310**. The material source **316** can also include one or more blades (e.g., doctor blades, recoater blades) that smooth the deposited curable material **306** into a relatively thin, uniform layer. For example, the curable material **306** can be formed into a layer having a thickness within a range from 200 microns to 300 microns, or any other desired thickness.

[0138] The curable material **306** can be conveyed by the carrier film **310** toward the build platform **308**. In the illustrated embodiment, the build platform **308** is located below the printer assembly **302**. In other embodiments, however, the build platform **308** can be positioned at a different location in the printer assembly **302**. The distance between the carrier film **310** and build platform **308** can be adjustable so that the curable material **306** at can be brought into direct contact with the surface of the build platform **308** (when printing the initial layer of the object **304**) or with the surface of the object **304** (when printing subsequent layers of the object **304**). For example, the build platform **308** can include or be coupled to a motor (not shown) that raises and/or lowers the build platform **308** to the desired height during the manufacturing process. Alternatively or in combination, the printer assembly **302** can include or be coupled to a motor (not shown) that raises and/or lowers the printer assembly **302** relative to the build platform **308**.

[0139] The printer assembly **302** includes an energy source **318** (e.g., a projector, light engine, laser) that outputs energy **320** (e.g., light, such as UV light) having a wavelength configured to partially or fully cure the curable material **306**. The carrier film **310** can be partially or completely transparent to the wavelength of the energy **320** to allow the energy **320** to pass through the carrier film **310** and onto the portion of the curable material **306** above the build platform **308**. Optionally, a transparent plate **322** can be disposed between the energy source **318** and the carrier film **310** to guide the carrier film **310** into a specific position (e.g., height) relative to the build platform **308**. During operation, the energy **320** can be patterned or scanned in a suitable pattern onto the curable material **306**, thus forming a layer of cured material onto the build platform **308** and/or on a previously formed portion of the object **304**. The geometry of the cured material can correspond to the desired cross-sectional geometry for the object **304**. The parameters for operating the energy source **318** (e.g., energy intensity, energy dosage, exposure time, exposure pattern, exposure wavelength, energy density, power density) can be set based on instructions from a controller **324**, as described in further detail below.

[0140] In some embodiments, the energy **320** is applied to the curable material **306** while the carrier film **310** moves to circulate the curable material **306** through the exposure zone of the energy source **318**. To maintain zero or substantially zero relative velocity between the curable material **306** and the build platform **308**, the printer assembly **302** can concurrently move horizontally relative to the build platform **308** opposite the direction of the motion of the carrier film **310** at the exposure zone. The energy **320** output by the energy source **318** can be coordinated with the movement of the carrier film **310** and build platform **308** so that the layer of cured material is formed with the correct geometry. For example, the energy source **318** can be a scrolling light engine (e.g., a scrolling digital light processing engine) that outputs an energy pattern that varies over time to match the motion of the printer assembly **302** and carrier film **310**. In other embodiments, however, the printer assembly **302** can be a stationary device that does not move relative to the build platform **308** while the energy **320** is being applied to the curable material **206**.

[0141] The newly formed layer of cured material can be separated from the carrier film **310** and the remaining curable material **306** at or after the exposure zone. In some embodiments, the separation occurs at least in part due to peel-off forces produced by the carrier film **310** wrapping around the roller **312** immediately downstream of the exposure zone. Peel-off forces can alternatively or additionally be provided by movements of the build platform **308** and/or printer assembly **302** (e.g., raising the printer assembly **302** away from the build platform **308**, moving the printer assembly **302** laterally away from the build platform **308**); use of a roller, blade, or other mechanism to facilitate separation of the cured material from the carrier film **310**; and/or other parameters of the printer assembly such as movement speed of the carrier film **310**.

[0142] The remaining curable material **306** can be carried by the carrier film **310** away from the build platform **308** and back toward the material source **316**. The material source **316** can deposit additional curable material **306** onto the carrier film **310** and/or smooth the curable material **306** to re-form a uniform layer of curable material **306** on the carrier film **310**. The curable material **306** can then be recirculated back to the build platform **308** to fabricate an additional layer of the object **304**. This process can be repeated to iteratively build up individual object layers on the build platform **308** until the object **304** is complete. The object **304** and build platform **308** can then be removed from the system **300** for post-processing.

[0143] In some embodiments, the system **300** is used in a high temperature lithography process utilizing a highly viscous curable material **306** (e.g., a highly viscous resin). Accordingly, the printer assembly **302** can include one or more heat sources (heating plates, infrared lamps, etc.) for heating the curable material **306** to lower the viscosity to a range suitable for additive manufacturing. For example, the printer assembly **302** can include a first heat source **326***a* positioned against the segment of the carrier film **310** before the build platform **308**, and a second heat source **326***b* positioned against the segment of the carrier film **310** after the build platform **308**. Alternatively or in combination, the printer assembly **302** can include heat sources at other locations.

[0144] The system **300** also includes a controller **324** (shown schematically) that is operably coupled to the printer assembly **302** and build platform **308** to control the operation thereof. The controller **324** can be or include a computing device including one or more processors and memory storing instructions for performing the additive manufacturing operations described herein. For example, the controller **324** can receive a digital data set (e.g., a 3D model) representing the object **304** to be fabricated, determine a plurality of object cross-sections to build up the object **304** from the curable material **306**, and can transmit instructions to the energy source **318** to output energy **320** to form the object cross-sections. The controller **324** can control the energy application parameters of the energy source **318**, such as the energy intensity, energy dosage, exposure time, exposure pattern, energy wavelength, and/or energy type of the energy **320** applied to the curable material **306**. Optionally, the controller **324** can also determine and control other operational parameters, such as the positioning of the build platform **308** (e.g., height) relative to the carrier film **310**, the movement speed and direction of the carrier film **310**, the amount of curable material

306 deposited by the material source **316**, the thickness of the material layer on the carrier film **310**, and/or the amount of heating applied to the curable material **306**.

[0145] Although FIG. **3** illustrates a representative example of a system **300** for additive manufacturing, this is not intended to be limiting, and the methods described herein can be implemented using other types of additive manufacturing systems, such as vat-based systems, material jetting systems, binder jetting systems, material extrusion systems, powder bed fusion systems, sheet lamination systems, or directed energy deposition systems.

B. Modular Build Platforms and Associated Systems and Methods

[0146] FIG. **4** is a side view of a modular build substrate **400** for additive manufacturing, in accordance with embodiments of the present technology. The modular build substrate **400** can be used to support a plurality of objects **402** during additive manufacturing and/or post-processing of the objects **402**, such as during any of the processes described in Section II.A above. For example, the modular build substrate **400** can be used during a lithography-based additive manufacturing process for fabricating the objects **402**.

[0147] The modular build substrate **400** includes a carrier **404** and a plurality of build platforms **406** attached to the carrier **404**. The carrier **404** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platforms **406** during the entire additive manufacturing process. The carrier **404** can be configured to hold any suitable number of build platforms **406**, such as two, three, four, five, 10, 20, 50, or more build platforms **406**. The build platforms **406** can be arranged on the carrier **404** in a linear array (e.g., a row), a 2D array (e.g., a regular grid), or any other suitable configuration, e.g., as described further below in connection with FIGS. **7**A-**10**B. The arrangement of the build platforms **406** can be varied based on the number and/or geometry of the objects **402** to be fabricated. The build platforms **406** can collectively form a build plane **408** that defines the total area available for printing the objects **402**. In some embodiments, the total area of the build plane **408** formed by the build platforms **406** is at least 1000 cm.sup.2, 1500 cm.sup.2, 2000 cm.sup.2, 2500 cm.sup.2, 3000 cm.sup.2, 3500 cm.sup.2, 4000 cm.sup.2, 4500 cm.sup.2, or 5000 cm.sup.2. In some embodiments, the total area of the build plane is larger than 5000 cm.sup.2 to accommodate larger objects and/or faster printing speeds. The build plane **408** can have a length greater than or equal to 20 cm, 50 cm, 100 cm, 150 cm, or 200 cm; and/or a width greater than or equal to 10 cm, 20 cm, 30 cm, 40 cm, or 50 cm.

[0148] The build platforms **406** can each be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one object **402**. Each build platform **406** can independently support any suitable number of objects **402**, such as one, two, three, four, five, 10, 20, 50, or more objects **402**. Each build platform **406** can be made of any suitable material (e.g., metal (such as aluminum or steel), polymer, ceramic (such as alumina-based ceramics), glass, wood, paper) that exhibits sufficient adhesion to the cured material to support the object **402** during the entire additive manufacturing process. In some embodiments, the build platform **406** is made partially or entirely out of a metal to allow for inductive heating of the build platform **406**, e.g., in embodiments where the additive manufacturing process is a high temperature lithography process. Aluminum may be advantageous for facilitating thermal homogeneity, while steel may be advantageous for providing enhanced strength. Composite materials are also contemplated, such as two or more materials arranged in layers or other configurations. The composite materials can be or include includes filled systems, such as silica filled plastics or fiber filled plastics. In some embodiments, the build platforms **406** are injection molded, recyclable, and/or compostable. The surface of the build platform **406** that contacts the object **402** can also be configured to promote adhesion. For instance, each build platform **406** can have a smooth, polished surface, a rough surface, a structured or patterned surface, etc., and/or may include various surface finishes. In embodiments where the build platform **406** is made out of a metal, the surface of the build platform **406** may or may not be anodized.

[0149] In some embodiments, the material of the build platform **406** is selected to withstand forces that may be applied to the build platform **406** during post-processing. For example, in embodiments where the build platform **406** and object **402** are centrifuged to remove residual curable material, the build platform **406** can be sufficiently stiff to resist forces arising from centrifugation (e.g., the build platform **406** exhibits little or no bending when subjected to centrifugation). Optionally, the lower surface of the build platform **406** can include ridges and/or other reinforcement features formed therein to enhance the stiffness of the build platform **406** while reducing the overall weight of the build platform **406**.

[0150] The geometry of each build platform **406** can be varied as desired. For instance, each build platform **406** can have any suitable shape, such as rectangular, square, triangular, trapezoidal, oval, circular, or any other polygonal or non-polygonal shape. Some or all of the build platforms **406** can have the same shape, or some or all of the build platforms **406** can have different shapes. Moreover, each build platform **406** can have any suitable size, such as an area within a range from 100 cm.sup.2 to 1000 cm.sup.2, 100 cm.sup.2 to 500 cm.sup.2, 100 cm.sup.2 to 200 cm.sup.2, 200 cm.sup.2 to 1000 cm.sup.2, 200 cm.sup.2 to 500 cm.sup.2, or 500 cm.sup.2 to 1000 cm.sup.2. For example, an individual build platform **406** can have a length and/or width that is less than or equal to 50 cm, 40 cm, 30 cm, 25 cm, 20 cm, 15 cm, or 10 cm. Some or all of the build platforms **406** can have the same size, or some or all of the build platforms **406** can have different sizes.

[0151] The build platforms **406** can be releasably coupled to the carrier **404** so that each build platform **406** is independently removable from the carrier **404**. The coupling between each build platform **406** and the carrier **404** can be configured to maintain the build platform **406** in a stable position and orientation during the additive manufacturing process, while also allowing for easy detachment of the build platform **406** from the carrier **404** upon completion of the additive manufacturing process and/or in preparation for post-processing of the objects **402**. Examples of attachment mechanisms that may be used to releasably couple the build platforms **406** to the carrier **404** include vacuum, mechanical fixation (e.g., interference fit, snap fit, interlocking features, fasteners, form-fitting inserts, clamps, springs, hinged features), electromagnetic fixation, magnetic fixation, and combinations thereof. Additional examples of attachment mechanisms are provided below in connection with FIGS. **14-29**.

[0152] In some embodiments, when the build platforms **406** are coupled to the carrier **404**, the upper surfaces of the build platforms **406** are level with each other to define a flat build plane **408**. For example, the maximum difference between the vertical positions of the upper surfaces of the build platforms **406** (also referred to herein as the “vertical deviation of the build plane **408**”) can be no more than 1 mm, 500 μm, 400 μm, 300 μm, 200 μm, 100 μm, or 50 μm; and/or within a range from 50 μm to 500 μm, or 500 μm to 1 mm. The maximum vertical deviation of the build plane **408** can be less than or equal to the thickness of an individual layer of the object **402** (e.g., the layer thickness can be within a range from 50 μm to 150 μm). In some embodiments, a flat build plane **408** is significant for ensuring proper adhesion of the initial layer(s) of the object **402**. A flat build plane **408** can be achieved, for example, via attachment mechanisms that avoid buckling of the build plane **408** when coupled to the carrier **404**, e.g., as discussed below in connection with FIGS. **14-29**.

[0153] The build platforms **406** can be “modular” build platforms **406** in that each build platform **406** can be independently selected, positioned on the carrier **404**, and/or removed from the carrier **404**. The use of such modular build platforms **406** can provide various advantages, such as providing a larger build plane **408** (e.g., for higher throughput printing) that can be segmented into smaller individual build platforms **406** that are compatible with post-processing devices (e.g., centrifuges, solvent cleaning devices (solvent baths such as ultrasonic solvent baths), UV or thermal furnaces, ovens (such as curing and/or annealing ovens), chillers, laser marking devices, laser cutting devices, ink jetting devices, powder coating devices, dip coating devices, spraying devices, conveyor belts, drying devices (dryers such as air blade dryers)); allowing build platforms **406** to be individually removed and replaced (e.g., if the build platform **406** becomes damaged or fouled); and/or allowing the type, geometry, and/or arrangement of the build platforms **406** to be varied in a modular fashion to accommodate different printing operations (e.g., different sizes, shapes, and/or arrangements of objects **402**).

[0154] In some embodiments, some or all of the build platforms **406** can include recesses (e.g., cavities, holes, pores, hollows, gaps, grooves), and the curable material can flow through and/or into the recesses. The curable material that flows through and/or into the recesses can be collected below the build platforms **406** for reuse in subsequent printing operations. For instance, the carrier **404** can include or be coupled to a container for collecting the curable material. Moreover, some or all of the build platforms **406** can include features configured to create a gap between the lower surface of the build platforms **406** and the upper surface of the carrier **404**, such excess curable material may accumulate within the gap without disrupting the flat build plane **408**, e.g., as described further below in connection with FIGS. **13**A-**13**C.

[0155] Optionally, the carrier **404** and/or the build platforms **406** can include an identifier. The identifier can be an electronic tag that is coupled to the carrier **404** and/or build platforms **406**, such as an RFID chip. Alternatively or in combination, the identifier can be a machine-readable marking (e.g., a camera-readable marking) such as a barcode, QR code, or other pattern that is engraved or otherwise formed in the carrier **404** and/or build platforms **406**. The identifier can be used to automatically identify the carrier **404** and/or the individual build platforms **406**, thereby allowing the objects **402** on the carrier **404** and/or build platforms **406** to be tracked throughout the manufacturing operation.

[0156] FIG. **5** is a partially schematic diagram of an additive manufacturing system **500** including the modular build substrate **400**, in accordance with embodiments of the present technology. The system **500** can be a 3D printer that is used to fabricate a plurality of 3D objects **402** using any suitable additive manufacturing technique, such as a lithography-based additive manufacturing process in which the objects **402** are fabricated from a curable material in a layer-by-layer manner, as described herein.

[0157] In the illustrated embodiment, the modular build substrate **400** is coupled to and held down onto a stationary base **502** (also known as a “base carrier”) of the system **500**. The base **502** can be a tray, plate, film, sheet, stage, table, or other generally planar substrate suitable for coupling to and supporting the modular build substrate **400** during the additive manufacturing process. As shown in FIG. **5**, the carrier **404** of the modular build substrate **400** can be coupled to the base **502** via an attachment mechanism **504** (also known as a “fixing means”). The attachment mechanism **504** can utilize any suitable releasable coupling technique, such as vacuum, mechanical fixation (e.g., interference fit, snap fit, interlocking features, form-fitting inserts, clamps, springs, hinged features), electromagnetic fixation, magnetic fixation, or combinations thereof. In this way, the carrier **404** can be separated from the base **502**, thus allowing the entire modular build substrate **400** to be automatically or manually removed from the system **500** after the additive manufacturing process has been completed.

[0158] As described herein, the manufacturing cycle for the objects **402** can include additive manufacturing of the objects **402** using the system **500**, at least one post-processing operation, and the subsequent removal of the objects **402** from the associated build platform **406**. In the illustrated embodiment, the objects **402** are additively manufactured in a layer-by-layer manner by a movable printer assembly **506** (also known as a “print head”) including an energy source **508** (e.g., a light engine), which travels over and across the build plane **408** formed by the build platforms **406** (e.g., along the directions indicated by arrows **510**). The components and operation of the printer assembly **506** can be identical or generally similar to those of the printer assembly **302** of FIG. **3**. For instance, the movable printer assembly **506** can be configured to apply a layer of a curable material (e.g., a light-polymerizable resin) onto an endless movable carrier film **512** (also known as a “carrier foil”), which can move in a circulating loop trajectory around the energy source **508** (e.g., in a counter-clockwise direction as shown in FIG. **5**). The carrier film **512** can convey the layer of curable material into an exposure zone of the energy source **508**, such that the curable material is exposed and cured while the printer assembly **506** is driven to move across the plurality of build platforms **406** (e.g., along the direction indicated by arrow **514**, which represents the printing direction for the illustrated movement direction of the carrier film **512**). Accordingly, a layer of cured material corresponding to a cross-section of each object **402** can be formed on the corresponding build platform **406**. The modular build substrate **400** can then then be lowered (e.g., via an elevator mechanism, which may be coupled to or part of the base **502**—not shown) and/or the movable printer assembly **506** can be raised in preparation for forming the next layer of cured material. This process can be repeated to build up the objects **402** from a plurality of sequential layers of cured material.

[0159] After the additive manufacturing process, the build platforms **406** with the respective fabricated objects **402** can be individually removed from the carrier **404**. The process of the build platforms **406** from the carrier **404** can be performed outside or inside the system **500**. Optionally, the carrier **404** can be removed from the base **502** before removing the build platforms **406** from the carrier **404**. Thereafter, the individual build platforms **406** with the respective fabricated objects **402** arranged thereon can be subjected to at least one post-processing operation. For instance, the post-processing operation can include removing residual curable material from the objects object **402**, removing solvents from the objects **402**, applying pressurized air to the objects **402**, drying the objects **402**, removing support structures from the objects **402**, post-curing the objects **402**, and/or performing surface modifications to the objects **402**. Some or all of these processes can be performed while the objects **402** remain on their respective build platforms **406**. Moreover, because the build platforms **406** can be separated from each other, objects **402** on different build platforms **406** can be processed at different types and/or using different post-processing techniques.

[0160] The configuration of the system **500** can be varied in many ways. For instance, the base **502** can be omitted, such that the carrier **404** serves as directly as the stationary base during additive manufacturing. Conversely, the carrier **404** may be omitted, such that the base **502** serves as the component that directly supports the build platforms **406**, in which case any of the features described herein as being part of the carrier **404** (e.g., attachment mechanisms) may instead be part of the base **502**. The system **500** can include additional components not shown in FIG. **5**, such as a controller including one or more processors and memory storing instructions for controlling the operation of the system **500**. In some embodiments, the build platforms **406** are cooled (e.g., via one or more cooling devices such as thermoelectric coolers, cold plates, cooled fluids, etc.). For instance, some or all of the build platforms **406** may be cooled to a surface temperature at the build plane **408** within a range from 20° C. to −150° C., or from 20° C. to −20° C. In other embodiments, some or all of the build platforms **406** may be heated (e.g., via one or more heating devices such as thermoelectric heaters, heat sinks, heating plates, heat lamps, heated fluid, inductive heaters, etc.). For instance, some or all of the build platforms **406** may be heated to a surface temperature at the build plane **408** within a range from 20° C. to 250° C., or from 30° C. to 90° C. Accordingly, the system **500** can include cooling and/or heating devices at any suitable location, such as between the build platforms **406** and carrier **404**, between the carrier **404** and the base **502**, within the build platforms **406**, within the carrier **404**, within the base **502**, etc.

[0161] FIG. **6** is a partially schematic diagram of another additive manufacturing system **500** including the modular build substrate **400**, in accordance with embodiments of the present technology. The system **600** can be a 3D printer that is used to fabricate a plurality of 3D objects **402** using any suitable additive manufacturing technique, such as a lithography-based additive manufacturing process in which the objects **402** are fabricated from a curable material in a layer-by-layer manner, as described herein. In the illustrated embodiment, the system **600** is configured for a vat-based additive manufacturing process, in which the modular build substrate **400**, including the carrier **404** with the build platforms **406** thereon, is immersed into a volume (e.g., a bath) of curable material **602** (e.g., light-polymerizable resin) within a reservoir **604** (e.g., a vat). One or more movable energy sources **606** (e.g., light engines) can locally irradiate a thin layer of the curable material **602** that extends above the build platforms **406** and/or the partly built objects **402**. In this way, the objects **402** can be built layer-by-layer. Each time a layer has been cured, the modular build substrate **400** can be lowered (e.g., via an elevator mechanism—not shown) in preparation for curing the next layer.

[0162] After the additive manufacturing process, the build platforms **406** with the respective fabricated objects **402** can be individually removed from the carrier **404**, which may be performed outside or inside the system **600**. The carrier **404** can be removed from the reservoir **604** before removing the build platforms **406** from the carrier **404**, or the build platforms **406** can be removed from the carrier **404** while the carrier **404** is within the reservoir **604**. Thereafter, the individual build platforms **406** with the respective fabricated objects **402** arranged thereon can be subjected to at least one post-processing step, as described elsewhere herein.

[0163] The configuration of the system **600** can be varied in many ways. For example, the system **600** can include additional components not shown in FIG. **6**, such as a controller including one or more processors and memory storing instructions for controlling the operation of the system **600**. The system **600** can optionally include a base that is releasably coupled to the carrier **404** to support the modular build substrate **400** (e.g., similar to the base **502** of FIG. **5**). In such embodiments, the base can be a movable stage that is positioned within the reservoir **604** to raise and lower the modular build substrate **400** during the additive manufacturing process. Moreover, in some embodiments, the build platforms **406** are cooled (e.g., via one or more cooling devices such as thermoelectric coolers, cold plates, cooled fluids, etc.), while in other embodiments, the build platforms may be heated (e.g., via one or more heating devices such as heat sinks, heating plates, heat lamps, heated fluid, etc.). Accordingly, the system **600** can include cooling and/or heating devices at any suitable location, such as between the build platforms **406** and carrier **404**, between the carrier **404** and the base, within the build platforms **406**, within the carrier **404**, within the base, within the reservoir **604**, coupled to the reservoir **604**, etc.

[0164] It will be appreciated that the modular build substrates and build platforms described herein can be used in other types of additive manufacturing processes besides the embodiments illustrated in FIGS. **5** and **6**. For example, the build platforms described herein can also be used as a valuable tool to facilitate post-processing in bottom-up, top-down, or volumetric radiation-curing vat polymerization processes as well as static film lamination processes. For example, a coating of light-polymerizable resin can be applied on a carrier film and can be irradiated with a static light engine, thereby building up one or more 3D objects on the build platforms arranged on the carrier.

[0165] FIGS. **7**A-**12** illustrate additional features of modular build substrates and build platforms configured in accordance with embodiments of the present technology. Any of the embodiments described in connection with FIGS. **7**A-**12** can be combined with each other and/or with the embodiments described in connection with FIGS. **4-6**. Moreover, the modular build substrates and build platforms described in connection with FIGS. **7**A-**12** can be generally similar to the modular build substrate **400** and build platform **406** of FIG. **4**, such that like numbers (e.g., build platform **406** versus build platform **706**) are used to identify similar or identical components, and the following discussion of FIGS. **7**A-**12** will be limited to those features that differ from the embodiments described in connection with FIG. **4**.

[0166] FIGS. **7**A-**10**B illustrate modular build substrates with various configurations of build platforms. For example, FIGS. **7**A and **7**B are side and top views of a modular build substrate **700** configured in accordance with embodiments of the present technology. The modular build substrate **700** includes a carrier **704** and a plurality of build platforms **706** releasably coupled to the carrier **704**. In the illustrated embodiment, the build platforms **706** each have a rectangular shape and are arranged in a line configuration (e.g., a linear array such as a row) on the carrier **704**. The build platforms **706** can be seamlessly and/or tightly packed next to each other so there is little or no gap between adjacent build platforms **706** (e.g., the distance between adjacent build platforms **706** is no more than 200 μm, 100 μm, 50 μm, 25 μm, or 10 μm). Accordingly, the build platforms **706** can collectively define a single continuous build plane **708** for fabricating one or more objects.

[0167] FIGS. **8**A and **8**B are side and top views of another modular build substrate **800** configured in accordance with embodiments of the present technology. The modular build substrate **800** includes a carrier **804** and a plurality of build platforms **806** releasably coupled to the carrier **804**. In the illustrated embodiment, the build platforms **806** each have a rectangular shape and are arranged in a grid configuration (e.g., a 2D array including a plurality of rows and a plurality of columns) on the carrier **804**. The build platforms **806** can be seamlessly and/or tightly packed next to each other so there is little or no gap between adjacent build platforms **806** (e.g., the distance between adjacent build platforms **806** is no more than 200 μm, 100 μm, 50 μm, 25 μm, or 10 μm). Accordingly, the build platforms **806** can collectively define a single continuous build plane **808** for fabricating one or more objects.

[0168] FIGS. **9**A and **9**B are side and top views of yet another modular build substrate **900** configured in accordance with embodiments of the present technology. The modular build substrate **900** includes a carrier **904** and a plurality of build platforms **906** releasably coupled to the carrier **904**. In the illustrated embodiment, the build platforms **906** each have a rectangular shape and are arranged in a grid configuration (e.g., a 2D array including a plurality of rows and a plurality of columns) on the carrier **904**. The build platforms **906** can be spaced apart from each other, such that there are gaps between adjacent build platforms **906** through which the carrier **904** is exposed. Accordingly, the build plane **908** of the modular build substrate **900** can be segmented into a plurality of discrete regions, each corresponding to a respective build platform **906**.

[0169] FIGS. **10**A and **10**B are side and top views of another modular build substrate **1000** configured in accordance with embodiments of the present technology. The modular build substrate **1000** includes a carrier **1004** and a plurality of build platforms **1006** releasably coupled to the carrier **1004**. In the illustrated embodiment, the build platforms **1006** each have a triangular shape and are arranged in a line configuration (e.g., a linear array such as a row) on the carrier **1004**. The build platforms **1006** can be seamlessly and/or tightly packed next to each other so there is little or no gap between adjacent build platforms **1006** (e.g., the distance between adjacent build platforms **1006** is no more than 200 μm, 100 μm, 50 μm, 25 μm, or 10 μm). Accordingly, the build platforms **1006** can collectively define a single continuous build plane **708** for fabricating one or more objects.

[0170] FIGS. **11**A-**11**G illustrate a build platform **1106** including recesses **1110** configured in accordance with embodiments of the present technology. The build platform **1106** can be a modular build platform that is part of a modular build substrate, as described herein. For example, the build platform **1106** can be releasably coupled to a carrier during an additive manufacturing process, and can be separated from the carrier during post-processing.

[0171] Referring first to FIGS. **11**A (side view) and **11**B (top view) together, the build platform **1106** includes a plurality of recesses **1110** formed therein. As best seen in FIG. **11**A, the recesses **1110** can be indentations, cavities, cutouts, etc., that are formed in the upper surface of the build platform **1106**. The recesses **1110** can extend partially or entirely through the thickness of the build platform **1106** toward the lower surface of the build platform **1106**. Although the illustrated embodiment includes six recesses **1110**, in other embodiments, the build platform **1106** can include a different number of recesses **1110**, such as one, two, three, four, five, 10, 20, or more recesses **1110**. Moreover, although the recesses **1110** are depicted as being arranged in a grid configuration, the recesses **1110** can alternatively be arranged in other configurations, such as in a line, cluster, or any other regular or irregular pattern. Some or all of the recesses **1110** can have the same geometry (e.g., the same size and/or shape), or some or all of the recesses **1110** can have different geometries (e.g., different sizes and/or shapes).

[0172] Referring next to FIGS. **11**C (side view) and **11**D (top view) together, the recesses **1110** of the build platform **1106** are each configured to receive a respective prefabricated element **1112.** The shape of the prefabricated element **1112** can be identical or similar to the shape of the corresponding recess **1110** such that the prefabricated element **1112** is received within the recess **1110** in a form-fitting manner. In the illustrated embodiment, for example, the prefabricated element **1112** and recess **1110** each have a square shape. In other embodiments, however, the prefabricated element **1112** and recess **1110** can have any other suitable shape, such as rectangular, circular, oval, triangular, polygonal, non-polygonal, etc.

[0173] The prefabricated element **1112** can be releasably coupled to the build platform **1106**. For example, the releasable coupling can include a mechanical connection, such as via one or more fasteners (e.g., screws, bolts) and/or a mechanical fit (e.g., interference fit, snap fit). Alternatively or in combination, the releasable coupling can involve a chemical connection (e.g., gluing, crosslinking, curing) and/or physical attachment mechanisms (e.g., vacuum, capillary forces, magnetic forces). Accordingly, the build platform **1106** can act as an adapter structure or adapter plate for the prefabricated elements **1112**.

[0174] Referring next to FIGS. **11**E (side view) and **11**F (top view) together, each prefabricated element **1112** can be arranged so that its upper surface is flush with the build plane **1108** of the build platform **1106**. Accordingly, an additively manufactured object **1102** can be built on top of the prefabricated element **1112**. In some embodiments, at least a portion of the object **1102** (e.g., the initial layer of the object **1102** deposited onto the prefabricated element **1112**) adheres to the prefabricated element **1112** so that the prefabricated element **1112** is integrated into and becomes part of the object **1102**. The prefabricated element **1112** can be made out of a different material than the object **1102**, such that the final product is a multi-material 3D structure including both the object **1102** and the prefabricated element **1112**. Alternatively, the prefabricated element **1112** can be made out of the same material as the object **1102**.

[0175] This approach can be used to incorporate various types of prefabricated elements **1112** into an object **1102**. For instance, in embodiments where the object **1102** is a dental appliance, the prefabricated element **1112** can be a functional component to be integrated into the dental appliance, such as an electronics module, battery, sensor, wire, bracket, elastic, block, etc. In some embodiments, the object **1102** is a first portion of the dental appliance that is made out of a first material, and the prefabricated element **1112** is a second portion of the dental appliance that is made out of a second, different material.

[0176] Referring next to FIGS. **11**G (side view) and **11**H (top view), once the additive manufacturing process is complete, the object **1102** can be removed from the build platform **1106** together with the prefabricated element **1112**, leaving the recesses **1110** empty. The build platform **1106** can then be reused for subsequent additive manufacturing operations.

[0177] Although FIGS. **11**A-**11**H illustrate a build platform **1106** with recesses **1110** for receiving prefabricated elements **1112**, in other embodiments, the recesses **1110** can be omitted, such that the prefabricated elements **1112** are placed directly on the upper surface of the build platform **1106**.

[0178] FIG. **12** is a side view of a build platform **1206** including an interface layer **1220**, in accordance with embodiments of the present technology. The interface layer **1220** can be a film, membrane, sheet, barrier, etc., that is positioned on an upper surface of the build platform **1206** to facilitate separation of an additively manufactured object **1202** from the build platform **1206** (e.g., after post-processing of the object **1202** is complete).

[0179] The interface layer **1220** can be made of a material that differs from the material used to fabricate the object **1202**. For instance, the material of the interface layer **1220** can differ from the material of the object **1202** with respect to at least one physical and/or chemical property, such as melting point, boiling point, solubility, etc. In some embodiments, the interface layer **1220** is made partially or entirely from a material that is polymerizable and, in its polymerized and/or pre-polymerized state, can be dissolved or swollen in a solvent. Such a material can include or consist of at least one polymerizable group, such as an acrylate, methacrylate, acrylamide, vinyl ether, vinyl ester, maleimide, cyclic ether, isocyanate, amine, or other polymerizable unsaturated or saturated group. The material can optionally further comprise at least one hydrophilic or oleophilic group. The polymerized and/or pre-polymerized material may be dissolvable or swellable in a solvent, such as water, alcohol, oil, or other organic solvents. Such materials can include, for example, hydroxyl, carbonyl, or carboxyl groups, and/or derivatives with other electronegative heteroatoms, amines, ionic liquids, and/or salts. For example, the material can be or include hydroxyethylacrylate (HEA), hydroxyethylmethacrylate (HEMA), 2-(2-ethoxyethoxy)ethyl acrylate (EOEOEA), acryloyl morpholine (ACMO), polyethylene glycol derivates, polyethers, hydroxy ethylene, and/or lauryl acrylates.

[0180] The interface layer **1220** can be applied to the build platform **1206** before the object **1202** is fabricated on the build platform **1206**. For instance, the interface layer **1220** can be applied via spray coating, dip coating, spin coating, adhesives, chemical deposition techniques, and/or other suitable coating techniques. The object **1202** can then be fabricated on the interface layer **1220** via an additive manufacturing process. The material of the object **1202** can exhibit sufficient adhesion to the interface layer **1220** so the object **1202** can be stably supported on the build platform **1206** during additive manufacturing.

[0181] After the additive manufacturing process, the interface layer **1220** can be removed from the build platform **1206** via a post-processing operation to destabilize, eliminate, or otherwise remove the interface layer **1220**, thereby causing the object **1202** to be detached from the build platform **1206**. For example, the interface layer **1220** can be removed by subjecting the interface layer **1220** to a physical and/or chemical process that causes the interface layer **1220** to disintegrate or otherwise lose its stability, such as dissolving in a solvent, etching, melting, etc. Alternatively or in combination, the interface layer **1220** can be removed by peeling, scraping, or otherwise physically separating the interface layer **1220** from the build platform **1206**.

[0182] In some embodiments, the object **1202** is removed from the build platform **1206** together with the interface layer **1220**, and is subsequently separated from the interface layer **1220** in a separate process step via physical and/or chemical processes. Alternatively, the removal of the interface layer **1220** from the build platform **1206** can simultaneously cause separation of the object **1202** from the interface layer **1220**, e.g., if the interface layer **1220** is dissolved, melted, etched, etc.

[0183] FIGS. **13**A-**13**C illustrate modular build substrates with features to accommodate excess material, in accordance with embodiments of the present technology. Any of the embodiments described in connection with FIGS. **13**A-**13**C can be combined with each other and/or with any of the embodiments described in connection with FIGS. **4-12**.

[0184] FIG. **13**A is a side cross-sectional view of a portion of a modular build substrate **1300** including a carrier **1302** and a build platform **1304**, in accordance with embodiments of the present technology. The carrier **1302** and build platform **1304** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-12**). For instance, the carrier **1302** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platform **1304** during the additive manufacturing process. The build platform **1304** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0185] In the illustrated embodiment, the build platform **1304** includes an upper surface **1306** configured to support at least one additively manufactured object, and a lower surface **1308** opposite the upper surface **1306**. The build platform **1304** can be positioned on an upper surface **1310** of the carrier **1302** such that the lower surface **1308** of the build platform **1304** faces the upper surface **1310** of the carrier **1302**. One or more protrusions **1312** can be formed in the lower surface **1308** of the build platform **1304** to create a gap **1314** between the lower surface **1308** of the build platform **1304** and the upper surface **1310** of the carrier **1302**. The gap **1314** can allow material from the additive manufacturing process (e.g., excess curable material and/or debris) to accumulate between the build platform **1304** and the carrier **1302** without disrupting the vertical alignment between the build platform **1304** and neighboring build platforms on the carrier **1302**.

This configuration can be advantageous, for example, in embodiments where the additive manufacturing process uses a liquid curable material (e.g., a resin) that may seep into spaces between the build platform **1304** and the carrier **1302**.

[0186] The gap **1314** can be sufficiently large to accommodate and trap material therein, and can also be sufficiently small to allow efficient heat transfer between the carrier **1302** and the build platform **1304** (e.g., in embodiments where the carrier **1302** includes or is thermally coupled to a heating device for heating the build platform **1304**). In some embodiments, the size of the gap **1314** (denoted by distance D.sub.1 in FIG. **13**A, which may correspond to the height of the protrusions **1312**), is within a range from 0.1 mm to 3 mm, 0.1 mm to 1 mm, 0.1 mm to 0.5 mm, 0.25 mm to 0.75 mm, 0.5 mm to 1 mm, or 1 mm to 3 mm.

[0187] The protrusions **1312** can be shaped in a variety of ways to provide sufficient clearance between the build platform **1304** and the carrier **1302**. In some embodiments, the protrusions **1312** are shaped as struts, ribs, rods, pins, posts, cylinders, cones, polyhedrons, balls, bumps, etc. The protrusions **1312** can have a uniform diameter and/or width, or may have a variable diameter and/or width. The protrusions **1312** can all have the same shape and/or size, or some or all of the protrusions **1312** can have different shapes and/or sizes (e.g., depending on the geometry of the build platform **1304**, carrier **1302**, and/or the desired size of the gap **1314**). In some embodiments, some or all of the protrusions **1312** are discrete structures that are spaced apart from each other. In other embodiments, some or all of the protrusions **1312** may be interconnected with each other. For example, the protrusions **1312** can include a plurality of elongate members (e.g., ribs) that are interconnected to form an ‘H’-shape, ‘I’-shape, ‘L’-shape, ‘T’-shape, ‘X’-shape, ‘Z’-shape, a grid or lattice, etc.

[0188] The build platform **1304** can have any suitable number of protrusions **1312** extending therefrom, such as one, two, three, four, five, ten, 20, 50, or more protrusions **1312**. In some embodiments, the build platform **1304** includes a single protrusion **1312**, e.g., a protrusion **1312** coupled to a central portion of the lower surface **1308** of the build platform **1304**. Alternatively, the build platform **1304** can include a plurality of protrusions **1312**, which can have any of a variety of distributions on the build platform **1304**, such as in a grid, lattice, or other shape (e.g., honeycomb, circular, oval, square, rectangular, triangular) or in a random distribution. In some embodiments, the protrusions **1312** are located at or near the corners of the build platform **1304**. For example, in embodiments where the build platform **1304** has four corners, a protrusion **1312** can be located at or near each corner. In such cases, the protrusions **1312** can have a consistent spacing (e.g., equidistant) from one or more edges of the build platform **1304**. For instance, the protrusions **1312** can have a separation distance of no more than 1 mm, 5 mm, 1 cm, 2 cm, 3 cm, etc. from the one or more edges of the build platform **1304**.

[0189] In some embodiments, the protrusions **1312** are integrally formed with the build platform **1304** and made from the same material as the build platform **1304**. For instance, the build platform **1304** and protrusions **1312** can be made partially or entirely out of a relative high modulus and/or stiff material (e.g., steel, aluminum, or another metal). Alternatively, the protrusions **1312** can be made from a different material than the build platform **1304**. For example, the protrusions **1312** can be made partially or entirely out of a relatively low modulus and/or deformable material (e.g., silicone, rubber, or another polymer), while the build platform **1304** can be made partially or entirely out of a relatively high modulus and/or stiff material (e.g., steel, aluminum, or another metal).

[0190] While the protrusions **1312** have been discussed herein with reference to the build platform **1304**, in some embodiments, the protrusions **1312** can additionally or alternatively be formed in the carrier **1302**. For example, the protrusions **1312** can extend from the upper surface **1310** of the carrier **1302** and contact the lower surface **1308** of the build platform **1304**, creating the gap **1314**. The protrusions **1312** of the carrier **1302** can be generally similar to the protrusions **1312** of the build platform **1304**. In some embodiments, a first plurality of protrusions **1312** can be formed in the carrier **1302** and a second plurality of protrusions **1312** can be formed in the build platform **1304**, where both the first plurality of protrusions **1312** and the second plurality of protrusions **1312** create the gap **1314**.

[0191] An example of the protrusions **1312** will now be discussed with reference to FIG. **13**B. FIG. **13**B is a bottom view of the build platform **1304** with a plurality of pins **1316**, in accordance with embodiments of the present technology. The pins **1316** can be used as the protrusions **1312** to form the gap **1314** between the build platform **1304** and the carrier **1302**, as discussed above. The pins **1316** can be discrete structures that are spaced apart from each other. The pins **1316** can be arranged on the lower surface **1308** of the build platform **1304** in many different ways, such as in a grid, lattice, or other suitable distribution. In the illustrated embodiment, for example, the pins **1316** are arranged such that there are four pins **1316** near each corner of the build platform **1304**, a pin **1316** at the center of the build platform **1304**, a pin **1316** near the center of each of the four quadrants of the build platform **1304**, and additional pins **1316** formed along the vertical and horizontal axes of the build platform **1304**. In other embodiments, the pins **1316** may instead be distributed randomly along the bottom surface **1308**.

[0192] As another example, FIG. **13**C is a bottom view of the build platform **1304** with a plurality of ribs **1318**, in accordance with embodiments of the present technology. The ribs **1318** can be used as the protrusions **1312** to form the gap **1314** between the build platform **1304** and the carrier **1302**, as discussed above. The ribs **1318** can be elongate members that extend along the lower surface **1308** of the build platform **1304**. In the illustrated embodiment, the build platform **1304** includes three parallel ribs **1318** extending from a first edge **1320** of the build platform **1304** to a second, opposite edge **1322** of the build platform **1304**. The ribs **1318** can be spaced apart from each other by a uniform distance, e.g., a distance of at least 1 cm, 5 cm, 10 cm, 20 cm, etc. In other embodiments, however, the number and arrangement of the ribs **1318** can modified, e.g., the build platform **1304** can include a different number of ribs **1318** (e.g., one, two, four, five, or more ribs **1318**), some or all of the ribs **1318** may not be parallel to each other, the spacing between the ribs **1318** may be variable, the build platform **1304** can alternatively or additionally include ribs **1318** that extend from a third edge **1324** (left edge) to a fourth edge **1326** (right edge) of the build platform **1304**, etc.

[0193] The geometry (e.g., length and/or width) of the ribs **1318** can be varied as desired. In the illustrated embodiment, the length of each rib **1318** is equal to the length of the build platform **1304** from the first edge **1320** to the second edge **1322**. However, in other embodiments, the length of the ribs **1318** may be less than the length of the build platform **1304** from the first edge **1320** to the second edge **1322**, e.g., no more than 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, etc., of the length of the build platform **1304** between the first edge **1320** and the second edge **1322**. In some embodiments, each rib **1318** has a width within a range from 0.1 mm to 10 mm, or from 1 mm to 5 mm.

[0194] In some embodiments, the ribs **1318** mechanically reinforce the build platform **1304** to increase its overall stiffness. For example, the ribs **1318** can be made of one or more materials that are stiffer than the material of the build platform **1304**. In other embodiments, the ribs **1318** can be made of the same material as the build platform **1304** and can increase the stiffness of the build platform **1304** by increasing the thickness of the build platform **1304** at selected locations. In some embodiments, the arrangement of the ribs **1318** can also increase the stiffness of the build platform **1304**, such as by having multiple intersecting ribs **1318**.

[0195] In some embodiments, the present technology provides attachment mechanisms for releasably coupling modular build platforms to a carrier. The attachment mechanisms can be configured to couple the build platforms to the carrier to form a flat build plane, e.g., a build plane having a vertical deviation no more than 500 μm, 400 μm, 300 μm, 200 μm, 100 μm, or 50 μm, and/or a vertical deviation within a range from a range from 50 μm to 500 μm, or 500 μm to 1 mm. Moreover, the attachment mechanisms can secure the build platforms to the carrier to prevent the build platforms from moving relative to the carrier during additive manufacturing, e.g., due to forces applied to the build platforms by the printer assembly. The attachment mechanisms described herein can use many different types of couplings, such as mechanical couplings, magnetic couplings, vacuum couplings, or suitable combinations thereof.

[0196] In some embodiments, the attachment mechanism includes at least one mechanical coupling device, such as a block, wedge, clamp, clip, latch, spring, ratchet, cam lock, etc. For example, a mechanical coupling device can include at least one spring component (e.g., a spring clip) that uses a spring force to secure the build platform to the carrier (e.g., the spring force may bias the spring component into engagement with the build platform). As another example, a mechanical coupling device can include at least one movable component that is movable between an open configuration (e.g., in which the component is disengaged from the build platform and the build platform is removable from the carrier) and a closed configuration (e.g., in which the component engages the build platform to secure the build platform to the carrier). The movable component can be a rotatable component (e.g., a hinged clamp), a translatable component (e.g., a sliding latch) or a combination thereof. In a further example, a mechanical coupling device can include a fixed, non-movable component that secures the build platform to the carrier by obstructing movement of the build platform relative to the carrier.

[0197] In some embodiments, the attachment mechanism includes at least one magnetic coupling device. The magnetic coupling device can include, for example, a first magnet located on or within the build platform, and a second magnet located on or within the carrier. The attractive force between the first magnet and the second magnet can secure the build platform to the carrier. Alternatively or in combination, the carrier can be formed partially or entirely out of a magnetic material (e.g., a ferromagnetic material), and the carrier can be configured to engage one or more magnets located on or within the build platform. Alternatively or in combination, the build platform can be formed partially or entirely out of a magnetic material (e.g., a ferromagnetic material), and the build platform can be configured to engage one or more magnets located on or within the carrier. The magnetic coupling devices disclosed herein can include permanent magnets and/or electrically actuated magnets (e.g., electromagnets).

[0198] An attachment mechanism of the present technology can include one or more flexible coupling devices, one or more rigid coupling devices, or a combination thereof. A flexible coupling device can allow for limited movement of the coupled portion of the build platform relative to the carrier. For instance, a flexible coupling device can include a flexible material that is deformable and/or deflectable when engaged with the build platform, such as a spring material. Examples of spring materials include metals (e.g., spring steel), polymers, and composites. Alternatively or in combination, the flexible coupling device can have a flexible structure that permits some degree of deformation and/or deflection when engaged with the build platform, such as a telescopic mechanism, ratchet mechanism, compliant mechanism, etc. A rigid coupling device can prevent movement of the coupled portion of the build platform relative to the carrier. A rigid coupling device can be made out of a rigid material that does not substantially deform or deflect when engaged with the build platform and/or can have a rigid structure that does not permit deformation or deflection when engaged with the build platform.

[0199] In some embodiments, the attachment mechanisms described herein include at least one flexible coupling device that is configured to allow for some degree of lateral movement of the build platform relative to the underlying carrier while inhibiting vertical movement of the build platform relative to the carrier, which may disrupt the flat build plane for printing. For instance, in embodiments where the build platforms are heated or cooled during additive manufacturing, thermal expansion/shrinkage effects may cause the build platform to expand/shrink relative to the carrier, particularly if the build platform is made of a material having a different coefficient of thermal expansion than the carrier. The use of a flexible coupling device at one or both ends of the build platform can accommodate small lateral dimensional changes of the build platform due to thermal effects, while preserving the flat build plane and keeping the build platform secured to the carrier. In contrast, rigid coupling devices at both ends of the build platform may result in buckling and/or bending of the build platform if the build platform expands relative to the carrier, or may result in decoupling of the build platform if the build platform shrinks relative to the carrier. In other embodiments, however, an attachment mechanism may include only rigid coupling devices, e.g., if no significant thermal expansion/shrinkage effects are expected to occur during the additive manufacturing process.

[0200] The attachment mechanisms described herein can include any suitable number of coupling devices for each build platform on the carrier, such as one, two, three, four, five, or more coupling devices per build platform. For example, in embodiments where the build platform has a quadrilateral shape (e.g., a square or rectangular shape), the build platform can be coupled to the carrier via one or more coupling devices located at one side, two sides, three sides, or all four sides of the build platform. In some embodiments, the opposite ends of the build platform are secured to the carrier by respective coupling devices, e.g., a first coupling device is located at and coupled to a first end of the build platform, and a second coupling device is located at and coupled to a second, opposite end of the build platform. The sides of the build platform extending between the first and second ends may also be coupled to respective coupling devices, or may not be coupled to any coupling devices (e.g., the sides may instead be held in place via contact with the sides of neighboring build platforms).

[0201] In embodiments where the attachment mechanism includes multiple coupling devices, some or all of the coupling devices may use the same type of coupling, or some or all of the coupling devices may use different types of couplings. For instance, the attachment mechanism can include a first coupling device that couples to a first portion (e.g., a first end) of a build platform, and a second coupling device that couples to a second portion (e.g., a second end) of a build platform. The first coupling device may use the same type of coupling as the second coupling device (e.g., both are mechanical coupling devices, both are magnetic coupling devices, both are flexible coupling devices), or the first coupling device may use a different type of coupling than the second coupling device (e.g., one is a mechanical coupling device and the other is a magnetic coupling device, one is a flexible coupling device and the other is a rigid coupling device).

[0202] The number, type, and location of the coupling devices of the attachment mechanism can be varied as desired, e.g., depending on the number of build platforms, geometry of the build platforms, arrangement of the build platforms on the carrier, material properties of the build platforms, material properties of the carrier, type of additive manufacturing process, directionality of the additive manufacturing process, type of precursor material used in the additive manufacturing process, and/or other relevant considerations. Moreover, although certain embodiments herein are illustrated and described in connection with attachment mechanisms that are located on the carrier, any of the attachment mechanisms can alternatively or additionally be located on the build platform.

[0203] FIGS. **14-29** illustrate representative examples of attachment mechanisms and coupling devices for coupling modular build platforms to a carrier, in accordance with embodiments of the present technology. Any of the embodiments described in connection with FIGS. **14-29** can be combined with each other and/or with any of the embodiments described in connection with FIGS. **4-13**C.

[0204] FIG. **14** is a perspective view of a modular build substrate **1400** including a carrier **1402**, a plurality of build platforms **1404**, and an attachment mechanism including a plurality of blocks **1406**, in accordance with embodiments of the present technology. The carrier **1402** and the build platforms **1404** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-13**C). For instance, the carrier **1402** can be a generally planar substrate for coupling to and supporting the build platforms **1404** during an additive manufacturing process, and the build platforms **1404** can each be a generally planar substrate suitable for coupling to and

supporting at least one additively manufactured object thereon.
[0205] The blocks **1406** can serve as rigid coupling devices to inhibit movement of the build platforms **1404** relative to the carrier **1402**, e.g., due to forces applied to the build platforms **1404** by a printer assembly (e.g., the printer assembly **302** of FIG. **3** or the printer assembly **506** of FIG. **5**). In the illustrated embodiment, for example, the build platforms **1404** are arranged in a linear array along the direction of motion of the printer assembly (e.g., direction X), and the blocks **1406** are positioned on the carrier **1402** at opposite ends of the linear array. The blocks **1406** can contact and engage the edges of the build platforms **1404** at the ends of the linear array (e.g., the leftmost and rightmost build platforms **1404**) to obstruct movement of the build platforms **1404** relative to the carrier **1402** along the X direction.
[0206] Although FIG. **14** illustrates four blocks **1406**, with two blocks **1406** at each end of the linear array of build platforms **1404**, in other embodiments, a different number of blocks **1406** can be used, such as one, three, four, five, or more blocks **1406** at each end of the array. Moreover, the location of the blocks **1406** can be varied as desired, e.g., the blocks **1406** may be located at one end of the array only, such as at the left end only or the right end only; and/or additional blocks **1406** may be positioned at other locations, such as the upper or lower edges of the build platforms **1404**. Further, the blocks **1406** can have any suitable shape (e.g., pins, posts, wedges).
[0207] FIG. **15** is a side cross-sectional view of a modular build substrate **1500** including a carrier **1502**, a build platform **1504**, and an attachment mechanism including an external spring clip **1506**, in accordance with embodiments of the present technology. The carrier **1502** and the build platform **1504** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-13**C). For instance, the carrier **1502** can be a generally planar substrate for coupling to and supporting the build platform **1504** during an additive manufacturing process, and the build platform **1504** can be a generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.
[0208] The spring clip **1506** can serve as a flexible coupling device for securing the build platform **1504** to the carrier **1502**. The spring clip **1506** can be made partially or entirely out of a spring material (e.g., a spring metal) that is deformable and/or deflectable when engaged by the build platform **1504**, and the deformation and/or deflection of the spring material can produce a spring force that is directed against the build platform **1504** to retain the build platform **1504** on the carrier **1502**. The flexibility of the spring clip **1506** can accommodate some lateral movement of the build platform **1504** relative to the carrier **1502**, e.g., due to thermal expansion/shrinkage effects as discussed elsewhere herein. In some embodiments, the spring clip **1506** is configured to engage an external portion of the build platform **1504**. The external portion can be any portion of the build platform **1504** that is accessible without having to decouple the build platform **1504** from the carrier **1502**. For instance, as shown in FIG. **15**, the external portion can be at or proximate to a lateral side of the build platform **1504**.
[0209] The spring clip **1506** can include a base portion **1508**, an engagement portion **1510**, and an optional handle portion **1512**. The base portion **1508** of the spring clip **1506** can be coupled to the carrier **1502**, e.g., via a fastener, actuatable gripper, adhesive, bonding, welding, or a suitable combination thereof. For example, a fastener **1514** (e.g., a screw) can be positioned through the base portion **1508** to couple the spring clip **1506** to the carrier **1502** through the base portion **1508**. In some embodiments, the fastener **1514** is inserted into a hole in the base portion **1508** and is received by a milled hole or recess in the carrier **1502**.
[0210] The engagement portion **1510** can be connected to the base portion **1508** and can extend laterally inward toward the build platform **1504**. The engagement portion **1510** is configured to contact the build platform **1504** to couple the build platform **1504** to the carrier **1502**. In some embodiments, the engagement portion **1510** is configured to engage an external projection **1516** in the build platform **1504**. The projection **1516** can be a lip, flange, shoulder, etc., that extends laterally from a side of the build platform **1504**, e.g., by a distance of at least 1 mm, 2 mm, 5 mm,

or 10 mm. In some embodiments, the projection **1516** includes an elongate body having a first upper surface **1518** that is lower than a second upper surface **1520** of the rest of the build platform **1504**. For example, the first upper surface **1518** can be lower than the second upper surface **1520** by at least 5 mm, 1 cm, 2 cm, etc. This configuration can ensure that the spring clip **1506** remains below the second upper surface **1520** of the build platform **1504**, e.g., to avoid interfering with the additive manufacturing process.

[0211] The engagement portion **1510** of the spring clip **1506** can contact the first upper surface **1518** of the projection **1516** to couple the projection **1516**, and thus the build platform **1504**, to the carrier **1502**. For instance, when the build platform **1504** is placed on the carrier **1502**, the projection **1516** can displace the engagement portion **1510** of the spring clip **1506** upward and away from the carrier **1502**, and the engagement portion **1510** can resist the displacement to apply a spring force downward against the first upper surface **1518** of the projection **1516**. In some embodiments, the projection **1516** may be shaped to complement the spring clip **1506**. For example, although the first upper surface **1518** of the projection **1516** is depicted as being horizontal, the first upper surface **1518** can alternatively be angled, e.g., in embodiments where the engagement portion **1510** of the spring clip **1506** is also angled.

[0212] The handle portion **1512** can be connected to the engagement portion **1510** to allow the spring clip **1506** to be retracted, e.g., during placement of the build platform **1504** on the carrier **1502** and/or removal of the build platform **1504** from the carrier **1502**. For example, a user can pull upward on the handle portion **1512** to temporarily retract the spring clip **1506** upward and away from the carrier **1502**, thus allowing the build platform **1504** to be placed on the carrier **1502**. The handle portion **1512** can then be released to allow the spring clip **1506** to move downward toward the build platform **1504** so that the engagement portion **1510** contacts and applies a downward force against the projection **1516**. To remove the build platform **1504** from the carrier **1502**, the handle portion **1512** can be lifted upward to separate the engagement portion **1510** from the projection **1516**, thereby releasing the build platform **1504** from the spring clip **1506**.

[0213] Although FIG. **15** illustrates a single spring clip **1506** that engages a single portion of the build platform **1504**, in other embodiments, the modular build substrate **1500** can include a plurality of spring clips **1506** on the carrier **1502** to couple to different portions of the build platform **1504** (e.g., different sides of the build platform **1504**). Moreover, the build platform **1504** can include any of the features described in connection with the other embodiments herein, such as a plurality of projections **1522** (e.g., pins, ribs) for creating a gap between the build platform **1504** and carrier **1502**.

[0214] FIGS. **16**A-**16**C illustrate a modular build substrate **1600** including a carrier **1602**, a build platform **1604**, and an attachment mechanism including a plurality of internal spring clips **1606**, in accordance with embodiments of the present technology. Specifically, FIG. **16**A is a side cross-sectional view of the build platform **1604** coupled to the carrier **1602** via an individual spring clip **1606**, FIG. **16**B is a bottom view of the build platform **1604**, and FIG. **16**C is a top view of the carrier **1602** and a plurality of spring clips **1606**. The carrier **1602** and the build platform **1604** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4**-**13**C). For instance, the carrier **1602** can be a generally planar substrate for coupling to and supporting the build platform **1604** during an additive manufacturing process, and the build platform **1604** can be a generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0215] Referring to FIGS. **16**A and **16**C together, the spring clips **1606** can serve as flexible coupling devices for securing the build platform **1604** to the carrier **1602**. The spring clips **1606** can be generally similar to the spring clip **1506** of FIG. **15**. For example, the spring clips **1606** can each be made partially or entirely out of a spring material that applies a spring force to the build platform **1604** to retain the build platform **1604** on the carrier **1602**, and the flexibility of the spring material can permit some degree of lateral movement of the build platform **1604** relative to the carrier **1602**.

[0216] In some embodiments, as best seen in FIG. **16**A, the spring clips **1606** are each configured to engage an internal portion **1608** of the build platform **1604**. The internal portion **1608** can be any portion of the build platform **1604** that is not accessible without first decoupling the build platform **1604** from the carrier **1602**. For instance, the internal portion **1608** can be at or proximate to a lower surface of the build platform **1604**. As depicted, the internal portion **1608** can define a recess **1610** in the lower surface that is configured to receive the spring clip **1606** therein when the build platform **1604** is placed on the carrier **1602**. In some embodiments, the use of spring clips **1606** that engage the internal portions **1608** of the build platform **1604** allows for a greater build area (e.g., a larger unobstructed upper surface of the build platform **1604**). Moreover, the spring clip **1606** may be protected from residual curable material (e.g., resin) during the additive manufacturing process.

[0217] Each spring clip **1606** can include a base portion **1612**, an engagement portion **1614**, and an optional handle portion **1616**. The base portion **1612** of the spring clip **1606** can be coupled to the carrier **1602** (e.g., via a fastener, actuatable gripper, adhesive, bonding, welding, or a suitable combination thereof). For example, at least one fastener **1618** (e.g., a screw) can be inserted into at least one hole in the base portion **1612** and received by a milled hole and/or recess in the carrier **1602**.

[0218] The engagement portion **1614** can be connected to the base portion **1612** and can be configured to couple to an internal projection **1620** in the build platform **1604**. The projection **1620** can be a lip, flange, shoulder, etc., that extends into the recess **1610**, e.g., by a distance of at least 1 mm, 2 mm, 5 mm, or 10 mm. The engagement portion **1614** of the spring clip **1606** can directly contact the projection **1620** to couple the projection **1620**, and thus the build platform **1604**, to the carrier **1602**. For instance, when the build platform **1604** is placed on the carrier **1602**, the projection **1620** can displace the engagement portion **1614** of the spring clip **1606** upward and away from the carrier **1602**, and the engagement portion **1614** can resist the displacement to apply a spring force downward against the projection **1620** to constrain vertical and/or lateral movement of the build platform **1604** relative to the carrier **1602**. Although the upper surface of the projection **1620** is depicted as being angled (e.g., to pull the build platform **1604** downward against the carrier **1602**), in other embodiments, the upper surface of the projection **1620** can instead be horizontal.

[0219] Referring to FIGS. **16**B and **16**C together, the carrier **1602** can include a plurality of spring clips **1606** and the build platform **1604** can include a corresponding plurality of recesses **1610**, such that each spring clip **1606** is received within a corresponding recess **1610**. Although the illustrated embodiment shows four spring clips **1606** and four recesses **1610**, with each spring clip **1606** and recess **1610** being located near a respective corner of the build platform **1604**, in other embodiments, a different number of spring clips **1606** and recesses **1610** can be used (e.g., one, two, three, five, or more spring clips **1606** and recesses **1610**), and/or the spring clips **1606** and recesses **1610** can be positioned at different locations relative to the build platform **1604**.

[0220] Referring again to FIG. **16**A, the build platform **1604** may optionally include an overhanging edge **1622**. The overhanging edge **1622** can extend laterally past an edge of the carrier **1602** such that excess material (e.g., excess resin and/or debris) on the build platform **1604** can flow along and drip off the edge **1622** to a collection reservoir (not depicted), as opposed to accumulating on the carrier **1602** and/or between the carrier **1602** and the build platform **1604**. The collection reservoir can be a groove, channel, gutter, etc., formed in or coupled to the carrier **1602**, or can be separate from the carrier (e.g., a separate bucket, vat, drip tray). In some embodiments, the overhanging edge **1622** is perpendicular to the upper surface **1624** of the build platform **1604**. In other embodiments, the overhanging edge **1622** is angled, e.g., having a drop-off of 0 to 10 degrees, 20 to 30 degrees, 30 to 40 degrees, 40 to 50 degrees, 50 to 60 degrees, 60 to 70 degrees, 70 to 80 degrees, 80 to 90 degrees, etc.

[0221] In some embodiments, the build platform **1604** includes a notch **1626** formed in the lower surface near the edge of the build platform **1604**. The notch **1626** can optionally be configured to receive a portion of a removal tool that decouples the build platform **1604** from the carrier **1602**. In such cases, the removal tool may be used to apply an upward force against the build platform **1604**, and the upward force can disengage the projection **1620** from the spring clip **1606** to release the build platform **1604** from the carrier **1602**.

[0222] Furthermore, the build platform **1604** can optionally include a plurality of protrusions **1628**. The protrusions **1628** can be generally similar to the protrusions **1312** of FIG. **13** and be configured to provide a separation distance between the build platform **1604** and the carrier **1602**. In some embodiments, the protrusions **1628** are pins arranged in a grid, e.g., as depicted in FIG. **16**B.

[0223] FIG. **17** is a side cross-sectional view of a modular build substrate **1700** including a carrier **1702**, a build platform **1704**, and an attachment mechanism including a plurality of spring clips **1706***a*, **1706**, in accordance with embodiments of the present technology. The carrier **1702** and the build platform **1704** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-13**C). For instance, the carrier **1702** can be a generally planar substrate for coupling to and supporting the build platform **1704** during an additive manufacturing process, and the build platform **1704** can be a generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0224] The plurality of spring clips **1706***a*, **1706***b* can include a first spring clip **1706***a* and a second spring clip **1706***b* (collectively, “spring clips **1706**”). The spring clips **1706** can be generally similar to the spring clip **1506** of FIG. **15** and/or the spring clip **1606** of FIG. **16**, except that the spring clips **1706** are oriented vertically rather than laterally. For example, the spring clips **1706** can each be made partially or entirely out of a spring material that applies a spring force to the build platform **1704** to retain the build platform **1704** on the carrier **1702**, and that permits some degree of lateral movement of the build platform **1704** relative to the carrier **1702**. In some embodiments, the spring clips **1706** each include a base portion **1708**, an engagement portion **1710**, and an optional handle portion **1712**, which may be generally similar to the corresponding components of the embodiments of FIGS. **15** and **16**.

[0225] The spring clips **1706** can each be affixed to a respective side of the carrier **1702**. For example, as depicted, the first spring clip **1706***a* is attached to a first (e.g., left) side **1702***a* of the carrier **1702** and the second spring clip **1706***b* is attached to a second (e.g., right) side **1702***b* of the carrier **1702**, e.g., via respective fasteners **1714**. The spring clips **1706** can be oriented vertically such that the engagement portions **1710** and handle portions **1712** extend upward from the base portions **1708** and toward the build platform **1704**. In some embodiments, at least one end of each of the spring clips **1706** surpasses the total height of the carrier **1702**.

[0226] The spring clips **1706** can be configured to releasably couple the build platform **1704** to the carrier **1702**. For instance, the first spring clip **1706***a* can be configured to couple to a first (e.g., left) side **1702***a* of the build platform **1704**, and the second spring clip **1706***b* can be configured to couple to a second (e.g., right side) **1702***b* of the build platform **1704**. In the illustrated embodiment, the spring clips **1706** are each configured to engage an external portion of the build platform **1704**, such as external projections **1716** formed in the lateral surfaces of the build platform **1704**. The projections **1716** can be lips, flanges, shoulders, etc., that extend laterally from sides of the build platform **1704**, e.g., by a distance of at least 1 mm, 2 mm, 5 mm or 10 mm.

[0227] The projections **1716** can have respective upper surfaces **1718** configured to engage the engagement portions **1710** of the spring clips **1706**. When the build platform **1704** is placed on the carrier **1702**, the projections **1716** can displace the engagement portions **1710** of the spring clips **1706** upward and/or outward, and the engagement portions **1710** can resist the displacement to impart one or more spring forces onto the upper surfaces **1718** of the projections **1716** to constrain vertical and/or lateral movement of the build platform **1704** relative to the carrier **1702**. The upper surfaces **1718** can have a geometry that is complementary to the engagement portions **1710** of the spring clips **1706**. In some embodiments, the upper surfaces **1718** have a first angle that is complementary to a second angle of the engagement portions **1710**. In other embodiments, however, the upper surfaces **1718** can instead be horizontal.

[0228] In some embodiments, the build platform **1704** further includes overhanging edges **1720** that, similar to the overhanging edges **1622** of FIG. **16**, are configured to direct excess material into one or more collection reservoirs (not depicted). In some embodiments, the overhanging edges **1720** extend laterally beyond the spring clips **1706**, e.g., by at least 2 mm, 5 mm, 1 cm, 2 cm, or 5 cm, to protect the spring clips **1706** from the excess material.

[0229] The modular build substrate **1700** can further include registration features to align the build platform **1704** to the carrier **1702** in a predetermined position and/or orientation. For instance, the carrier **1702** can include a first registration feature **1722**, and the build platform **1704** can include a second registration feature **1724** that mates with the first registration feature **1722** to affix the position and/or orientation of the build platform **1704** relative to the carrier **1702**. In the illustrated embodiment, the first registration feature **1722** is a protrusion in the carrier **1702** (e.g., a post, pin, bump), and the second registration feature **1724** is a recess in the build platform **1704** (e.g., a hole, channel, aperture) that receives the first registration feature **1722**. In other embodiments, the first registration feature **1722** can be a recess in the build platform **1704** and the second registration feature **1724** can be a protrusion in the carrier **1702**. Alternatively, or in combination, other types of components can be used for the first registration feature **1722** and the second registration features **1724**, such as fasteners, magnets, hooks, teeth, snap-fit elements, and/or other components that mate or otherwise engage each other to align the carrier **1702** to the build platform **1704**. Furthermore, although FIG. **17** illustrates a single registration feature **1722** in the carrier **1702** and a single registration feature **1724** in the build platform **1704**, the carrier **1702** and build platform **1704** can alternatively include a plurality of registration features.

[0230] FIG. **18** is a side cross-sectional view of a modular build substrate **1800** including a carrier **1802**, a build platform **1804**, and an attachment mechanism including a plurality of spring clips **1806**, **1808**. The carrier **1802** and the build platform **1804** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-13**C). For instance, the carrier **1802** can be a generally planar substrate for coupling to and supporting the build platform **1804** during an additive manufacturing process, and the build platform **1804** can be a generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0231] The attachment mechanism can include an internal spring clip **1806** configured to engage an internal portion of the build platform **1804**, and an external spring clip **1808** configured to engage an external portion of the build platform **1804**. The internal spring clip **1806** can be generally similar to the spring clip **1606** of FIG. **16**, and the external spring clip **1808** can be generally similar to the spring clip **1706** of FIG. **17**. For example, the spring clips **1806**, **1808** can each be made partially or entirely out of a spring material that applies a spring force to the build platform **1804** to retain the build platform **1804** on the carrier **1802**, and that permits some degree of lateral movement of the build platform **1804** relative to the carrier **1802**. The internal spring clip **1806** can have a first base portion **1810**, a first engagement portion **1812**, and an optional first handle portion **1814**; and the external spring clip **1808** can have a second base portion **1816**, a second engagement portion **1818**, and a second optional handle portion **1820**.

[0232] In some embodiments, the internal spring clip **1806** and the external spring clip **1808** are used in tandem to secure the build platform **1804** to the carrier **1802**. For instance, as shown in FIG. **18**, the internal spring clip **1806** and the external spring clip **1808** can both be located at the same side of the build platform **1804**, and thus can be used to collectively couple the side of the build platform **1804** to the carrier **1802**. However, in other embodiments, the internal spring clip **1806** and the external spring clip **1808** may be used independently from one another, e.g., the attachment mechanism may include the internal spring clip **1806** only or the external spring clip **1808** only. Moreover, in other embodiments, the internal spring clip **1806** may be located at a different (e.g., opposite) side of the build platform **1804** than the external spring clip **1808**.

[0233] In some embodiments, the build platform **1804** includes an internal portion **1822** defining a recess **1824** for receiving the internal spring clip **1806** therein when the build platform **1804** is placed on the carrier **1802**. The internal portion **1822** and recess **1824** may be generally similar to the internal portion **1608** and recess **1610** of FIGS. **16**A-**16**C. For example, the internal portion **1822** can further include an internal projection **1826**, such that the first engagement portion **1812** of the internal spring clip **1806** is configured to directly contact and couple to the internal projection **1826**. When the build platform **1804** is placed on the carrier **1802**, the internal projection **1826** can displace the first engagement portion **1812** of the internal spring clip **1806** upward and away from the carrier **1802**, and the first engagement portion **1812** can resist the displacement to apply a spring force downward against the internal projection **1826** to constrain vertical and/or lateral movement of the build platform **1604** relative to the carrier **1602**. The internal projection **1826** can be generally similar to the projection **1620** of FIG. **16**. In some embodiments, the internal projection **1826** is a lip, flange, shoulder, etc., that extends laterally into the recess **1824**, e.g., by a distance of at least 1 mm, 2 mm, 5 mm, or 10 mm. The upper surface of the internal projection **1826** can have a first angle that is complementary to a second angle of the first engagement portion **1812** of the internal spring clip **1806**. In other embodiments, the upper surface of the internal projection **1826** can instead be horizontal.
[0234] The external spring clip **1808** can be affixed to a side of the carrier **1802**, e.g., via a fastener. The external spring clip **1808** can be oriented vertically such that the second engagement portion **1818** and second handle portion **1820** extend upward from the second base portion **1816** and toward the build platform **1804**. The second engagement portion **1818** can be configured to engage an external projection **1828** formed in the lateral surface of the build platform **1804**. When the build platform **1804** is placed on the carrier **1802**, the external projection **1828** can displace the second engagement portion **1818** of the external spring clip **1808** upward and/or outward, and the second engagement portion **1818** can resist the displacement to impart one or more spring forces onto the upper surface of the external projection **1828** to constrain vertical and/or lateral movement of the build platform **1804** relative to the carrier **1802**. The external projection **1828** can be generally similar to the projection **1716** of FIG. **17**. For instance, the external projection **1828** can be a lip, flange, shoulder, etc., that extends laterally from the build platform **1804**, e.g., by a distance of at least 1 mm, 2 mm, 5 mm or 10 mm. In some embodiments, the upper surface of the external projection **1828** includes a third angle that is complementary to a fourth angle of the second engagement portion **1818** of the external spring clip **1808**. In other embodiments, however, the upper surface of the external projection **1828** can instead be horizontal.
[0235] In some embodiments, the build platform **1804** further includes an overhanging edge **1830**. The overhanging edge **1830** can be generally similar to the overhanging edges **1720** of FIG. **17** and can be configured to direct excess material (e.g., resin) into a collection reservoir and/or away from the recess **1824**. In some embodiments, the overhanging edge **1830** extends laterally beyond the external spring clip **1808**, e.g., by at least 2 mm, 5 mm, 1 cm, 2 cm, or 5 cm, to protect the external spring clip **1808** from the excess material.
[0236] In some embodiments, the build platform **1804** is mounted onto the carrier **1802** using a placement tool. For example, the bottom surface **1832** of the carrier **1802** can include a first recess **1834** formed therein for receiving a portion of the placement tool. The placement tool can include one or more elongate members that exert forces on the carrier **1802** via the first recess **1834**, which in turn cause the build platform **1804** to engage with the carrier **1802**. Further details of an example placement process are described below in connection with FIGS. **19**A and **19**B.
[0237] In some embodiments, the build platform **1804** is removed from the carrier **1802** using a removal tool. For example, the lateral surface **1836** of the carrier **1802** can include a second recess **1838** formed therein for receiving a portion of the removal tool. The removal tool can include one or more elongate members that exert forces on the carrier **1802** via the second recess **1838**, which in turn causes the build platform **1804** to disengage from the carrier **1802**. Further details of an

example removal process are described below in connection with FIGS. **20**A and **20**B.

[0238] In other embodiments, however, the build platform **1804** may be placed on the carrier **1802** and/or removed from the carrier **1802** without aid of any tools. In such embodiments, the first recess **1834** and/or the second recess **1838** are optional and may be omitted.

[0239] FIGS. **19**A and **19**B illustrate a process for coupling the build platform **1804** to the carrier **1802** using a placement tool **1902**, in accordance with embodiments of the present technology. Specifically, FIG. **19**A is a side cross-sectional view of the carrier **1802**, build platform **1804**, and placement tool **1902** in a first configuration before the build platform **1804** is coupled to the carrier **1802**, and FIG. **19**B is a side cross-sectional view of the carrier **1802**, build platform **1804** and placement tool **1902** in a second configuration after the build platform **1804** is coupled to the carrier **1802**.

[0240] Turning now to FIG. **19**A, the placement tool **1902** can include an elongate body **1904** that acts as a lever for applying forces to the carrier **1802** and the build platform **1804** to couple these components to each other, e.g., by causing the spring clips **1806**, **1808** on the carrier **1802** to engage the corresponding portions of the build platform **1804**. The elongate body **1904** can have a suitable length and can provide a handle for a user to grasp and rotate the placement tool **1902** via the elongate body **1904**. For example, the elongate body **1904** can have a length of at least 5 cm, 10 cm, 15 cm, 20 cm, etc. The elongate body **1904** can be made out of a relatively rigid and/or stiff material, such as a metal (e.g., steel, aluminum, brass, copper, titanium), a ceramic, a polymer (e.g., thermoformed or thermoset polymer), a composite, or suitable combinations thereof.

[0241] The elongate body **1904** can have a first arm **1906** configured to contact and apply force to the build platform **1804**, and a second arm **1908** configured to contact and apply force to the carrier **1802**. The first arm **1906** can include a roller **1910** (e.g., a wheel or rotating cylinder) that is configured to slide and/or roll along the upper surface of the build platform **1804** during the placement process to distribute forces along the build platform **1804**. The second arm **1908** can include a tip **1912** that is configured to fit at least partially into the first recess **1834** of the carrier **1802** to create leverage for the roller **1910** to press the build platform **1804** downward against the carrier **1802**. In some embodiments, the first arm **1906** and the second arm **1908** are connected by a bridge **1914**. The curvature and the arc length of the bridge **1914** can be adjusted, e.g., based on the dimensions of the build platform **1804** and carrier **1802**. For example, the bridge **1914** can be an extendable member configured to adapt to the height of the build platform **1804** and carrier **1802**.

[0242] During the placement process, the build platform **1804** can be placed onto the carrier **1802**, with the internal spring clip **1806** and external spring clip **1808** on the carrier **1802** initially disengaged from the internal projection **1826** and external projection **1828**, respectively, of the build platform **1804**. The tip **1912** of the second arm **1908** can then be inserted into the first recess **1834** of the carrier **1802**, and the roller **1910** of the first arm **1906** can be placed against the upper surface of the build platform **1804**. The user can then rotate the elongate body **1904**, e.g., in a counterclockwise direction with the tip **1912** of the second arm **1908** serving as the center of rotation. The rotation of the elongate body **1904** can produce a downward force that is transmitted through the first arm **1906** and roller **1910** onto the build platform **1804**. The roller **1910** can slide along the upper surface of the build platform **1804** as the elongate body **1904** rotates to propagate the applied forces from the edge of the build platform **1804** toward the center of the build platform **1804**. The forces applied to the build platform **1804** can cause the internal projection **1826** and external projection **1828** to engage with the internal spring clip **1806** and external spring clip **1808**, respectively, thus coupling the build platform **1804** to the carrier **1802**, as depicted in FIG. **19**B. The placement tool **1902** can then be removed.

[0243] FIGS. **20**A and **20**B illustrate a process for decoupling the build platform **1804** from the carrier **1802** using a removal tool **2002**, in accordance with embodiments of the present technology. Specifically, FIG. **20**A is a side cross-sectional view of the carrier **1802**, build platform **1804**, and removal tool **2002** in a first configuration while the build platform **1804** is coupled to the carrier

1802, and FIG. **20**B is a side cross-sectional view of the carrier **1802**, build platform **1804** and removal tool **2002** in a second configuration after the build platform **1804** is decoupled from the carrier **1802** (the internal spring clip **1806**, internal portion **1822**, recess **1824**, and internal projection **1826** are omitted from FIGS. **20**A and **20**B merely for purposes of simplicity).

[0244] Turning now to FIG. **20**A, the removal tool **2002** can include an elongate body **2004** that acts as a lever for applying forces to the carrier **1802** and the build platform **1804** to decouple these components from each other, e.g., by causing the spring clips **1806**, **1808** on the carrier **1802** to disengage the corresponding portions of the build platform **1804**. The elongate body **2004** can have a suitable length and can provide a handle for a user to grasp and rotate the removal tool **2002** via the elongate body **2004**. For example, the elongate body **2004** can be made out of a relatively rigid and/or stiff material, such as a metal (e.g., steel, aluminum, brass, copper, titanium), a ceramic, a polymer (e.g., thermoformed or thermoset polymer), a composite, or suitable combinations thereof. The elongate body **2004** can have an arm **2006** configured to contact and apply force to the carrier **1802** and/or the build platform **1804**. The arm **2006** can engage the carrier **1802** and/or the build platform **1804** via a tip **2008** that is configured to fit into the second recess **1838** of the carrier **1802** to create leverage for the arm **2006** to press the build platform **1804** upward away from the carrier **1802**.

[0245] The build platform **1804** can initially be coupled to the carrier **1802**, with the internal spring clip **1806** and the external spring clip **1808** on the carrier **1802** initially engaged with the internal projection **1826** and the external projection **1828**, respectively, of the build platform **1804**. The tip **2008** of the arm **2006** can then be inserted into the second recess **1838** of the carrier **1802**. The user can then rotate the elongate body **2004**, e.g., in a clockwise direction with the tip **2008** of the arm **2006** serving as the center of rotation. The rotation of the elongate body **2004** can produce an upward force that is transmitted through the arm **2006** onto the build platform **1804**. The force applied to the build platform **1804** can cause the internal projection **1826** and external projection **1828** to disengage from the internal spring clip **1806** and external spring clip **1808**, respectively, thus decoupling the build platform **1804** from the carrier **1802**, as depicted in FIG. **20**B. The removal tool **2002** can then be removed.

[0246] FIGS. **21**A-**21**C illustrate a modular build substrate **2100** including a carrier **2102**, one or more build platforms **2104**, and an attachment mechanism including one or more spring clips **2106** and one or more rotatable clips **2108**, in accordance with embodiments of the present technology. Specifically, FIG. **21**A is a perspective view of the modular build substrate, FIG. **21**B is a side view of a portion of the modular build substrate including the one or more spring clips **2106**, and FIG. **21**C is a perspective view of another portion of the modular build substrate including the one or more rotatable clips **2108**.

[0247] Referring now to FIG. **21**A, the carrier **2102** and the build platforms **2104** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-13**C). For instance, the carrier **2102** can be a generally planar substrate for coupling to and supporting the build platforms **2104** during an additive manufacturing process, and the build platforms **2104** can each be a generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon. As depicted, the build platforms **2104** can be arranged in a linear array with a separation distance D.sub.2 between neighboring build platforms **2104**. For instance, the separation distance D.sub.2 can be at least 1 mm, 2 mm, 5 mm, 1 cm, 2 cm, or more. The separation distance D.sub.2 can ensure adequate space to accommodate the attachment mechanism (e.g., the spring clips **2106** and the rotatable clips **2108**), which may be positioned between the build platforms **2104** as shown in FIG. **21**A. Additionally, or alternatively, the separation distance D.sub.2 can help ensure that the build platforms **2104** can be independently and easily removed from the carrier **2102**.

[0248] Each build platform **2104** can be coupled to the carrier **2102** via one or more spring clips **2106** and one or more rotatable clips **2108**. The spring clips **2106** can be positioned proximate to and configured to engage a first side **2104***a* of the build platform **2104**, and the rotatable clips **2108** can be positioned proximate to and configured to engage a second, opposite side **2104***b* of the build platform **2104**. The first side **2104***a* and second side **2104***b* can correspond to the edges of the build platform **2104** that are orthogonal to the direction of motion of the printer assembly (e.g., direction X). In some embodiments, the spring clips **2106** serve as flexible coupling devices that allow some lateral movement of the first side **2104***a* of the build platform **2104** relative to the carrier **2102**, while the rotatable clips **2108** serve as rigid coupling devices that prevent lateral movement of the second side **2104***b* of the build platform **2104** relative to the carrier **2102**. This configuration can be advantageous to accommodate dimensional changes of the build platform **2104** due to thermal expansion/shrinkage effects, as described elsewhere herein.

[0249] In some embodiments, the spring clips **2106** are arranged on the carrier **2102** in a linear array (e.g., a row). The spring clips **2106** may collectively releasably couple the first side **2104***a* of the build platform **2104** to the carrier **2102**. Each of the spring clips **2106** can be generally similar to the spring clips **1506** of FIG. **15**. For example, as shown in FIG. **21**B, each spring clip **2106** can be made partially or entirely out of a spring material and can include a base portion **2110**, an engagement portion **2112**, and an optional handle portion **2114**. The base portion **2110** can be coupled to the carrier **2102**, e.g., via a fastener, actuatable gripper, adhesive, bonding, welding, or a suitable combination thereof. The engagement portion **2112** can be configured to engage an external portion of the build platform **2104**, such as an external projection **2116**, which may be generally similar to the projection **1516** of FIG. **15**. For example, the external projection **2116** can extend laterally from the first side **2104***a* of the build platform **2104** and can have a lower upper surface than the build platform **2104**. When the build platform **2104** is placed on the carrier **2102**, the external projection **2116** can displace the engagement portion **2112** of the spring clip **2106** upward and away from the carrier **2102**, and the engagement portion **2112** can resist the displacement to apply a spring force downward against the external projection **2116** to secure the first side **2104***a* of the build platform **2104** to the carrier **2102**.

[0250] The rotatable clips **2108** can be arranged on the carrier **2102** in a linear array (e.g., in a row). The rotatable clips **2108** may collectively releasably couple the second side **2104***b* of the build platform **2104** to the carrier **2102**. The rotatable clips **2108** can be configured to engage and lock the second side **2104***b* of the build platform **2104** to the carrier **2102** in a manner that inhibits movement of the second side **2104***b* of build platform **2104** relative to the carrier **2102**.

[0251] Referring now to FIG. **21**C, the rotatable clips **2108** can include a base portion **2118** and an engagement portion **2120** coupled to the base portion **2118** via a hinge **2122**. The base portion **2118** can be coupled to the carrier **2102**, e.g., via fasteners, actuatable grippers, adhesives, bonding, welding, or a suitable combination thereof. The engagement portion **2120** can be configured as a generally flat structure, such as a sheet, strip, tab, etc. The engagement portion **2120** can be made out of a relatively rigid and/or stiff material, such as metal (e.g., steel, aluminum, brass, copper, titanium), a ceramic, a polymer (e.g., thermoformed or thermoset polymer), a composite, or suitable combinations thereof. The hinge **2122** can allow the engagement portion **2120** to rotate in a vertical direction relative to the base portion **2118**. For example, the engagement portion **2120** can rotate in a range from 0 to 30 degrees, 30 to 60 degrees, 60 to 90 degrees, 90 to 120 degrees, 120 to 150 degrees, or 150 to 180 degrees.

[0252] The rotatable clips **2108** can be configured to rotate between an open configuration in which the engagement portions **2120** are disengaged from the second side **2104***b* of the build platform **2104**, and a closed configuration in which the engagement portions **2120** are engaged with the second side **2104***b* of the build platform. In the closed configuration, the engagement portions **2120** can directly contact an external portion of the second side **2104***b* of the build platform, such as one or more external projections **2124** of the build platform **2104**, which can be identical or generally similar to the external projection **2116**. The contact between the engagement portions **2120** and the external projections **2124** can obstruct movement of the second side **2104***b* of the build platform

2104 relative to the carrier **2102**. Moreover, when in the closed configuration, the rotatable clips **2108** can be entirely below the upper surface of the build platform **2104**, e.g., to avoid interfering with the additive manufacturing process.

[0253] The rotatable clips **2108** can optionally be disengaged from the build platform **2014** by a release mechanism **2126**. In some embodiments, the release mechanism **2126** is a tab extending vertically that, when retracted, cause the engagement portions **2120** of the rotatable clips **2108** to rotate away from the build platform **2104**. For example, a user may pull back the release mechanism **2126** away from the build platform **2104** to free the second side **2104***b* of the build platform **2104** from the carrier **2102**. In other embodiments, the release mechanism **2126** can have a different form factor, such as a strut, pin, switch, cantilever, string, etc.

[0254] Although FIGS. **21**A-**21**C illustrate an embodiment in which the first side **2104***a* of the build platforms **2104** are coupled to spring clips **2106** and the second side **2104***b* of the build platforms **2104** are coupled to rotatable clips **2108**, other configurations can also be used, e.g., the spring clips **2106** can be used for both the first side **2104***a* and the second side **2104***b* of the build platforms **2104**, the rotatable clips **2108** can be used for both the first side **2104***a* and the second side **2104***b* of the build platforms **2104**, or the spring clips **2106** and/or rotatable clips **2108** can each independently be substituted with any of the other coupling devices described herein. Moreover, although the remaining sides of the build platforms **2014** (e.g., corresponding to the edges of the build platforms **2104** that are parallel to direction X) are depicted as not being coupled to the carrier **2102** by any coupling devices, in other embodiments, one or both of the remaining sides can be coupled to the carrier **2102** via spring clips **2106**, rotatable clips **2108**, and/or any of the other coupling devices described herein.

[0255] FIGS. **22**A-**22**C illustrate a modular build substrate **2200** including a carrier **2202**, one or more build platforms **2204**, and an attachment mechanism including a plurality of laterally rotatable clips **2206**, in accordance with embodiments of the present technology. Specifically, FIG. **22**A is a perspective view of the modular build substrate, FIG. **22**B is a perspective view of the modular build substrate with the laterally rotatable clips **2206** in an open configuration, and FIG. **22**C is a perspective view of the modular build substrate with the laterally rotatable clips **2206** in a closed configuration.

[0256] Referring now to FIG. **22**A, the carrier **2202** and the build platforms **2204** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-13**C). For instance, the carrier **2202** can be a generally planar substrate for coupling to and supporting the build platforms **2204** during an additive manufacturing process, and the build platforms **2204** can each be a generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon (a single build platform **2204** is shown in FIG. **22**A merely for purposes of simplicity). The build platforms **2204** can be arranged on the carrier **2202** in an array (e.g., a row or grid).

[0257] Each build platform **2204** can be coupled to the carrier **2202** via one or more laterally rotatable clips **2206** and one or more vertically rotatable clips **2208**. The laterally rotatable clips **2206** can be positioned proximate to and configured to engage a first side **2204***a* of the build platform **2204**, and the vertically rotatable clips **2208** can be positioned proximate to and configured to engage a second, opposite side **2204***b* of the build platform **2204**. The first side **2204***a* and second side **2204***b* can correspond to the edges of the build platform **2204** that are parallel to the direction of motion of the printer assembly (e.g., direction X).

[0258] The laterally rotatable clips **2206** can be arranged on the carrier **2202** in a linear array (e.g., in a row). The laterally rotatable clips **2206** can each include an axle **2210** and a body **2212** that is rotatably coupled to the axle **2210**. The axle **2210** can be attached to a base portion **2214** (e.g., a plate, strip, sheet). The base portion **2214** can be affixed to the carrier **2202**, e.g., via a fastener **2216**. For example, the base portion **2214** can be affixed to the carrier **2202** using a screw that is inserted into the base portion **2214** and received by a milled hole and/or recess in the carrier **2202**.

In other embodiments, however, the base portion **2214** may be omitted such that the laterally rotatable clips **2206** may be mounted directly to the carrier **2202** (e.g., the axle **2210** can be attached directly to the carrier **2202**).

[0259] The body **2212** can be an elongate member having a first end **2212***a* and a second end **2212***b* opposite the first end **2212***a*. One or both of the first ends **2212***a*, **2212***b* can serve as engagement portions for coupling to the build platform **2204**. In the illustrated embodiment, the body **2212** has a generally rectangular shape (e.g., rounded rectangular shape). In other embodiments, the body **2212** can have a different shape, such as square, circular, oval, oblong, etc. The body **2212** can be rotated laterally about the axle **2210** through a plurality of different orientations. In some embodiments, the body **2212** is rotatable about the axle **2210** over an angular range of 360 degrees. In other embodiments, the body **2212** may be rotatable about the axle **2210** over an angular range less than 360 degrees, e.g., an angular range less than or equal to 30 degrees, 60 degrees, 90 degrees, 120 degrees, 150 degrees, etc.

[0260] The laterally rotatable clips **2206** can be configured to rotate between an open configuration in which the first end **2212***a* and/or second end **2212***b* are disengaged from one or more build platforms **2204** (e.g., FIG. **22**B), and a closed configuration in which the first end **2212***a* and/or second end **2212***b* are engaged with the one or more build platforms **2204** (e.g., FIG. **22**C). As shown in FIG. **22**B, in the open configuration, the laterally rotatable clips **2206** are in an initial orientation in which the first end **2212***a* and second end **2212***b* are spaced apart from the build platforms **2204**. As shown in FIG. **22**C, in the closed configuration, the laterally rotatable clips **2206** have been rotated approximately 90 degrees from their initial position, such that the first end **2212***a* and/or the second end **2212***b* are engaged with an external portion of one or more build platforms **2204**. Specifically, the first end **2212***a* and/or the second end **2212***b* can be positioned over and/or in direct contact with one or more external projections **2218** of one or more build platforms **2204** to obstruct vertical and/or lateral movement of the build platforms **2204** relative to the carrier **2202**. The external projections **2218** may be generally similar to the other external projections described herein, e.g., the projection **1516** of FIG. **15**.

[0261] Referring back to FIG. **22**A, the carrier **2202** can include one or more vertically rotatable clips **2208** opposite of the laterally rotatable clips **2206** to couple to the second side **2204***b* of the build platform **2204**. The vertically rotatable clips **2208** can be generally similar to the rotatable clips **2108** of FIG. **21**. For example, the vertically rotatable clips **2208** can include a hinged engagement portion that rotates vertically downward to secure the second side **2204***b* of the build platform **2104** to the carrier **2202**.

[0262] Although FIGS. **22**A-**22**C illustrate an embodiment in which the first side **2204***a* of the build platforms **2204** are coupled to laterally rotatable clips **2206** and the second side **2204***b* of the build platforms **2204** are coupled to vertically rotatable clips **2208**, other configurations can also be used, e.g., the laterally rotatable clips **2206** can be used for both the first side **2204***a* and the second side **2204***b* of the build platforms **2204**, the vertically rotatable clips **2208** can be used for both the first side **2204***a* and the second side **2204***b* of the build platforms **2204**, or the laterally rotatable clips **2206** and/or the vertically rotatable clips **2208** can each independently be substituted with any of the other coupling devices described herein. Moreover, although the remaining sides of the build platforms **2204** (e.g., corresponding to the edges of the build platforms **2204** that are orthogonal to direction X) are depicted as not being coupled to the carrier **2202** by any coupling devices, in other embodiments, one or both of the remaining sides can be coupled to the carrier **2202** via laterally rotatable clips **2206**, vertically rotatable clips **2208**, and/or any of the other coupling devices described herein.

[0263] FIG. **23** is a side cross-sectional view of modular build substrate **2300** including a carrier **2302**, a build platform **2304**, and an attachment mechanism including a fixed clip **2306** and a rotatable clip **2308**, in accordance with embodiments of the present technology. The carrier **2302** and build platform **2304** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-12**). For instance, the carrier **2302** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platform **2304** during an additive manufacturing process. The build platform **2304** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0264] The build platform **2304** can be coupled to the carrier **2302** via at least one fixed clip **2306** and at least one rotatable clip **2308**. In the illustrated embodiment, the fixed clip **2306** is configured to engage a first side **2310***a* of the build platform **2304**, and the rotatable clip **2308** is configured to engage a second side **2310***b* of the build platform **2304** opposite the first side **2310***a*. The fixed clip **2306** can serve as a rigid coupling device for securing the first side **2310***a* of the build platform **2304** to the carrier **2302**. As shown in FIG. **23**, the fixed clip **2306** can include a base portion **2312** and an engagement portion **2314**. In some embodiments, the base portion **2312** and the engagement portion **2314** are integrally formed with each other and made of a rigid material that does not substantially deform and/or deflect when the fixed clip **2306** engages the build platform **2304**. The base portion **2312** can be configured to be mounted to the carrier **2302**, e.g., via a fastener, actuatable gripper, adhesive, bonding, welding, or a suitable combination thereof. The engagement portion **2314** can be connected to the base portion **2312**, and can extend upward and laterally inward from the base portion **2312** toward the first side **2310***a* of the build platform **2304**.

[0265] In the illustrated embodiment, the engagement portion **2314** of the fixed clip **2306** is configured to engage an external projection **2316***a* formed in the first side **2310***a* of the build platform **2304** to rigidly couple the first side **2310***a* to the carrier **2302**. The projection **2316** may be generally similar to the other external projections described herein, e.g., the projection **1516** of FIG. **15**. For example, the projection **2316***a* can extend laterally outward from the first side **2310***a* of the build platform **2304**. The projection **2316***a* can be a lip, flange, shoulder, or other suitable member for coupling to the fixed clip **2306**. The projection **2316***a* can be adjacent to the bottom surface of the build platform **2304** and/or positioned below the upper surface of the build platform **2304** such that when the fixed clip **2306** engages the projection **2316***a*, the fixed clip **2306** remains below the upper surface of the build platform **2304**, e.g., to avoid interfering with the additive manufacturing process. Optionally, the build platform **2304** can include an additional projection **2318***a* formed in the first side **2310***a* of the build platform **2304** and positioned above the projection **2316***a*, e.g., to prevent material from dripping onto the projection **2316***a* during the additive manufacturing process.

[0266] The engagement portion **2314** of the fixed clip **2306** can include a lower surface **2320** that contacts an upper surface **2322** of the projection **2316***a*, thereby constraining lateral and vertical movement of the build platform **2304** relative to the carrier **2302**. The lower surface **2320** can optionally be angled toward the carrier **2302** to pull the projection **2316***a*, and thus the build platform **2304**, downward toward the carrier **2302**. In other embodiments, however, the lower surface **2320** may not be angled, and may instead be substantially parallel to the surface of the carrier **2302**. Moreover, although the upper surface **2322** of the projection **2316***a* is depicted as being substantially parallel to the surface of the carrier **2302**, the upper surface **2322** can alternatively be angled.

[0267] The rotatable clip **2308** can serve as a flexible coupling device for securing the second side **2310***b* of the build platform **2304** to the carrier **2302**. As shown in FIG. **23**, the rotatable clip **2308** can include a base portion **2324** and an engagement portion **2326**. The base portion **2324** can be configured to be mounted to the carrier **2302** via a rotatable coupling (e.g., a hinge, pivot, ball joint) that permits rotation of the rotatable clip **2308** in a vertical direction, e.g., as indicated by the double-headed arrow in FIG. **23**. For instance, the rotatable clip **2308** can be rotated between a closed configuration in which the rotatable clip **2308** engages the second side **2310***b* of the build platform **2304** (shown in FIG. **23**), and an open configuration in which the rotatable clip **2308** is spaced apart from the second side **2310***b* of the build platform **2304**.

[0268] When the rotatable clip **2308** is in the closed configuration, the engagement portion **2326** of the rotatable clip **2308** can engage a projection **2316***b* formed in the second side **2310***b* of the build platform **2304**. The projection **2316***b* can be identical or generally similar to the projection **2316***a*, e.g., the projection **2316***b* can be a lip, flange, shoulder, etc., that extends laterally outward from the second side **2310***b* and is positioned below the upper surface of the build platform **2304**. Optionally, the build platform **2304** can include an additional projection **2318***b* formed in the second side **2310***b* of the build platform **2304** above the projection **2316***b*, e.g., to prevent material from dripping onto the projection **2316***b* during the additive manufacturing process.

[0269] The engagement portion **2326** of the rotatable clip **2308** can include a lower surface **2328** that contacts an upper surface **2330** of the projection **2316***b*, thereby constraining lateral and vertical movement of the build platform **2304** relative to the carrier **2302**. Although the lower surface **2328** of the engagement portion **2326** and the upper surface **2330** of the projection **2316***b* are both depicted as being substantially parallel to the surface of the carrier **2302**, in other embodiments, the lower surface **2328** and/or the upper surface **2330** can alternatively be angled. In some embodiments, the rotatable clip **2308** is made partially or entirely out of a flexible material (e.g., a spring material such as a spring metal) such that the rotatable clip **2308** can accommodate some degree of movement of the second side **2310***b* of the build platform **2304** relative to the carrier **2302**, e.g., due to thermal expansion/shrinkage effects as discussed elsewhere herein. In other embodiments, however, the rotatable clip **2308** can instead be configured as a rigid coupling device and thus can be made out of a rigid material that is not deformable and/or deflectable when engaged with the second side **2310***b* of the build platform **2304**.

[0270] The configuration of the modular build substrate **2300** shown in FIG. **23** can provide various advantages. For instance, the build platform **2304** can be placed into and removed from the carrier **2302** simply by rotating the rotatable clip **2308** between the closed and open configurations, respectively, thereby avoiding the need for specialized placement and removal tools. Moreover, the fixed clip **2306** and rotatable clip **2308** can be used to precisely fix the build platform **2304** at a desired location (e.g., x- and y-position) relative to the carrier **2302**, without requiring additional registration features to align the build platform **2304** on the carrier **2302**.

[0271] FIGS. **24**A-**24**G illustrate a modular build substrate **2400** including a carrier **2402**, a plurality of build platforms **2404**, and an attachment mechanism including a plurality of fixed clips **2406** and a plurality of rotatable clips **2408**, in accordance with embodiments of the present technology. Specifically, FIG. **24**A is a top view of the modular build substrate **2400**, FIG. **24**B is a perspective view of a fixed clip **2406**, FIG. **24**C is a perspective view of a portion of a build platform **2404**, FIG. **24**D is a top view of a rotatable clip **2408** and the build platform **2404**, FIG. **24**E is a perspective view of the rotatable clip **2408** and the build platform **2404**, FIG. **24**F is a perspective view of the rotatable clip **2408** in an open configuration, and FIG. **24**G is a perspective view of the rotatable clip **2408** in a closed configuration.

[0272] Referring first to FIG. **24**A, the carrier **2402** and build platforms **2404** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-12**). For instance, the carrier **2402** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platforms **2404** during an additive manufacturing process. The build platforms **2404** can each be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0273] In the illustrated embodiment, the build platforms **2404** are arranged in a linear array on the carrier **2402**. Each build platform **2404** is coupled to the carrier **2402** by a fixed clip **2406** and a rotatable clip **2408**. The fixed clip **2406** can be coupled to a first side **2410***a* of each build platform **2404**, and the rotatable clip **2408** can be coupled to a second side **2410***b* of the build platform **2404** opposite the first side **2410***a* (reference numbers are shown for a single build platform **2404** merely for purposes of simplicity). The first side **2410***a* and second side **2410***b* can correspond to the edges of the build platform **2204** that are parallel to the direction of motion of the printer assembly (e.g., direction X).

[0274] Referring to FIGS. **24**A and **24**B together, the fixed clip **2406** can serve as a rigid coupling device for securing the first side **2410***a* of the build platform **2404** to the carrier **2402**. The fixed clip **2406** can include a base portion **2412** and a pair of engagement portions **2414**. In some embodiments, the base portion **2412** and the engagement portions **2414** are integrally formed with each other and made of a rigid material that does not substantially deform and/or deflect when the fixed clip **2406** engages the build platform **2404**. The base portion **2412** can be configured to be mounted to the carrier **2402**, e.g., via fasteners, actuatable grippers, adhesives, bonding, welding, or a suitable combination thereof. The engagement portions **2414** can be connected to the base portion **2412**, and can extend laterally inward from the base portion **2412** toward the first side **2410***a* of the build platform **2304**.

[0275] Referring next to FIGS. **24**B and **24**C together, each engagement portion **2414** can be configured to engage a respective projection **2416** formed in the first side **2410***a* of the build platform **2404** to rigidly couple the first side **2410***a* to the carrier **2402**. The projections **2416** can each be a lip, flange, shoulder, etc., that extends laterally outward from the first side **2410***a* of the build platform **2404** and/or may be integrally formed with the build platform **2404**. The projections **2416** can be adjacent to the bottom surface of the build platform **2404** and/or positioned below the upper surface of the build platform **2404** such that when the fixed clip **2406** engages the projections **2416**, the fixed clip **2406** remains below the upper surface of the build platform **2404**. Optionally, the build platform **2404** can include an additional projection **2418** located between and above the projections **2416**, e.g., to serve as a handle for removing the build platform **2404** from the carrier **2402**. In such embodiments, the fixed clip **2406** can include a notch **2420** between the engagement portions **2414** to accommodate the additional projection **2418**.

[0276] In some embodiments, the engagement portions **2414** of the fixed clip **2406** each include a lower surface **2422** that contacts an upper surface **2424** of the respective projection **2416** on the build platform **2404**, thereby constraining lateral and vertical movement of the build platform **2404** relative to the carrier **2402**. The lower surface **2422** can optionally be angled toward the carrier **2402** to pull the projection **2416**, and thus the build platform **2404**, downward toward the carrier **2402**. In other embodiments, however, the lower surface **2422** may not be angled, and may instead be substantially parallel to the surface of the carrier **2402**. Moreover, although the upper surfaces **2424** of the projections **2416** are depicted as being substantially parallel to the surface of the build platform **2404**, the upper surfaces **2424** can alternatively be angled.

[0277] Referring next to FIGS. **24**D and **24**E together, the rotatable clip **2408** can serve as a flexible coupling device for securing the second side **2410***b* of the build platform **2404** to the carrier **2402**. The rotatable clip **2408** can include a base portion **2426** and a pair of engagement portions **2428**. In some embodiments, the base portion **2426** and the engagement portions **2428** are integrally formed with each other and made of a rigid material that does not substantially deform and/or deflect when the rotatable clip **2408** engages the build platform **2404**. However, the base portion **2426** can be coupled to the carrier **2402** via a flexible coupling that is deformable and/or deflectable to accommodate some degree of movement of the second side **2410***b* of the build platform **2404** relative to the carrier **2402**, e.g., due to thermal expansion/shrinkage effects. For instance, the base portion **2426** can be coupled to a spring member **2430** that is coupled to the carrier **2402**. As best seen in FIG. **24**A, the spring member **2430** can be an elongate element (e.g., a wire, rod, shaft) that extends along the length of the carrier **2402** and is coupled to the base portion **2426** of each rotatable clip **2408** on the carrier **2402**. The spring member **2430** can terminate in an arm **2432** that serves as a handle for rotating the spring member **2430**. The spring member **2430** can be rotatably coupled to the carrier **2402** and rigidly coupled to the rotatable clips **2408**, such that vertical rotation of the spring member **2430** causes each of the rotatable clips **2408** to rotate together in a vertical direction relative to the carrier **2402** and build platforms **2404** (e.g., as

indicated by the double-headed arrows in FIG. **24**E). For instance, the rotatable clips **2408** can each be rotated between a closed configuration in which each rotatable clip **2408** engages the second side **2410***b* of its respective build platform **2404** (FIGS. **24**D and **24**F) and an open configuration in which each the rotatable clip **2408** is spaced apart from the second side **2310***b* of the build platform **2304** (FIG. **24**G).
[0278] As best seen in FIG. **24**D, the pair of engagement portions **2428** of the rotatable clip **2408** can be configured to engage a pair of projections **2416** formed in the second side **2410***b* of the build platform **2404** to couple the second side **2410***b* to the carrier **2402**. The projections **2416** in the second side **2410***b* can be identical to the projections **2416** in the first side **2410***a*, and accordingly can include any of the features discussed above with respect to FIG. **24**C. Moreover, the build platform **2404** can optionally include an additional projection **2418** formed in the second side **2410***b*, which can be identical to the projection **2418** in the first side **2410***a*, e.g., as previously described with respect to FIG. **24**C.
[0279] Each engagement portion **2428** of the rotatable clip **2408** can include a lower surface **2434** configured to contact an upper surface **2424** of the corresponding projection **2416**, thereby constraining lateral and vertical movement of the build platform **2404** relative to the carrier **2302**. The lower surface **2434** can optionally be angled toward the carrier **2402** to pull the projection **2416**, and thus the build platform **2404**, downward toward the carrier **2402**. In other embodiments, however, the lower surface **2434** may not be angled, and may instead be substantially parallel to the surface of the carrier **2402**. Moreover, although the upper surfaces **2424** of the projections **2416** are depicted as being substantially parallel to the surface of the build platform **2404**, the upper surfaces **2424** can alternatively be angled.
[0280] Referring to FIG. **24**F, when placing a build platform **2404** on the carrier **2402** or removing the build platform **2404** from the carrier **2402**, the spring member **2430** can be rotated in a first direction (e.g., by pulling the arm **2432** upward away from the carrier **2402**) to place the rotatable clip **2408** in an open configuration. In the open configuration, the engagement portions **2428** of the rotatable clip **2408** can be spaced apart from the projections **2416** at the second side **2410***b* of the build platform **2404**.
[0281] Referring next to FIG. **24**G, to secure the build platform **2404** to the carrier **2402**, the spring member **2430** can be rotated in a second, opposite direction (e.g., by pushing the arm **2432** downward toward the carrier **2402**) to place the rotatable clip **2408** in a closed configuration. In the closed configuration, the engagement portions **2428** of the rotatable clip **2408** can directly contact the projections **2416** at the second side **2410***b* of the build platform **2404**, thereby coupling the second side **2410***b* of the build platform **2404** to the carrier **2402**. Optionally, the arm **2432** can be temporarily secured in the downward position by a lock member **2436**, which can be a block, plate, post, or other overhanging element that engages the arm **2432** to keep the rotatable clips **2408** closed. However, due to the elastic properties of the spring member **2430**, the spring member **2430** can still exhibit some degree of deflection and/or deformation even when the rotatable clips **2408** are closed, which can be beneficial for accommodating changes in the dimensions of the build platforms **2404** (e.g., due to thermal expansion/shrinkage effects).
[0282] Although FIGS. **24**A-**24**G illustrate an embodiment in which the first side **2410***a* of the build platforms **2404** are coupled to fixed clips **2406** and the second side **2410***b* of the build platforms **2404** are coupled to rotatable clips **2208**, other configurations can also be used, e.g., the fixed clips **2406** can be used for both the first side **2410***a* and the second side **2410***b* of the build platforms **2404**, the rotatable clips **2208** can be used for both the first side **2410***a* and the second side **2410***b* of the build platforms **2404**, or the fixed clips **2406** and/or the rotatable clips **2408** can each independently be substituted with any of the other coupling devices described herein. Moreover, although the remaining sides of the build platforms **2404** (e.g., corresponding to the edges of the build platforms **2404** that are orthogonal to direction X) are depicted as not being coupled to the carrier **2402** by any coupling devices, in other embodiments, one or both of the

remaining sides can be coupled to the carrier **2402** via fixed clips **2406**, rotatable clips **2408**, and/or any of the other coupling devices described herein.

[0283] FIG. **25** is a side cross-sectional view of a modular build substrate **2500** including a carrier **2502**, a build platform **2504**, and an attachment mechanism including a plurality of magnets **2506***a*-**2506***d*, in accordance with embodiments of the present technology. The carrier **2502** and build platform **2504** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-12**). For instance, the carrier **2502** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platform **2504** during an additive manufacturing process. The build platform **2504** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0284] The build platform **2504** can be coupled to the carrier **2502** via the plurality of magnets **2506***a*-**2506***b* (collectively, "magnets **2506**"). In some embodiments, the carrier **2502** includes a first set of magnets (e.g., magnets **2506***a* and **2506***b*), and the build platform **2504** can include a second set of magnets (e.g., magnets **2506***c* and **2506***d*). The magnets **2506** can be situated such that the first set of magnets and the second set of magnets are attracted to one another, causing a coupling force between the carrier **2502** and the build platform **2504** to secure the build platform **2504** to the carrier **2502**. For instance, the first set of magnets of the carrier **2502** can have a first polarity (e.g., the south/positive pole is oriented upward toward the build platform **2504**, and the second set of magnets of the build platform **2504** can have a second, opposite polarity (e.g., the north/negative pole is oriented downward toward the carrier **2502**). Each magnet in the carrier **2502** can be paired with a corresponding magnet in the build platform **2504** with the opposite polarity, thereby generating an attractive force between the magnets that pulls the carrier **2502** and build platform **2504** toward each other.

[0285] The magnets **2506** can be electromagnets or permanent magnets (e.g., made of a magnetic material). The magnets **2506** can emit a suitable magnetic field strength, e.g., at least 1 mT, 10 mT, or 100 mT. In some embodiments, multiple magnets may be employed (e.g., an assembly of magnetic pairs) to increase the overall attractive forces between the build platform **2504** and the carrier **2502**. As depicted, the build platform **2504** can optionally include a plurality of protrusions **2508** defining a gap **2510** between the carrier **2502** and the build platform **2504**, and the gap **2510** can be sufficiently small to maintain sufficient attractive forces between the magnets **2506** to secure the build platform **2504** to the carrier **2502**.

[0286] The modular build substrate **2500** can further include registration features to align the build platform **2504** to the carrier **2502** in a predetermined position and/or orientation. In some embodiments, the carrier **2502** includes a first registration feature **2512**, and the build platform **2504** includes a second registration feature **2514** that mates with the first registration feature **2512** to affix the position and/or orientation of the build platform **2504** relative to the carrier **2502**. The first registration feature **2512** can be generally similar to the first registration feature **1722** of FIG. **17**, and the second registration feature **2514** can be generally similar to the second registration feature **1724** of FIG. **17**. For example, the first registration feature **2512** can be a protrusion extending from the carrier **2502** (e.g., a post, pin, bump), and the second registration feature **2514** can be a recess in the build platform **2504** (e.g., a hole, channel, aperture) that receives the first registration feature **2512**. In other embodiments, the first registration feature **2512** can be a recess in the build platform **2504** and the second registration feature **2514** can be a protrusion in the carrier **2502**.

[0287] In some embodiments, the configuration of FIG. **25** provides various advantages, such as providing a larger build area on the build platform **2504**, providing easy to clean surfaces of the build platform **2504** and the carrier **2502**, and/or case of manufacture. In some embodiments, the magnets **2506** can serve as flexible coupling devices to accommodate thermal expansion/shrinkage effects, e.g., caused by heating of one or more of the carrier **2502**, build platform **2504**, or objects

carried thereon as described elsewhere herein.
[0288] FIG. **26** is a side cross-sectional view of a modular build substrate **2600** including a carrier **2602**, a build platform **2604**, and an attachment mechanism including a spring clip **2606** and a plurality of magnets **2608***a* and **2608***b*, in accordance with embodiments of the present technology. The carrier **2602** and build platform **2604** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-12**). For instance, the carrier **2602** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platform **2604** during an additive manufacturing process. The build platform **2604** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.
[0289] A first side **2604***a* of the build platform **2604** can be coupled to the carrier **2602** via the spring clip **2606**. The spring clip **2606** can serve as a flexible coupling device for securing the first side **2604***a* of the build platform **2604** to the carrier **2602**. The spring clip **2606** can be generally similar to the internal spring clip **1606** of FIGS. **16**A-**16**C and/or the internal spring clip **1806** of FIG. **18**. In the illustrated embodiment, for example, the spring clip **2606** is configured to engage an internal portion **2610** of the build platform **2604**. The internal portion **2610** can be at or proximate to a lower surface of the build platform **2604**, and can define a recess **2612** that is configured to receive the spring clip **2606** therein when the build platform **2604** is placed on the carrier **2602**.
[0290] In some embodiments, the spring clip **2606** includes a base portion **2614** and an engagement portion **2616**. In some embodiments, the base portion **2614** and the engagement portion **2616** are integrally formed with each other and are each made of a spring material that is deformable and/or deflectable. The base portion **2614** can be coupled to the carrier **2602**, and the engagement portion **2616** can be configured to engage an internal projection **2618** of the internal portion **2610** of the build platform **2604**. The internal projection **2618** can be generally similar to the internal projection **1620** of FIGS. **16**A-**16**C and/or the internal projection **1826** of FIG. **18**. For instance, the internal projection **2618** can be a lip, flange, shoulder, or other suitable member for coupling to the spring clip **2606**. The engagement portion **2616** of the spring clip **2606** can include a lower surface **2620** that directly contacts an upper surface **2622** of the internal projection **2618**, thereby constraining lateral and vertical movement of the build platform **2604** relative to the carrier **2602**. The upper surface **2622** of the internal projection **2618** can be angled toward the carrier **2602** to pull the internal projection **2618**, and thus the build platform **2604**, downward toward the carrier **2602**. In other embodiments, however, the upper surface **2622** can instead be horizontal.
[0291] A second side **2604***b* of the build platform **2604** can be coupled to the carrier **2602** via the plurality of magnets **2608**. In some embodiments, the carrier **2602** includes a first magnet **2608***a*, and the build platform includes a second magnet **2608***b*. The magnets **2608** can be situated such that the first magnet **2608***a* and the second magnet **2608***b* are attracted to one another, causing a coupling force between the carrier **2602** and the build platform **2604** to secure the second side **2604***b* of the build platform **2604** to the carrier **2602**. For instance, the first magnet **2608***a* of the carrier **2502** can have a first polarity (e.g., the south/positive pole is oriented upward toward the build platform **2604**, and the second magnet **2608***b* of the build platform **2604** can have a second, opposite polarity (e.g., the north/negative pole is oriented downward toward the carrier **2602**).
[0292] In some embodiments, the second magnet **2608***b* is laterally offset from the first magnet **2608***a*. For example, the second magnet **2608***b* may be laterally closer to the spring clip **2606** than the first magnet **2608***a*. In some embodiments, this lateral offset imparts a magnetic force that biases the build platform **2604** to the left to bring the second magnet **2608***b* into closer alignment with the first magnet **2608***a*. The biasing of the build platform **2604**, and thereby the internal projection **2618**, can enhance the engagement of the spring clip **2606** with the internal projection **2618**. For example, the internal projection **2618** can be pushed leftward against the spring clip **2606**, thereby increasing the spring force exerted onto the internal projection **2618** by the spring

clip **2606**. The effect of this biasing can be controlled by varying the strength of the first magnet **2608***a*, the second magnet **2608***b*, or both.

[0293] Alternatively, or in combination, the carrier **2602** can include an external spring clip (e.g., similar to the external spring clip **1506** of FIG. **15**, the external spring clip **1706** of FIG. **17**, and/or the external spring clip **1808** of FIG. **18**) that is configured to engage an external portion of the second side **2604***b* of the build platform **2604** (e.g., an external projection similar to the external projection **1516** of FIG. **15**, the external projection **1716** of FIG. **17**, and/or the external projection **1828** of FIG. **18**). In such embodiments, the lateral offset between the first magnet **2606***a* and the second magnet **2606***b* may be reversed (e.g., the first magnet **2606***a* may be closer to the external spring clip than the second magnet **2606***b*). The resultant magnetic force can bias the build platform **2604** to the right, which may further enhance the engagement between the external projection and the external spring clip. For instance, the external projection can be pushed rightward against the external spring clip, thereby increasing the spring force exerted onto the external projection by the external spring clip.

[0294] FIG. **27** is a side cross-sectional view of a portion of a modular build substrate **2700** including a carrier **2702**, a build platform **2704**, and an attachment mechanism including a bolt **2706**, in accordance with embodiments of the present technology. The carrier **2702** and build platform **2704** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-12**). For instance, the carrier **2702** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platform **2704** during an additive manufacturing process. The build platform **2704** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0295] One or both sides of the build platform **2704** can be secured to the carrier **2702** via a bolt **2706**. In the illustrated embodiment, the bolt **2706** is not coupled directly to the build platform **2704**, but is instead coupled to a bar **2708** (e.g., a strip of sheet metal) which engages the build platform **2704**. For instance, the bar **2708** can be configured to push down against a projection **2710** extending from a lateral surface of the build platform **2704** to couple the build platform **2704** to the carrier **2702**. This configuration can secure the build platform **2704** without reducing the build area of the build platform **2704**.

[0296] FIG. **28** is a side cross-sectional view of a portion of a modular build substrate **2800** including a carrier **2802**, a build platform **2804**, and an attachment mechanism including a bolt **2806**, in accordance with embodiments of the present technology. The carrier **2802** and build platform **2804** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-12**). For instance, the carrier **2802** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platform **2804** during an additive manufacturing process. The build platform **2804** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0297] As shown in FIG. **28**, the bolt **2806** can extend through the build platform **2804** and the carrier **2802** to directly couple the build platform **2804** to the carrier **2802**. Although the bolt **2806** is depicted as being located at the central portion of the build platform **2804**, in other embodiments, the bolt **2806** can alternatively or additionally be located near the edges of the build platform **2804**.

[0298] FIG. **29** is a side cross-sectional view of a modular build substrate **2900** including a carrier **2902**, a build platform **2904**, and registration features, in accordance with embodiments of the present technology. The carrier **2902** and build platform **2904** can be generally similar to the other embodiments described herein (e.g., in connection with FIGS. **4-12**). For instance, the carrier **2902** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting the build platform **2904** during an additive manufacturing process. The build platform **2904** can be a tray, plate, film, sheet, or other generally planar substrate suitable for coupling to and supporting at least one additively manufactured object thereon.

[0299] The modular build substrate **2900** can include one or more registration features to align the build platform **2904** to the carrier **2902** in a predetermined position and/or orientation. In some embodiments, the carrier **2902** includes one or more first registration features **2906***a*, **2906***b*, and the build platform **2904** includes one or more second registration features **2908***a*, **2908***b* that each mate with the corresponding first registration feature to affix the position and/or orientation of the build platform **2904** relative to the carrier **2902**. For example, the first registration features **2906***a*, **2906***b* can each be a protrusion extending from the carrier **2902** (e.g., a post, pin, bump), and the second registration features **2908***a*, **2908***b* can each be a recess in the build platform **2904** (e.g., a hole, channel, aperture) that receives the corresponding first registration feature. In some embodiments, the build platform **2904** is coupled to the carrier **2902** by aligning the registration features to each other, then sliding the build platform **2904** laterally onto the carrier **2902** (e.g., along the direction into the plane of the page).

[0300] The shape of the registration features may be varied as desired. For instance, in the illustrated embodiment, the first registration feature **2906***a* and the second registration feature **2908***a* each have a rectangular shape, whereas the first registration feature **2906***b* and the second registration feature **2908***b* each have a trapezoidal and/or dovetailed shape. Other shapes are also possible, such as a square shape, circular shape, oval shape, triangular shape, T-shape, notched shape, etc. Some or all of the registration features may have the same shape, or some or all of the registration features may have different shapes. Moreover, any suitable number of registration features may be used, such as one, two, three, four, five or more registration features. Optionally, a bayonet fitting mechanism can be incorporated into the carrier **2902** and build platform **2904** to further improve case of use.

[0301] FIG. **30** is a flow diagram illustrating a method **3000** for fabricating additively manufactured objects, in accordance with embodiments of the present technology. The method **3000** can be performed by any of the systems and devices described herein, such as any of the embodiments of FIGS. **1-29**. In some embodiments, some or all of the processes of the method **3000** are implemented as computer-readable instructions (e.g., program code) that are configured to be executed by one or more processors of a computing device, such as a controller of an additive manufacturing system and/or a post-processing system. The method **3000** can be combined with any of the other methods described herein, such as the method **100** of FIG. **1**.

[0302] The method **3000** can begin at block **3002** with coupling a plurality of build platforms to a carrier. The build platforms can be modular build platforms that are releasably coupled to the carrier to form a modular build substrate for an additive manufacturing process, as described herein in connection with FIGS. **4-29**. For example, the build platforms can be coupled to the carrier via a releasable attachment technique, such as via vacuum, mechanical fixation (e.g., interference fit, snap fit, interlocking features, fasteners, form-fitting inserts, clamps, springs, hinged features), electromagnetic fixation, magnetic fixation, form-fitting inserts, or suitable combinations thereof. When coupled to the carrier, the build platforms can collectively define a build plane for an additive manufacturing process. The build plane can have an area of at least 1000 cm.sup.2, 1500 cm.sup.2, 2000 cm.sup.2, 2500 cm.sup.2, 3000 cm.sup.2, 3500 cm.sup.2, 4000 cm.sup.2, 4500 cm.sup.2, or 5000 cm.sup.2. The build plane can have a maximum vertical deviation that is less than or equal to 500 μm, 400 μm, 300 μm, 200 μm, 100 μm, or 50 μm, and/or within a range from a range from 50 μm to 500 μm, or 500 μm to 1 mm.

[0303] At block **3004**, the method **3000** can include forming a plurality of objects on the plurality of build platforms using an additive manufacturing process. The objects can be dental appliances, such as aligners, retainers, brackets and wires, whitening trays, mouth guards, night guards, anti-bruxing or anti-grinding devices, tongue thrust devices, palatal expanders, sleep apnea devices, anti-snoring devices, attachment templates, mandibular advancement devices, prefabricated attachment templates, etc. The additive manufacturing process can use any of the additive manufacturing techniques and systems described herein. For instance, the additive manufacturing process can be a lithography-based additive manufacturing process in which the objects are fabricated from a curable material in a layer-by-layer manner. Each build platform can receive and support one or more of the objects during the additive manufacturing process. For instance, a single build platform can receive one, two, three, four, five, 10, 20, or more objects. During the additive manufacturing process, the plurality of build platforms can remain coupled to the carrier, such that the carrier acts as a fixed support for the build platforms and the objects thereon.

[0304] At block **3006**, the method **3000** can continue with removing the plurality of build platforms from the carrier. The removal can be performed after the additive manufacturing process is complete. In some embodiments, the build platforms are removed manually by a human operator, while in other embodiments, the build platforms are removed automatically (e.g., by a robotic assembly or other automated mechanism). The build platforms can be removed while the carrier remains within the additive manufacturing system, or the carrier can be removed from the additive manufacturing system before removing the build platforms from the carrier.

[0305] At block **3008**, the method **3000** can include performing post-processing of the plurality of objects. The post-processing can include any of the operations described herein, such as removing residual material from the objects, post-curing the objects, washing the object, trimming support structures from the objects, etc. The objects can remain attached to their respective build platforms during post-processing, such that the build platforms are used to support and/or manipulate the objects during post-processing. In some embodiments, the process of block **3008** involves placing the build platforms with the attached objects into one or more post-processing devices, such as centrifuges, solvent baths, post-curing ovens, furnaces, etc. The individual build platforms can be sufficiently small to fit within the post-curing device(s). For instance, each build platform can have an area no greater than 1000 cm.sup.2, 900 cm.sup.2, 800 cm.sup.2, 700 cm.sup.2, 600 cm.sup.2, 500 cm.sup.2, 400 cm.sup.2, 300 cm.sup.2, 200 cm.sup.2, or 100 cm.sup.2.

[0306] At block **2710**, the method **3000** can include separating the plurality of objects from the respective build platforms. The separation can be performed using physical techniques, such as scraping, peeling, fracturing sacrificial portions of the objects, etc. Optionally, in embodiments where the build platform includes an interface layer between the surface of the build platform and the objects (e.g., as described above in connection with FIG. **12**), the separation can be performed by disintegrating, destabilizing, or otherwise removing the interface layer via physical and/or chemical techniques (e.g., scraping, peeling, dissolving, melting, etching). The separated objects can be subjected to additional post-processing and/or prepared for packaging and shipment. In some embodiments, the build platforms are reassembled onto the carrier for reuse in a subsequent additive manufacturing operation.

[0307] The method **3000** illustrated in FIG. **30** can be modified in many different ways. For example, the ordering of the processes shown in FIG. **30** can be varied. Some of the processes of the method **3000** can be omitted and/or the method **3000** can include additional processes not shown in FIG. **30**. For instance, the method **3000** can further include coupling a prefabricated element to at least one build platform, such that the prefabricated element becomes part of the object(s) formed on that build platform (e.g., as described above in connection with FIGS. **11**A-**11**H).

[0308] FIG. **31** is a flow diagram illustrating a method **3100** for fabricating additively manufactured objects, in accordance with embodiments of the present technology. The method **3100** can be performed by any of the systems and devices described herein, such as any of the embodiments of FIGS. **1-29**. In some embodiments, some or all of the processes of the method **3100** are implemented as computer-readable instructions (e.g., program code) that are configured to be executed by one or more processors of a computing device, such as a controller of an additive manufacturing system and/or a post-processing system. The method **3100** can be combined with any of the other methods described herein, such as the method **100** of FIG. **1** and/or the method

3000 of FIG. **30**.

[0309] The method **3100** can begin at block **3102**, in which a 3D printer comprising a carrier and a plurality of build platforms releasably fixed on the carrier is provided. One or more of the build platforms can define a build plane for building at least one 3D object thereon. The 3D printer may include a light engine for selectively curing layers of a light-polymerizable resin on the build platforms.

[0310] At block **3104**, a prefabricated element can be placed and/or mounted onto the build platform or into a recess of the build platform, and at least one of the layers of light-polymerizable resin is bonded to the prefabricated element during the printing of the 3D object.

[0311] At block **3106**, a plurality of 3D objects are built with the 3D printer. At least one of said plurality of 3D objects can be built on each build platform.

[0312] At block **3108**, the build platforms can be removed with said at least one 3D object placed thereon from the 3D printer.

[0313] At block **3110**, the 3D objects, while being arranged on their respective build platform, can be subjected to at least one post-processing step after the build platforms have been separated from the carrier.

[0314] The method **3100** illustrated in FIG. **31** can be modified in many different ways. For example, the ordering of the processes shown in FIG. **31** can be varied. Some of the processes of the method **3100** can be omitted (e.g., the process of block **3104**) and/or the method **3100** can include additional processes not shown in FIG. **31**.

III. Dental Appliances and Associated Methods

[0315] FIG. **32**A illustrates a representative example of a tooth repositioning appliance **3200** configured in accordance with embodiments of the present technology. The appliance **3200** can be manufactured using any of the systems, methods, and devices described herein. The appliance **3200** (also referred to herein as an “aligner”) can be worn by a patient in order to achieve an incremental repositioning of individual teeth **3202** in the jaw. The appliance **3200** can include a shell (e.g., a continuous polymeric shell or a segmented shell) having teeth-receiving cavities that receive and resiliently reposition the teeth. The appliance **3200** or portion(s) thereof may be indirectly fabricated using a physical model of teeth. For example, an appliance (e.g., polymeric appliance) can be formed using a physical model of teeth and a sheet of suitable layers of polymeric material. In some embodiments, a physical appliance is directly fabricated, e.g., using additive manufacturing techniques, from a digital model of an appliance.

[0316] The appliance **3200** can fit over all teeth present in an upper or lower jaw, or less than all of the teeth. The appliance **3200** can be designed specifically to accommodate the teeth of the patient (e.g., the topography of the tooth-receiving cavities matches the topography of the patient's teeth), and may be fabricated based on positive or negative models of the patient's teeth generated by impression, scanning, and the like. Alternatively, the appliance **3200** can be a generic appliance configured to receive the teeth, but not necessarily shaped to match the topography of the patient's teeth. In some cases, only certain teeth received by the appliance **3200** are repositioned by the appliance **3200** while other teeth can provide a base or anchor region for holding the appliance **3200** in place as it applies force against the tooth or teeth targeted for repositioning. In some cases, some, most, or even all of the teeth can be repositioned at some point during treatment. Teeth that are moved can also serve as a base or anchor for holding the appliance as it is worn by the patient. In some embodiments, no wires or other means are provided for holding the appliance **3200** in place over the teeth. In some cases, however, it may be desirable or necessary to provide individual attachments **3204** or other anchoring elements on teeth **3202** with corresponding receptacles **3206** or apertures in the appliance **3200** so that the appliance **3200** can apply a selected force on the tooth. Representative examples of appliances, including those utilized in the Invisalign® System, are described in numerous patents and patent applications assigned to Align Technology, Inc. including, for example, in U.S. Pat. Nos. 6,450,807, and 5,975,893, as well as on the company's website, which is accessible on the World Wide Web (see, e.g., the url “invisalign.com”). Examples of tooth-mounted attachments suitable for use with orthodontic appliances are also described in patents and patent applications assigned to Align Technology, Inc., including, for example, U.S. Pat. Nos. 6,309,215 and 6,830,450.

[0317] FIG. **32**B illustrates a tooth repositioning system **3210** including a plurality of appliances **3212**, **3214**, **3216**, in accordance with embodiments of the present technology. Any of the appliances described herein can be designed and/or provided as part of a set of a plurality of appliances used in a tooth repositioning system. Each appliance may be configured so a tooth-receiving cavity has a geometry corresponding to an intermediate or final tooth arrangement intended for the appliance. The patient's teeth can be progressively repositioned from an initial tooth arrangement to a target tooth arrangement by placing a series of incremental position adjustment appliances over the patient's teeth. For example, the tooth repositioning system **3210** can include a first appliance **3212** corresponding to an initial tooth arrangement, one or more intermediate appliances **3214** corresponding to one or more intermediate arrangements, and a final appliance **3216** corresponding to a target arrangement. A target tooth arrangement can be a planned final tooth arrangement selected for the patient's teeth at the end of all planned orthodontic treatment. Alternatively, a target arrangement can be one of some intermediate arrangements for the patient's teeth during the course of orthodontic treatment, which may include various different treatment scenarios, including, but not limited to, instances where surgery is recommended, where interproximal reduction (IPR) is appropriate, where a progress check is scheduled, where anchor placement is best, where palatal expansion is desirable, where restorative dentistry is involved (e.g., inlays, onlays, crowns, bridges, implants, veneers, and the like), etc. As such, it is understood that a target tooth arrangement can be any planned resulting arrangement for the patient's teeth that follows one or more incremental repositioning stages. Likewise, an initial tooth arrangement can be any initial arrangement for the patient's teeth that is followed by one or more incremental repositioning stages.

[0318] FIG. **32**C illustrates a method **3220** of orthodontic treatment using a plurality of appliances, in accordance with embodiments of the present technology. The method **3220** can be practiced using any of the appliances or appliance sets described herein. In block **3222**, a first orthodontic appliance is applied to a patient's teeth in order to reposition the teeth from a first tooth arrangement to a second tooth arrangement. In block **3224**, a second orthodontic appliance is applied to the patient's teeth in order to reposition the teeth from the second tooth arrangement to a third tooth arrangement. The method **3220** can be repeated as necessary using any suitable number and combination of sequential appliances in order to incrementally reposition the patient's teeth from an initial arrangement to a target arrangement. The appliances can be generated all at the same stage or in sets or batches (e.g., at the beginning of a stage of the treatment), or the appliances can be fabricated one at a time, and the patient can wear each appliance until the pressure of each appliance on the teeth can no longer be felt or until the maximum amount of expressed tooth movement for that given stage has been achieved. A plurality of different appliances (e.g., a set) can be designed and even fabricated prior to the patient wearing any appliance of the plurality. After wearing an appliance for an appropriate period of time, the patient can replace the current appliance with the next appliance in the series until no more appliances remain. The appliances are generally not affixed to the teeth and the patient may place and replace the appliances at any time during the procedure (e.g., patient-removable appliances). The final appliance or several appliances in the series may have a geometry or geometries selected to overcorrect the tooth arrangement. For instance, one or more appliances may have a geometry that would (if fully achieved) move individual teeth beyond the tooth arrangement that has been selected as the “final.” Such over-correction may be desirable in order to offset potential relapse after the repositioning method has been terminated (e.g., permit movement of individual teeth back toward their pre-corrected positions). Over-correction may also be beneficial to speed the rate of correction (e.g., an appliance with a geometry that is positioned beyond a desired intermediate or final position may shift the individual teeth toward the position at a greater rate). In such cases, the use of an appliance can be terminated before the teeth reach the positions defined by the appliance. Furthermore, over-correction may be deliberately applied in order to compensate for any inaccuracies or limitations of the appliance.

[0319] FIG. **33** illustrates a method **3300** for designing an orthodontic appliance, in accordance with embodiments of the present technology. The method **3300** can be applied to any embodiment of the orthodontic appliances described herein. Some or all of the steps of the method **3300** can be performed by any suitable data processing system or device, e.g., one or more processors configured with suitable instructions.

[0320] In block **3302**, a movement path to move one or more teeth from an initial arrangement to a target arrangement is determined. The initial arrangement can be determined from a mold or a scan of the patient's teeth or mouth tissue, e.g., using wax bites, direct contact scanning, x-ray imaging, tomographic imaging, sonographic imaging, and other techniques for obtaining information about the position and structure of the teeth, jaws, gums and other orthodontically relevant tissue. From the obtained data, a digital data set can be derived that represents the initial (e.g., pretreatment) arrangement of the patient's teeth and other tissues. Optionally, the initial digital data set is processed to segment the tissue constituents from each other. For example, data structures that digitally represent individual tooth crowns can be produced. Advantageously, digital models of entire teeth can be produced, including measured or extrapolated hidden surfaces and root structures, as well as surrounding bone and soft tissue.

[0321] The target arrangement of the teeth (e.g., a desired and intended end result of orthodontic treatment) can be received from a clinician in the form of a prescription, can be calculated from basic orthodontic principles, and/or can be extrapolated computationally from a clinical prescription. With a specification of the desired final positions of the teeth and a digital representation of the teeth themselves, the final position and surface geometry of each tooth can be specified to form a complete model of the tooth arrangement at the desired end of treatment.

[0322] Having both an initial position and a target position for each tooth, a movement path can be defined for the motion of each tooth. In some embodiments, the movement paths are configured to move the teeth in the quickest fashion with the least amount of round-tripping to bring the teeth from their initial positions to their desired target positions. The tooth paths can optionally be segmented, and the segments can be calculated so that each tooth's motion within a segment stays within threshold limits of linear and rotational translation. In this way, the end points of each path segment can constitute a clinically viable repositioning, and the aggregate of segment end points can constitute a clinically viable sequence of tooth positions, so that moving from one point to the next in the sequence does not result in a collision of teeth.

[0323] In block **3304**, a force system to produce movement of the one or more teeth along the movement path is determined. A force system can include one or more forces and/or one or more torques. Different force systems can result in different types of tooth movement, such as tipping, translation, rotation, extrusion, intrusion, root movement, etc. Biomechanical principles, modeling techniques, force calculation/measurement techniques, and the like, including knowledge and approaches commonly used in orthodontia, may be used to determine the appropriate force system to be applied to the tooth to accomplish the tooth movement. In determining the force system to be applied, sources may be considered including literature, force systems determined by experimentation or virtual modeling, computer-based modeling, clinical experience, minimization of unwanted forces, etc.

[0324] Determination of the force system can be performed in a variety of ways. For example, in some embodiments, the force system is determined on a patient-by-patient basis, e.g., using patient-specific data. Alternatively or in combination, the force system can be determined based on a generalized model of tooth movement (e.g., based on experimentation, modeling, clinical data, etc.), such that patient-specific data is not necessarily used. In some embodiments, determination of a force system involves calculating specific force values to be applied to one or more teeth to produce a particular movement. Alternatively, determination of a force system can be performed at a high level without calculating specific force values for the teeth. For instance, block **3304** can involve determining a particular type of force to be applied (e.g., extrusive force, intrusive force, translational force, rotational force, tipping force, torquing force, etc.) without calculating the specific magnitude and/or direction of the force.

[0325] The determination of the force system can include constraints on the allowable forces, such as allowable directions and magnitudes, as well as desired motions to be brought about by the applied forces. For example, in fabricating palatal expanders, different movement strategies may be desired for different patients. For example, the amount of force needed to separate the palate can depend on the age of the patient, as very young patients may not have a fully-formed suture. Thus, in juvenile patients and others without fully-closed palatal sutures, palatal expansion can be accomplished with lower force magnitudes. Slower palatal movement can also aid in growing bone to fill the expanding suture. For other patients, a more rapid expansion may be desired, which can be achieved by applying larger forces. These requirements can be incorporated as needed to choose the structure and materials of appliances; for example, by choosing palatal expanders capable of applying large forces for rupturing the palatal suture and/or causing rapid expansion of the palate. Subsequent appliance stages can be designed to apply different amounts of force, such as first applying a large force to break the suture, and then applying smaller forces to keep the suture separated or gradually expand the palate and/or arch.

[0326] The determination of the force system can also include modeling of the facial structure of the patient, such as the skeletal structure of the jaw and palate. Scan data of the palate and arch, such as X-ray data or 3D optical scanning data, for example, can be used to determine parameters of the skeletal and muscular system of the patient's mouth, so as to determine forces sufficient to provide a desired expansion of the palate and/or arch. In some embodiments, the thickness and/or density of the mid-palatal suture may be measured, or input by a treating professional. In other embodiments, the treating professional can select an appropriate treatment based on physiological characteristics of the patient. For example, the properties of the palate may also be estimated based on factors such as the patient's age—for example, young juvenile patients can require lower forces to expand the suture than older patients, as the suture has not yet fully formed.

[0327] In block **3306**, a design for an orthodontic appliance configured to produce the force system is determined. The design can include the appliance geometry, material composition and/or material properties, and can be determined in various ways, such as using a treatment or force application simulation environment. A simulation environment can include, e.g., computer modeling systems, biomechanical systems or apparatus, and the like. Optionally, digital models of the appliance and/or teeth can be produced, such as finite element models. The finite element models can be created using computer program application software available from a variety of vendors. For creating solid geometry models, computer aided engineering (CAE) or computer aided design (CAD) programs can be used, such as the AutoCAD® software products available from Autodesk, Inc., of San Rafael, CA. For creating finite element models and analyzing them, program products from a number of vendors can be used, including finite element analysis packages from ANSYS, Inc., of Canonsburg, PA, and SIMULIA (Abaqus) software products from Dassault Systèmes of Waltham, MA.

[0328] Optionally, one or more designs can be selected for testing or force modeling. As noted above, a desired tooth movement, as well as a force system required or desired for eliciting the desired tooth movement, can be identified. Using the simulation environment, a candidate design can be analyzed or modeled for determination of an actual force system resulting from use of the candidate appliance. One or more modifications can optionally be made to a candidate appliance, and force modeling can be further analyzed as described, e.g., in order to iteratively determine an appliance design that produces the desired force system.

[0329] In block **3308**, instructions for fabrication of the orthodontic appliance incorporating the design are generated. The instructions can be configured to control a fabrication system or device in order to produce the orthodontic appliance with the specified design. In some embodiments, the instructions are configured for manufacturing the orthodontic appliance using direct fabrication (e.g., stereolithography, selective laser sintering, fused deposition modeling, 3D printing, continuous direct fabrication, multi-material direct fabrication, etc.), in accordance with the various methods presented herein. In alternative embodiments, the instructions can be configured for indirect fabrication of the appliance, e.g., by thermoforming.

[0330] Although the above steps show a method **3300** of designing an orthodontic appliance in accordance with some embodiments, a person of ordinary skill in the art will recognize some variations based on the teaching described herein. Some of the steps may comprise sub-steps. Some of the steps may be repeated as often as desired. One or more steps of the method **3300** may be performed with any suitable fabrication system or device, such as the embodiments described herein. Some of the steps may be optional, e.g., the process of block **3304** can be omitted, such that the orthodontic appliance is designed based on the desired tooth movements and/or determined tooth movement path, rather than based on a force system. Moreover, the order of the steps can be varied as desired.

[0331] FIG. **34** illustrates a method **3400** for digitally planning an orthodontic treatment and/or design or fabrication of an appliance, in accordance with embodiments. The method **3400** can be applied to any of the treatment procedures described herein and can be performed by any suitable data processing system.

[0332] In block **3402**, a digital representation of a patient's teeth is received. The digital representation can include surface topography data for the patient's intraoral cavity (including teeth, gingival tissues, etc.). The surface topography data can be generated by directly scanning the intraoral cavity, a physical model (positive or negative) of the intraoral cavity, or an impression of the intraoral cavity, using a suitable scanning device (e.g., a handheld scanner, desktop scanner, etc.).

[0333] In block **3404**, one or more treatment stages are generated based on the digital representation of the teeth. The treatment stages can be incremental repositioning stages of an orthodontic treatment procedure designed to move one or more of the patient's teeth from an initial tooth arrangement to a target arrangement. For example, the treatment stages can be generated by determining the initial tooth arrangement indicated by the digital representation, determining a target tooth arrangement, and determining movement paths of one or more teeth in the initial arrangement necessary to achieve the target tooth arrangement. The movement path can be optimized based on minimizing the total distance moved, preventing collisions between teeth, avoiding tooth movements that are more difficult to achieve, or any other suitable criteria.

[0334] In block **3406**, at least one orthodontic appliance is fabricated based on the generated treatment stages. For example, a set of appliances can be fabricated, each shaped according to a tooth arrangement specified by one of the treatment stages, such that the appliances can be sequentially worn by the patient to incrementally reposition the teeth from the initial arrangement to the target arrangement. The appliance set may include one or more of the orthodontic appliances described herein. The fabrication of the appliance may involve creating a digital model of the appliance to be used as input to a computer-controlled fabrication system. The appliance can be formed using direct fabrication methods, indirect fabrication methods, or combinations thereof, as desired.

[0335] In some instances, staging of various arrangements or treatment stages may not be necessary for design and/or fabrication of an appliance. As illustrated by the dashed line in FIG. **34**, design and/or fabrication of an orthodontic appliance, and perhaps a particular orthodontic treatment, may include use of a representation of the patient's teeth (e.g., including receiving a digital representation of the patient's teeth (block **3402**)), followed by design and/or fabrication of an orthodontic appliance based on a representation of the patient's teeth in the arrangement represented by the received representation.

[0336] As noted herein, the techniques described herein can be used for the direct fabrication of dental appliances, such as aligners and/or a series of aligners with tooth-receiving cavities configured to move a person's teeth from an initial arrangement toward a target arrangement in accordance with a treatment plan. Aligners can include mandibular repositioning elements, such as those described in U.S. Pat. No. 10,912,629, entitled "Dental Appliances with Repositioning Jaw Elements," filed Nov. 30, 2015; U.S. Pat. No. 10,537,406, entitled "Dental Appliances with Repositioning Jaw Elements," filed Sep. 19, 2014; and U.S. Pat. No. 9,844,424, entitled "Dental Appliances with Repositioning Jaw Elements," filed Feb. 21, 2014; all of which are incorporated by reference herein in their entirety.

[0337] The techniques used herein can also be used to manufacture attachment placement devices, e.g., appliances used to position prefabricated attachments on a person's teeth in accordance with one or more aspects of a treatment plan. Examples of attachment placement devices (also known as "attachment placement templates" or "attachment fabrication templates") can be found at least in: U.S. application Ser. No. 17/249,218, entitled "Flexible 3D Printed Orthodontic Device," filed Feb. 24, 2021; U.S. application Ser. No. 16/366,686, entitled "Dental Attachment Placement Structure," filed Mar. 27, 2019; U.S. application Ser. No. 15/674,662, entitled "Devices and Systems for Creation of Attachments," filed Aug. 11, 2017; U.S. Pat. No. 11,103,330, entitled "Dental Attachment Placement Structure," filed Jun. 14, 2017; U.S. application Ser. No. 14/963,527, entitled "Dental Attachment Placement Structure," filed Dec. 9, 2015; U.S. application Ser. No. 14/939,246, entitled "Dental Attachment Placement Structure," filed Nov. 12, 2015; U.S. application Ser. No. 14/939,252, entitled "Dental Attachment Formation Structures," filed Nov. 12, 2015; and U.S. Pat. No. 9,700,385, entitled "Attachment Structure," filed Aug. 22, 2014; all of which are incorporated by reference herein in their entirety.

[0338] The techniques described herein can be used to make incremental palatal expanders and/or a series of incremental palatal expanders used to expand a person's palate from an initial position toward a target position in accordance with one or more aspects of a treatment plan. Examples of incremental palatal expanders can be found at least in: U.S. application Ser. No. 16/380,801, entitled "Releasable Palatal Expanders," filed Apr. 10, 2019; U.S. application Ser. No. 16/022,552, entitled "Devices, Systems, and Methods for Dental Arch Expansion," filed Jun. 28, 2018; U.S. Pat. No. 11,045,283, entitled "Palatal Expander with Skeletal Anchorage Devices," filed Jun. 8, 2018; U.S. application Ser. No. 15/831,159, entitled "Palatal Expanders and Methods of Expanding a Palate," filed Dec. 4, 2017; U.S. Pat. No. 10,993,783, entitled "Methods and Apparatuses for Customizing a Rapid Palatal Expander," filed Dec. 4, 2017; and U.S. Pat. No. 7,192,273, entitled "System and Method for Palatal Expansion," filed Aug. 7, 2003; all of which are incorporated by reference herein in their entirety.

EXAMPLES

[0339] The following examples are included to further describe some aspects of the present technology, and should not be used to limit the scope of the technology.

[0340] Example 1. An assembly for supporting 3D objects during an additive manufacturing process, the assembly comprising: [0341] a plurality of build platforms, each build platform configured to support one or more 3D objects during the additive manufacturing process; [0342] a carrier configured to support the plurality of build platforms; and [0343] an attachment mechanism configured to releasably couple the plurality of build platforms to the carrier during the additive manufacturing process such that the plurality of build platforms collectively form a build plane having a vertical deviation no greater than 500 μm.

[0344] Example 2. The assembly of Example 1, wherein each build platform comprises: [0345] an upper surface configured to support the one or more 3D objects, [0346] a lower surface opposite

the upper surface, and [0347] one or more protrusions formed on the lower surface such that when the build platform is releasably coupled to the carrier, a gap is formed between the lower surface of the build platform and an upper surface of the carrier.
[0348] Example 3. The assembly of Example 2, wherein the one or more protrusions each have a height within a range from 0.1 mm to 1 mm.
[0349] Example 4. The assembly of any one of Examples 1 to 3, wherein the attachment mechanism comprises, for each build platform: [0350] a first coupling device at or proximate to a first side of the build platform, and [0351] a second coupling device at or proximate to a second side of the build platform.
[0352] Example 5. The assembly of Example 4, wherein the first coupling device uses a different type of coupling than the second coupling device.
[0353] Example 6. The assembly of Example 5, wherein the first coupling device is a flexible coupling device and the second coupling device is a rigid coupling device.
[0354] Example 7. The assembly of Example 5 or 6, wherein the first coupling device comprises a spring material and the second coupling device does not comprise a spring material.
[0355] Example 8. The assembly of Example 4, wherein the first coupling device uses the same type of coupling as the second coupling device.
[0356] Example 9. The assembly of any one of Examples 1 to 8, wherein the attachment mechanism comprises a spring clip configured to releasably couple to a build platform of the plurality of build platforms.
[0357] Example 10. The assembly of Example 9, wherein the spring clip is configured to releasably couple to an external portion of the build platform.
[0358] Example 11. The assembly of Example 9, wherein the spring clip is configured to releasably couple to an internal portion of the build platform.
[0359] Example 12. The assembly of any one of Examples 9 to 11, wherein the spring clip is configured to releasably couple to a recess formed in the build platform.
[0360] Example 13. The assembly of Example 12, wherein the recess is formed in a lower surface of the build platform.
[0361] Example 14. The assembly of any one of Examples 1 to 13, wherein the attachment mechanism comprises a rotatable clip configured to releasably couple to a build platform of the plurality of build platforms.
[0362] Example 15. The assembly of Example 14, wherein the rotatable clip is rotatable between a closed configuration and an open configuration.
[0363] Example 16. The assembly of Example 15, wherein: [0364] when in the closed configuration, the rotatable clip contacts the build platform, and [0365] when in the open configuration, the rotatable clip is spaced apart from the build platform.
[0366] Example 17. The assembly of any one of Examples 1 to 16, wherein the attachment mechanism comprises a first magnet configured to releasably couple to a second magnet of a build platform of the plurality of build platforms.
[0367] Example 18. The assembly of any one of Examples 1 to 17, wherein the carrier comprises a first registration feature, a build platform of the plurality of build platforms comprises a second registration feature, and the first registration feature is configured to engage the second registration feature to align the build platform on the carrier in a fixed position and orientation.
[0368] Example 19. The assembly of any one of Examples 1 to 18, wherein the build plane is a single continuous build plane.
[0369] Example 20. The assembly of any one of Examples 1 to 18, wherein the build plane comprises a plurality of discrete regions.
[0370] Example 21. The assembly of any one of Examples 1 to 20, wherein the build plane has a total surface area of at least 1000 cm.sup.2.
[0371] Example 22. The assembly of any one of Examples 1 to 21, wherein the additive

manufacturing process comprises building up the one or more 3D objects of each build platform from a plurality of layers of a curable material.
[0372] Example 23. The assembly of any one of Examples 1 to 22, wherein the carrier includes or is thermally coupled to a heat source.
[0373] Example 24. The assembly of Example 23, wherein the heat source is configured to heat the build plane to a temperature within a range from 30° C. to 200° C.
[0374] Example 25. The assembly of any one of Examples 1 to 24, wherein the one or more 3D objects comprise one or more dental appliances.
[0375] Example 26. A system for manufacturing 3D objects, the system comprising: [0376] the assembly of any one of Examples 1 to 25; and [0377] a printer assembly configured to receive the assembly, wherein the printer assembly comprises: [0378] a source of a curable material, and [0379] an energy source configured to output energy toward the curable material to form the one or more 3D objects on each build platform of the plurality of build platforms of the assembly according to an additive manufacturing process.
[0380] Example 27. The system of Example 26, further comprising a stationary base, wherein the carrier of the assembly is releasably coupled to the stationary base.
[0381] Example 28. The system of Example 26 or 27, wherein the curable material comprises a polymerizable resin.
[0382] Example 29. The system of any one of Examples 26 to 28, wherein the printer assembly comprises a carrier film configured to convey a layer of the curable material toward the assembly.
[0383] Example 30. The system of Example 29, wherein the printer assembly is configured to move relative to the assembly while the energy source outputs energy toward the layer of curable material.
[0384] Example 31. The system of any one of Examples 26 to 28, wherein the source of the curable material comprises a reservoir of the curable material, and the assembly is positioned within the reservoir.
[0385] Example 32. The system of any one of Examples 26 to 31, further comprising at least one post-processing device configured to perform at least one post-processing operation on the one or more 3D objects of least one build platform while the one or more 3D objects remain on the at least one build platform and while the at least one build platform is separated from the carrier.
[0386] Example 33. The system of Example 32, wherein the at least one post-processing device comprises one or more of a centrifuge, a solvent bath, or a post-curing oven.
[0387] Example 34. A method comprising: [0388] coupling a plurality of build platforms to a carrier to form a build plane having a vertical deviation no greater than 500 μm; [0389] forming a plurality of 3D objects on the plurality of build platforms using an additive manufacturing process, wherein each build platform receives one or more 3D objects thereon; and [0390] removing the plurality of build platforms from the carrier after the additive manufacturing process.
[0391] Example 35. The method of Example 34, wherein each build platform comprises: [0392] an upper surface configured to support the one or more 3D objects, [0393] a lower surface opposite the upper surface, and [0394] one or more protrusions formed on the lower surface such that when the build platform is coupled to the carrier, a gap is formed between the lower surface of the build platform and an upper surface of the carrier.
[0395] Example 36. The method of Example 35, wherein the one or more protrusions each have a height within a range from 0.1 mm to 1 mm.
[0396] Example 37. The method of any one of Examples 34 to 36, wherein each build platform is coupled to the carrier by: [0397] coupling a first side of the build platform to the carrier via a first coupling device, and [0398] coupling a second side of the build platform to the carrier via a second coupling device.
[0399] Example 38. The method of Example 37, wherein the first coupling device uses a different type of coupling than the second coupling device.

[0400] Example 39. The method of Example 38, wherein the first coupling device is a flexible coupling device and the second coupling device is a rigid coupling device.
[0401] Example 40. The method of Example 38 or 39, wherein the first coupling device comprises a spring material and the second coupling device does not comprise a spring material.
[0402] Example 41. The method of Example 37, wherein the first coupling device uses the same type of coupling as the second coupling device.
[0403] Example 42. The method of any one of Examples 34 to 41, wherein coupling the plurality of build platforms to the carrier comprises engaging a build platform of the plurality of build platforms with a spring clip.
[0404] Example 43. The method of Example 42, wherein the spring clip engages an external portion of the build platform.
[0405] Example 44. The method of Example 42, wherein the spring clip engages an internal portion of the build platform.
[0406] Example 45. The method of any one of Examples 42 to 44, wherein the spring clip engages a recess formed in the build platform.
[0407] Example 46. The method of Example 45, wherein the recess is formed in a lower surface of the build platform.
[0408] Example 47. The method of any one of Examples 34 to 46, wherein coupling the plurality of build platforms to the carrier comprises engaging a build platform of the plurality of build platforms with a rotatable clip.
[0409] Example 48. The method of Example 47, wherein the rotatable clip is rotatable between a closed configuration and an open configuration.
[0410] Example 49. The method of Example 48, wherein: [0411] when in the closed configuration, the rotatable clip engages the build platform, and [0412] when in the open configuration, the rotatable clip is disengaged from the build platform.
[0413] Example 50. The method of any one of Examples 34 to 49, wherein coupling the plurality of build platforms to the carrier comprises engaging a first magnet of the carrier with a second magnet of a build platform of the plurality of build platforms.
[0414] Example 51. The method of any one of Examples 34 to 50, wherein coupling the plurality of build platforms to the carrier comprises aligning a build platform of the plurality of build platforms to the carrier using a first registration feature on the build platform and a second registration feature on the carrier.
[0415] Example 52. The method of any one of Examples 34 to 51, wherein the build plane is a single continuous build plane.
[0416] Example 53. The method of any one of Examples 34 to 51, wherein the build plane comprises a plurality of discrete regions.
[0417] Example 54. The method of any one of Examples 34 to 53, wherein the build plane has a total surface area of at least 1000 cm.sup.2.
[0418] Example 55. The method of any one of Examples 34 to 54, wherein the additive manufacturing process comprises building up the one or more 3D objects of each build platform from a plurality of layers of a curable material.
[0419] Example 56. The method of any one of Examples 34 to 55, further comprising heating the build plane during the additive manufacturing process.
[0420] Example 57. The method of Example 56, wherein the build plane is heated to a temperature within a range from 30° C. to 200° C.
[0421] Example 58. The method of any one of Examples 34 to 57, further comprising performing post-processing of the one or more 3D objects of at least one build platform while the one or more 3D objects remain on the at least one build platform.
[0422] Example 59. The method of Example 58, wherein the post-processing comprises one or more of removing residual curable material from the one or more 3D objects, washing the one or

more 3D objects in a solvent, or post-curing the one or more 3D objects.

[0423] Example 60. The method of Example 58 or 59, wherein performing the post-processing comprises placing the at least one build platform with the one or more 3D objects into a centrifuge, solvent bath, or post-curing oven.

[0424] Example 61. The method of any one of Examples 58 to 60, further comprising separating the one or more 3D objects from the at least one build platform after the post-processing.

[0425] Example 62. The method of any one of Examples 34 to 61, wherein the one or more 3D objects comprise one or more dental appliances.

CONCLUSION

[0426] Although many of the embodiments are described above with respect to systems, devices, and methods for manufacturing dental appliances, the technology is applicable to other applications and/or other approaches, such as manufacturing of other types of objects. Moreover, other embodiments in addition to those described herein are within the scope of the technology. Additionally, several other embodiments of the technology can have different configurations, components, or procedures than those described herein. A person of ordinary skill in the art, therefore, will accordingly understand that the technology can have other embodiments with additional elements, or the technology can have other embodiments without several of the features shown and described above with reference to FIGS. **1-34**.

[0427] The various processes described herein can be partially or fully implemented using program code including instructions executable by one or more processors of a computing system for implementing specific logical functions or steps in the process. The program code can be stored on any type of computer-readable medium, such as a storage device including a disk or hard drive. Computer-readable media containing code, or portions of code, can include any appropriate media known in the art, such as non-transitory computer-readable storage media. Computer-readable media can include volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage and/or transmission of information, including, but not limited to, random-access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), flash memory, or other memory technology; compact disc read-only memory (CD-ROM), digital video disc (DVD), or other optical storage; magnetic cassettes, magnetic tape, magnetic disk storage, or other magnetic storage devices; solid state drives (SSD) or other solid state storage devices; or any other medium which can be used to store the desired information and which can be accessed by a system device.

[0428] The descriptions of embodiments of the technology are not intended to be exhaustive or to limit the technology to the precise form disclosed above. Where the context permits, singular or plural terms may also include the plural or singular term, respectively. Although specific embodiments of, and examples for, the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology, as those skilled in the relevant art will recognize. For example, while steps are presented in a given order, alternative embodiments may perform steps in a different order. The various embodiments described herein may also be combined to provide further embodiments.

[0429] As used herein, the terms “generally,” “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

[0430] Moreover, unless the word “or” is expressly limited to mean only a single item exclusive from the other items in reference to a list of two or more items, then the use of “or” in such a list is to be interpreted as including (a) any single item in the list, (b) all of the items in the list, or (c) any combination of the items in the list. As used herein, the phrase “and/or” as in “A and/or B” refers to A alone, B alone, and A and B. Additionally, the term “comprising” is used throughout to mean including at least the recited feature(s) such that any greater number of the same feature and/or

additional types of other features are not precluded.
[0431] To the extent any materials incorporated herein by reference conflict with the present disclosure, the present disclosure controls.
[0432] It will also be appreciated that specific embodiments have been described herein for purposes of illustration, but that various modifications may be made without deviating from the technology. Further, while advantages associated with certain embodiments of the technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

Claims

1. An assembly for supporting 3D objects during an additive manufacturing process, the assembly comprising: a plurality of build platforms, each build platform configured to support one or more 3D objects during the additive manufacturing process; a carrier configured to support the plurality of build platforms; and an attachment mechanism configured to releasably couple the plurality of build platforms to the carrier during the additive manufacturing process such that the plurality of build platforms collectively form a build plane having a vertical deviation no greater than 500 μm.
2. The assembly of claim 1, wherein each build platform comprises: an upper surface configured to support the one or more 3D objects, a lower surface opposite the upper surface, and one or more protrusions formed on the lower surface such that when the build platform is releasably coupled to the carrier, a gap is formed between the lower surface of the build platform and an upper surface of the carrier.
3. The assembly of claim 1, wherein the attachment mechanism comprises, for each build platform: a first coupling device at or proximate to a first side of the build platform, and a second coupling device at or proximate to a second side of the build platform.
4. The assembly of claim 3, wherein the first coupling device uses a different type of coupling than the second coupling device.
5. The assembly of claim 4, wherein the first coupling device is a flexible coupling device and the second coupling device is a rigid coupling device.
6. The assembly of claim 4, wherein the first coupling device comprises a spring material and the second coupling device does not comprise a spring material.
7. The assembly of claim 1, wherein the attachment mechanism comprises a spring clip configured to releasably couple to a build platform of the plurality of build platforms.
8. The assembly of claim 7, wherein the spring clip is configured to releasably couple to an external portion of the build platform.
9. The assembly of claim 7, wherein the spring clip is configured to releasably couple to an internal portion of the build platform.
10. The assembly of claim 7, wherein the spring clip is configured to releasably couple to a recess formed in the build platform.
11. The assembly of claim 10, wherein the recess is formed in a lower surface of the build platform.
12. The assembly of claim 1, wherein the attachment mechanism comprises a rotatable clip configured to releasably couple to a build platform of the plurality of build platforms.
13. The assembly of claim 12, wherein: the rotatable clip is rotatable between a closed configuration and an open configuration, when in the closed configuration, the rotatable clip contacts the build platform, and when in the open configuration, the rotatable clip is spaced apart from the build platform.
14. The assembly of claim 1, wherein the build plane is a single continuous build plane.
15. The assembly of claim 1, wherein the build plane comprises a plurality of discrete regions.

16. The assembly of claim 1, wherein the build plane has a total surface area of at least 1000 cm.sup.2.
17. The assembly of claim 1, wherein the additive manufacturing process comprises building up the one or more 3D objects of each build platform from a plurality of layers of a curable material.
18. The assembly of claim 1, wherein the carrier includes or is thermally coupled to a heat source.
19. The assembly of claim 18, wherein the heat source is configured to heat the build plane to a temperature within a range from 30° C. to 200° C.
20. The assembly of claim 1, wherein the one or more 3D objects comprise one or more dental appliances.